

K318.2 Specific Requirements

K318.2.1 Prototype Tests. A prototype joint shall have been subjected to performance tests in accordance with [para. K304.7.2\(b\)](#) to determine the safety of the joint under test conditions simulating all expected service conditions. Testing shall include cyclic simulation.

K318.2.2 Prohibited Joints. Bell type and adhesive joints are not permitted.

PART 5 FLEXIBILITY AND SUPPORT

K319 FLEXIBILITY

Flexibility analysis shall be performed for each piping system. [Paragraphs 319.1](#) through [319.6](#) apply, except for [para. 319.4.1\(c\)](#). The computed displacement stress range shall be within the allowable displacement stress range in [para. K302.3.5](#) and shall also be included in the fatigue analysis in accordance with [para. K304.8](#).

K320 ANALYSIS OF SUSTAINED LOADS

K320.1 Basic Assumptions and Requirements

[Paragraph 320.1](#) applies, but refer to [para. K302.3.5\(c\)](#) instead of [para. 302.3.5\(c\)](#).

K320.2 Stress Due to Sustained Loads

[Paragraph 320.2](#) applies, except that references to expansion joints are not applicable.

K321 PIPING SUPPORT

Piping supports and methods of attachment shall be in accordance with [para. 321](#) except as modified below, and shall be detailed in the engineering design.

K321.1 General

K321.1.1 Objectives. [Paragraph 321.1.1](#) applies, but substitute "Chapter" for "Code" in [321.1.1\(a\)](#).

K321.1.4 Materials. [Paragraph 321.1.4](#) applies, but replace [321.1.4\(e\)](#) with the following:

(e) Attachments welded to the piping shall be of a material compatible with the piping and the service. Other requirements are specified in [paras. K321.3.2](#) and [K323.4.2\(b\)](#).

K321.3 Structural Attachments

K321.3.2 Integral Attachments. [Paragraph 321.3.2](#) applies, but substitute "K321.1.4(e)" for "321.1.4(e)" and "Chapter IX" for "Chapter V."

PART 6 SYSTEMS

K322 SPECIFIC PIPING SYSTEMS

K322.3 Instrument Piping

K322.3.1 Definition. Instrument piping within the scope of this Chapter includes all piping and piping components used to connect instruments to high pressure piping or equipment. Instruments, permanently sealed fluid-filled tubing systems furnished with instruments as temperature- or pressure-responsive devices, and control piping for air or hydraulically operated control apparatus (not connected directly to the high pressure piping or equipment) are not within the scope of this Chapter.

K322.3.2 Requirements. Instrument piping within the scope of this Chapter shall be in accordance with [para. 322.3.2](#) except that the design pressure and temperature shall be determined in accordance with [para. K301](#), and the requirements of [para. K310](#) shall apply. Instruments, and control piping not within the scope of this Chapter, shall be designed in accordance with [para. 322.3](#).

K322.6 Pressure-Relieving Systems

[Paragraph 322.6](#) applies, except for [para. 322.6.3](#).

K322.6.3 Overpressure Protection. Overpressure protection for high pressure piping systems shall conform to the following:

(a) The cumulative capacity of pressure relief devices shall be sufficient to prevent the pressure from rising more than 10% above the design pressure at the operating temperature during the relieving condition.

(b) System protection shall include one pressure relief device with a set pressure at or below the design pressure at the operating temperature for the relieving condition. The set pressure of additional pressure relief devices shall not exceed 105% of the design pressure.

PART 7 MATERIALS

K323 GENERAL REQUIREMENTS

(a) [Paragraph K323](#) states limitations and required qualifications for materials based on their inherent properties. Their use is also subject to requirements elsewhere in [Chapter IX](#) and in [Table K-1 \(Table K-1C\)](#).

(b) Specific attention should be given to the manufacturing process to ensure uniformity of properties throughout each piping component.

(c) See [para. K321.1.4](#) for support materials.

K323.1 Materials and Specifications

K323.1.1 Listed Materials

(a) Any material used in a pressure-containing piping component shall conform to a listed specification, except as provided in (b) below or in para. K323.1.2.

(b) Materials manufactured to specification editions different from those listed in Appendix E may be used, provided

(1) the requirements for chemical composition and heat-treatment condition in the edition of the specification to which the material was manufactured meet the requirements of the listed edition

(2) the specified minimum tensile and yield strengths, and, if applicable, the specified maximum tensile and yield strengths, required by the two editions of the specification are the same, and

(3) the material has been tested and examined in accordance with the requirements of the listed edition of the specification

A material that does not meet the requirements of (b)(1), (b)(2), and (b)(3) may be evaluated as an unlisted material in accordance with para. K323.1.2.

K323.1.2 Unlisted Materials. An unlisted material may be used, provided it conforms to a published specification covering chemistry, physical and mechanical properties, method and process of manufacture, heat treatment, and quality control, and otherwise meets the requirements of this Chapter. Allowable stresses shall be determined in accordance with the applicable allowable stress basis of this Chapter or a more conservative basis.

K323.1.3 Unknown Materials. Materials of unknown specification, type, or grade are not permitted.

K323.1.4 Reclaimed Materials. Reclaimed pipe and other piping components may be used provided they are properly identified as conforming to a listed specification, have documented service history for the material and fatigue life evaluation, and otherwise meet the requirements of this Chapter. Sufficient cleaning and inspection shall be made to determine minimum wall thickness and freedom from defects that would be unacceptable in the intended service.

K323.1.5 Product Analysis. Conformance of materials to the product analysis chemical requirements of the applicable specification shall be verified, and certification shall be supplied. Requirements for product analysis are defined in the applicable materials specification.

K323.1.6 Repair of Materials by Welding. A material defect may be repaired by welding, provided that all of the following criteria are met:

- (a) The material specification provides for weld repair.
- (b) The welding procedure and welders or welding operators are qualified as required by para. K328.2.

(c) The repair and its examination are performed in accordance with the material specification and with the owner's approval.

K323.2 Temperature Limitations

The designer shall verify that materials that meet other requirements of this Chapter are suitable for service throughout the operating temperature range.

K323.2.1 Upper Temperature Limits, Listed Materials. A listed material shall not be used at a temperature above that shown in the Max. Temp. column of Table K-1 (Table K-1C). (24)

K323.2.2 Lower Temperature Limits, Listed Materials

(a) The design minimum temperature for listed materials and their welds shall not be colder than the impact test temperature determined in accordance with para. K323.3.4(a), except as provided in (b) or (c) below.

(b) For listed materials and their welds subjected to neither longitudinal nor circumferential stresses greater than 41 MPa (6 ksi), the design minimum temperature shall not be colder than the lower of -48°C (-55°F) and the impact test temperature determined in accordance with para. K323.3.4(a).

(c) For listed materials and their welds exempted from Charpy impact testing by para. K323.3.1(a), the design minimum temperature shall not be colder than -198°C (-325°F) for fully austenitic stainless steels, or -48°C (-55°F) for other materials.

K323.2.3 Temperature Limits, Unlisted Materials. An unlisted material acceptable under para. K323.1.2 shall be qualified for service at all temperatures within a stated range from design minimum temperature to design (maximum) temperature, in accordance with para. K323.2.4. However, the upper temperature limit shall be less than the temperature for which an allowable stress, determined in accordance with para. 302.3.2, is governed by the creep or stress rupture provisions of that paragraph.

K323.2.4 Verification of Serviceability

(a) When an unlisted material is used, the designer is responsible for demonstrating the validity of the allowable stresses and other design limits, and of the approach taken in using the material, including the derivation of stress data and the establishment of temperature limits.

(b) Paragraph 323.2.4(b) applies except that allowable stress values shall be determined in accordance with para. K302.3.

K323.3 Impact Testing Methods and Acceptance Criteria

(24) K323.3.1 General

(a) Metallic piping components and their welds used in High Pressure Fluid Service shall be subjected to Charpy V-notch impact testing as described in (b), with the following exceptions:

- (1) nuts
- (2) washers
- (3) gaskets, including lens rings, cone rings, and other similar sealing devices

(4) piping components meeting the requirements of a standard listed in Table K326.1.1-1 that includes pressure-temperature ratings, and the design conditions are within these ratings. For such components, the impact test methods and acceptance criteria of para. 323.3 apply.

(5) other piping components and their welds where the maximum obtainable longitudinal Charpy V-notch specimen (see para. K323.3.3) has a width along the notch less than 2.5 mm (0.098 in.)

(b) Charpy impact testing shall be performed in accordance with Table K323.3.1-1 on representative samples using the testing methods described in paras. K323.3.2 through K323.3.4. Acceptance criteria are described in para. K323.3.5.

K323.3.2 Procedure. Paragraph 323.3.2 applies.

K323.3.3 Test Specimens

(a) Each set of impact test specimens shall consist of three specimen bars. Impact tests shall be made using standard 10 mm (0.394 in.) square cross section Charpy V-notch specimen bars oriented in the transverse direction.

(b) Where component size, shape, or both do not permit specimens as specified in (a) above, standard 10 mm square cross-section longitudinal Charpy specimens may be prepared.

(c) Where component size, shape, or both do not permit specimens as specified in (a) or (b) above, subsize longitudinal Charpy specimens may be prepared. Test temperature shall be reduced in accordance with Table 323.3.4-1.

(d) If necessary in (a), (b), or (c) above, corners of specimens parallel to and on the side opposite the notch may be as shown in Figure K323.3.3-1.

K323.3.4 Test Temperatures. For all Charpy impact tests, the test temperature criteria in (a) or (c) below shall be observed.

(a) Charpy impact tests shall be conducted at a temperature no warmer than the coldest of the following:

- (1) 21°C (70°F)
- (2) when the design minimum temperature is warmer than or equal to -48°C (-55°F), the lowest component temperature expected in service at which a material

or weld will be subjected to a longitudinal or circumferential stress greater than 41 MPa (6 ksi)

(3) when the design minimum temperature is colder than -48°C (-55°F), the design minimum temperature

(b) In specifying the temperatures in (a)(2) and (a)(3), the following shall be considered:

- (1) range of operating conditions
- (2) upset conditions
- (3) ambient temperature extremes
- (4) required leak test temperature

(c) Where the largest possible test specimen has a width along the notch less than the lesser of 80% of the material thickness or 8 mm (0.315 in.), the test shall be conducted at a reduced temperature in accordance with Table 323.3.4-1, considering the temperature as reduced below the test temperature required by (a).

K323.3.5 Acceptance Criteria

(a) *Minimum Energy Requirements for Materials Other Than Bolting.* The applicable minimum impact energy requirements for materials shall be those shown in Table K323.3.5-1. Lateral expansion shall be measured in accordance with ASTM A370 (for title see para. 323.3.2). The results shall be included in the impact test report.

(b) *Minimum Energy Requirements for Bolting Materials.* The applicable minimum energy requirements shall be those shown in Table K323.3.5-1 except as provided in Table K323.3.1-1.

(c) *Weld Impact Test Requirements.* Where two base metals having different required impact energy values are joined by welding, the impact test energy requirements shall equal or exceed the requirements of the base material having the lower required impact energy.

(d) Retests

(1) *Retest for Absorbed Energy Criteria.* When the average value of the three specimens equals or exceeds the minimum value permitted for a single specimen, and the value for more than one specimen is below the required average value, or when the value for one specimen is below the minimum value permitted for a single specimen, a retest of three additional specimens shall be made. The value for each of these retest specimens shall equal or exceed the required average value.

(2) *Retest for Erratic Test Results.* When an erratic result is caused by a defective specimen or uncertainty in the test, a retest will be allowed. The report giving test results shall specifically state why the original specimen was considered defective or which step of the test procedure was carried out incorrectly.

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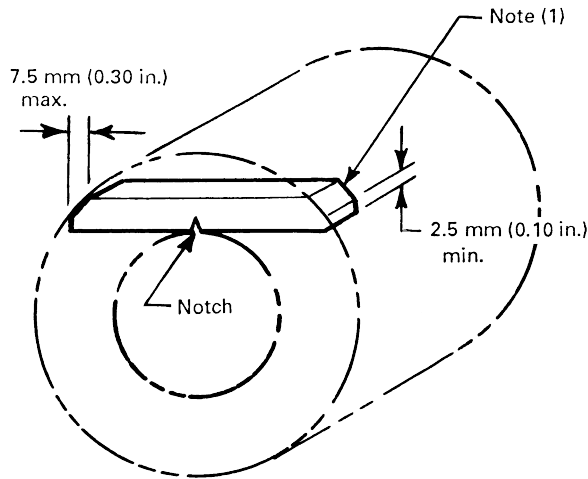
Table K323.3.1-1
Impact Testing Requirements

Test Characteristics		Column A Pipe, Tubes, and Components Made From Pipe or Tubes	Column B Other Components, Fittings, Etc.	Column C Bolts
Tests on Materials	Number of tests	As required by the material specification, or one test set per lot [Note (1)], whichever is greater, except as permitted by para. K323.3.1(a).		
	Location and orientation of specimens [Note (2)]	(a) Transverse to the longitudinal axis, with notch parallel to axis. [Note (3)] (b) Where component size, shape, or both do not permit specimens as specified in (a) above, paras. K323.3.3(b) through K323.3.3(d) apply as needed.	(a) Transverse to the direction of maximum elongation during rolling or to direction of major working during forging. Notch shall be oriented parallel to direction of maximum elongation or major working. [Note (3)] (b) If there is no single identifiable axis, e.g., for castings or triaxial forgings, specimens shall either meet the longitudinal values of Table K323.3.5-1, or three sets of orthogonal specimens shall be prepared, and the lowest impact values obtained from any set shall meet the transverse values of Table K323.3.5-1. (c) Where component size, shape, or both do not permit specimens as specified in (a) or (b) above, paras. K323.3.3(c) and K323.3.3(d) apply as needed.	(a) Bolts ≤52 mm (≤2 in.) nominal size made in accordance with ASTM A320 shall meet the impact requirements of that specification. (b) For all other bolts, longitudinal specimens shall be taken. The impact values obtained shall meet the transverse values of Table K323.3.5-1.
Tests on Welds in Fabrication or Assembly [Note (4)]	Procedure qualification thickness limits	Impact tests shall be valid only if the thickness of the test material meets the requirements of ASME BPVC, Section IX, Table QW-451.1 (including Notes), except that if the thickness is less than 16 mm ($\frac{5}{8}$ in.), the thickness of the test material shall be the minimum thickness qualified.		
	Testing requirements [Note (5)]	Impact tests shall be performed on the weld metal and the HAZ of the procedure qualification test coupon at a test temperature no warmer than that determined in accordance with para. K323.3.4.		
	Location and orientation of specimens	(1) Weld metal impact specimens shall be taken across the weld with the notch in the weld metal. Each specimen shall be oriented so that the notch axis is normal to the surface of the material and one face of the specimen shall be within 1.5 mm ($\frac{1}{16}$ in.) of the surface of the material. (2) Heat-affected zone impact specimens shall be taken across the weld and have sufficient length to locate the notch in the heat-affected zone, after etching. The notch shall be cut approximately normal to the material surface in such a manner as to include as much heat-affected zone material as possible in the resulting fracture.		
	Acceptance criteria	(1) Impact tests shall meet the acceptance criteria applicable to the base metal in accordance with para. K323.3.5. (2) The impact values obtained from both the weld metal and HAZ specimens shall meet the transverse values in Table K323.3.5-1.		

NOTES:

- (1) A lot shall consist of pipe or components of the same nominal size, made from the same heat of material, and heat treated together. If a continuous type furnace is used, pipe or components may be considered to have been heat treated together if they are processed during a single continuous time period at the same furnace conditions.
- (2) Impact tests shall be performed on a representative sample of material after completion of all heat treatment and forming operations involving plastic deformation, except that cold bends made in accordance with para. K304.2.1 need not be tested after bending.
- (3) For welded piping components, specimens shall be taken from the base metal, weld metal, and the heat-affected zone.
- (4) For welds in the fabrication or assembly of piping or components, including repair welds.
- (5) The test piece shall be large enough to permit preparing the number of specimens required by para. K323.3. If this is not possible, additional test pieces shall be prepared.

Figure K323.3.3-1
Example of an Acceptable Impact Test Specimen



GENERAL NOTE: This Figure illustrates how an acceptable transverse Charpy specimen can be obtained from a tubing or component shape too small for a full length standard specimen in accordance with ASTM A370. The corners of a longitudinal specimen parallel to and on the side opposite the notch may be as shown.

NOTE: (1) Corners of the Charpy specimen [see para. K323.3.3(d)] may follow the contour of the component within the dimension limits shown.

K323.4 Requirements for Materials

K323.4.1 General. Requirements in para. K323.4 apply to pressure-containing parts, not to materials used as supports, gaskets, packing, or bolting. See also Appendix F, para. F323.4.

K323.4.2 Specific Requirements

- (a) Ductile iron and other cast irons are not permitted.
- (b) Zinc-coated materials are not permitted for pressure containing components and may not be attached to pressure-containing components by welding.

(24) **K323.4.3 Metallic Clad and Lined Materials.** Materials with metallic cladding or lining may be used in accordance with the following provisions:

- (a) For metallic clad or lined piping components, the base metal shall be an acceptable material as defined in para. K323, and the thickness used in pressure design in accordance with para. K304 shall not include the thickness of the cladding or lining. The allowable stress used shall be that for the base metal at the design temperature. For such components, the cladding or lining may be any material that, in the judgment of the user, is suitable for the intended service and for the method of manufacture and assembly of the piping component.

(b) Fabrication by welding of clad or lined piping components and the inspection and testing of such components shall be done in accordance with applicable provisions of ASME BPVC, Section VIII, Division 1, Part UCL, UCL-30 through UCL-52, and the provisions of this Chapter.

(c) If a metallic liner also serves as a gasket or as part of the flange facing, the requirements and limitations in para. K308.4 apply.

K323.5 Deterioration of Materials in Service

Paragraph 323.5 applies.

K325 MISCELLANEOUS MATERIALS

Paragraph 325 applies.

PART 8 STANDARDS FOR PIPING COMPONENTS

K326 REQUIREMENTS FOR COMPONENTS

K326.1 Dimensional Requirements

K326.1.1 Listed Piping Components. Dimensional standards for piping components are listed in Table K326.1.1-1. Dimensional requirements contained in specifications listed in Appendix K shall also be considered requirements of this Chapter.

K326.1.2 Unlisted Piping Components. Piping components not listed in Table K326.1.1-1 or Appendix K shall meet the pressure design requirements described in para. K302.2.3 and the mechanical strength requirements described in para. K302.5.

K326.1.3 Threads. The dimensions of piping connection threads not otherwise covered by a governing component standard or specification shall conform to the requirements of applicable standards listed in Table K326.1.1-1 or Appendix K.

K326.2 Ratings of Components

K326.2.1 Listed Components. The pressure-temperature ratings of components listed in Table K326.1.1-1 are accepted for pressure design in accordance with para. K303.

K326.2.2 Unlisted Components. The pressure-temperature ratings of unlisted piping components shall conform to the applicable provisions of para. K304.

K326.3 Reference Documents

The documents listed in Table K326.1.1-1 contain references to codes, standards, and specifications not listed in Table K326.1.1-1. Such unlisted codes, standards, and

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Table K323.3.5-1
Minimum Required Charpy V-Notch Impact Values

Specimen Orientation	Pipe Wall or Component Thickness, mm (in.)	Number of Specimens [Note (1)]	Energy, J (ft-lbf), for Specified Minimum Yield Strength, MPa (ksi) [Note (2)]	
			≤932 (≤135)	>932 (>135)
Transverse	≤25 (≤1.0)	Average for 3	27 (20)	34 (25)
		Minimum for 1	20 (15)	27 (20)
	>25 and ≤51 (>1.0 and ≤2.0)	Average for 3	34 (25)	41 (30)
		Minimum for 1	27 (20)	33 (24)
	>51 (>2.0)	Average for 3	41 (30)	47 (35)
		Minimum for 1	33 (24)	38 (28)
Longitudinal	≤25 (≤1.0)	Average for 3	54 (40)	68 (50)
		Minimum for 1	41 (30)	54 (40)
	>25 and ≤51 (>1.0 and ≤2.0)	Average for 3	68 (50)	81 (60)
		Minimum for 1	54 (40)	65 (48)
	>51 (>2.0)	Average for 3	81 (60)	95 (70)
		Minimum for 1	65 (48)	76 (56)

NOTES:

- (1) See para. K323.3.5(c) for permissible retests.
- (2) Energy values in this table are for standard size specimens. For subsize specimens, these values shall be multiplied by the ratio of the actual specimen width to that of a full-size specimen, 10 mm (0.394 in.).

specifications shall be used only in the context of the listed documents in which they appear.

The design, materials, fabrication, assembly, examination, inspection, and testing requirements of this Chapter are not applicable to components manufactured in accordance with the documents listed in Table K326.1.1-1, unless specifically stated in this Chapter or in the listed document.

K326.4 Repair of Piping Components by Welding

A defect in a piping component may be repaired by welding subject to all of the following requirements:

- (a) The piping component specification provides for weld repair or, if not covered by a specification, the manufacturer allows for weld repair.
- (b) The welding procedure and welders or welding operators are qualified as required by para. K328.2.
- (c) The repair and its examination are performed in accordance with the piping component specification or, if not covered by a specification, the manufacturer's requirements.
- (d) The owner approves the weld repair.

PART 9 FABRICATION, ASSEMBLY, AND ERECTION

K327 GENERAL

Piping materials and components are prepared for assembly and erection by one or more of the fabrication processes covered in paras. K328, K330, K331, K332, and K333. When any of these processes is used in assembly or erection, requirements are the same as for fabrication.

K328 WELDING

Welding shall conform to the requirements of this Part and the applicable requirements of para. K311.

K328.1 Welding Responsibility

Each employer is responsible for

- (a) the welding performed by personnel of its organization
- (b) conducting the qualification tests required to qualify the welding procedure specifications used by personnel in its organization
- (c) conducting the qualification tests required to qualify its welders and welding operators

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**Table K326.1.1-1
 Component Standards**

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Designator	Title
Component Standards Having Established Ratings	
Fittings, Flanges, Gaskets, Line Blanks, and Valves	
ASME B16.5	Pipe Flanges and Flanged Fittings: NPS $\frac{1}{2}$ Through NPS 24 Metric/Inch Standard [Notes (1), (2)]
ASME B16.20	Metallic Gaskets for Pipe Flanges
ASME B16.34	Valves — Flanged, Threaded, and Welding End [Notes (1), (2)]
ASME B16.48	Line Blanks [Notes (1), (2)]
MSS SP-65	High-Pressure Chemical Industry Flanges and Threaded Stubs for Use with Lens Gaskets [Notes (1), (2)]
Component Standards Not Having Specific Ratings	
Fittings	
ASME B16.9	Factory-Made Wrought Butt welding Fittings [Notes (3), (4)]
ASME B16.11	Forged Fittings, Socket-Welding and Threaded [Notes (3), (4)]
Dimensional Standards	
Bolting	
ASME B18.2.1	Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)
ASME B18.2.2	Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)
Pipe	
ASME B36.10	Welded and Seamless Wrought Steel Pipe
ASME B36.19	Welded and Seamless Wrought Stainless Steel Pipe
Threads	
API 5B	Threading, Gauging, and Inspection of Casing, Tubing, and Line Pipe Threads
ASME B1.1	Unified Inch Screw Threads (UN, UNR, and UNJ Thread Forms)
ASME B1.20.1	Pipe Threads, General Purpose (Inch)
Welding End Preparation for Piping Components	
ASME B16.25	Butt welding Ends

GENERAL NOTE: The approved edition dates of these standards and specifications, along with the names and websites of the sponsoring organizations, are shown in [Appendix E](#).

NOTES:

- (1) Use of these standards does not require calculations in accordance with this Chapter. Where materials not listed in [Appendix K](#) are to be used, they need not be qualified as unlisted materials in accordance with [para. K323.1.2](#). However, materials shall not exceed the upper temperature limit in [para. K323.2.1](#) or [para. K323.2.3](#).
- (2) Piping components fully complying with these standards, including the pressure–temperature ratings contained therein, may be used, provided
 - (a) they meet the fluid service requirements in [Parts 3](#) and [4](#) of this Chapter.
 - (b) they are examined and leak tested in accordance with the requirements of [paras. K341](#) and [K345](#), respectively.
- (3) Use of these standards requires calculations in accordance with this Chapter. Where materials not listed in [Appendix K](#) are to be used, they shall be qualified as unlisted materials in accordance with [para. K323.1.2](#).
- (4) Piping components made in accordance with these standards may be used, provided
 - (a) their pressure design is in accordance with the requirements of [para. K302.2.2\(a\)](#).
 - (b) they meet the fluid service requirements in [Parts 3](#) and [4](#) of this Chapter.
 - (c) they are examined and leak tested in accordance with the requirements of [paras. K341](#) and [K345](#), respectively.
 - (d) they are Charpy V-notch impact tested in accordance with the methods described in [paras. K323.3.1](#) through [K323.3.4](#) and meet the acceptance criteria specified in [para. K323.3.5](#). Note that such impact testing may require the destruction of one component from the same lot as the component to be used in service [see [Table K323.3.1-1](#), Note (1)].
 - (e) the upper temperature limit in [para. K323.2.1](#) or [para. K323.2.3](#) is not exceeded.

K328.2 Welding Qualifications

K328.2.1 Qualification Requirements. Qualification of the welding procedures to be used and of the performance of welders and welding operators shall comply with the requirements of ASME BPVC, Section IX, except as modified herein.

(a) Impact tests shall be performed for all procedure qualifications in accordance with [para. K323.3](#).

(b) Except as provided in (f), test weldments shall be made using the same specification and type or grade of base metal(s), and the same specification and classification of filler metal(s) as will be used in production welding.

(c) Test weldments shall be subjected to essentially the same heat treatment, including cooling rate and cumulative time at temperature, as the production welds.

(d) When tensile specimens are required by ASME BPVC, Section IX, the yield strength shall also be determined, using the method required for the base metal. The yield strength of each test specimen shall be not less than the specified minimum yield strength at room temperature (S_Y) for the base metals joined. Where two base metals having different S_Y values are joined by welding, the yield strength of each test specimen shall be not less than the lower of the two S_Y values.

(e) Mechanical testing is required for all performance qualification tests.

(f) If the same specification and classification of filler metal(s) as will be used in production welding are employed, qualification on pipe or tubing shall also qualify for plate of the same type or grade, but qualification on plate does not qualify for pipe or tubing.

K328.2.2 Procedure Qualification by Others. Qualification of welding procedures by others is not permitted.

K328.2.3 Performance Qualification by Others. Welding performance qualification by others is not permitted.

K328.2.4 Qualification Records. [Paragraph 328.2.4](#) applies.

K328.3 Materials

(24) **K328.3.1 Electrodes and Filler Metal.** [Paragraph 328.3.1](#) applies, together with the following additional requirements:

(a) Welding electrodes and filler metals shall be specified in the engineering design.

(b) With the owner's approval, a welding electrode or filler metal not conforming to a specification listed in ASME BPVC, Section II, Part C may be used if a procedure qualification test, including an all-weld-metal test in accordance with AWS B4.0M or AWS B4.0⁸, is first successfully made.

⁸ AWS B4.0M or AWS B4.0. Standard Methods for Mechanical Testing of Welds. American Welding Society.

K328.3.2 Weld Backing Material. Backing rings shall not be used.

K328.3.3 Consumable Inserts. [Paragraph 328.3.3](#) applies, except that procedures shall be qualified as required by [para. K328.2](#).

K328.4 Preparation for Welding

K328.4.1 Cleaning. [Paragraph 328.4.1](#) applies.

K328.4.2 End Preparation

(a) *General*

(1) Butt weld end preparation is acceptable only if the surface is machined or ground to bright metal.

(2) Butt welding end preparation contained in ASME B16.25 or any other end preparation that meets the procedure qualification is acceptable. [For convenience, the basic bevel angles taken from ASME B16.25, with some additional J-bevel angles, are shown in [Figure 328.4.2-1](#), illustrations (a) and (b).]

(b) *Circumferential Welds*

(1) If components' ends are trimmed as shown in [Figure 328.4.2-1](#), illustration (a) or illustration (b) to accommodate consumable inserts, or as shown in [Figure K328.4.3-1](#) to correct internal misalignment, such trimming shall not result in a finished wall thickness before welding less than the required minimum wall thickness, t_m .

(2) It is permissible to size pipe ends of the same nominal size to improve alignment, if wall thickness requirements are maintained.

(3) Where necessary, weld metal may be deposited on the inside or outside of the component to permit alignment or provide for machining to ensure satisfactory seating of inserts.

(4) When a butt weld joins sections of unequal wall thickness and the thicker wall is more than $1\frac{1}{2}$ times the thickness of the other, end preparation and geometry shall be in accordance with acceptable designs for unequal wall thickness in ASME B16.5.

K328.4.3 Alignment

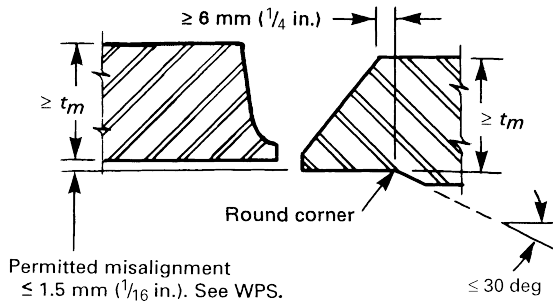
(a) *Girth Butt Welds*

(1) Inside diameters of components at the ends to be joined shall be aligned within the dimensional limits in the welding procedure and the engineering design, except that no more than 1.5 mm ($\frac{1}{16}$ in.) misalignment is permitted as shown in [Figure K328.4.3-1](#).

(2) If the external surfaces of the two components are not aligned, the weld shall be tapered between the two surfaces with a slope not steeper than 1:4.

(b) *Longitudinal Butt Joints.* Preparation for longitudinal butt welds (not made in accordance with a standard listed in [Table K-1](#) ([Table K-1C](#)) or [Table K326.1.1-1](#)) shall conform to the requirements of (a).

Figure K328.4.3-1
Pipe Bored for Alignment:
Trimming and Permitted Misalignment



(c) Branch Connection Welds

(1) The dimension m in Figure K328.5.4-1 shall not exceed $\pm 1.5 \text{ mm}$ ($1/16 \text{ in.}$).

(2) The dimension g in Figure K328.5.4-1 shall be specified in the engineering design and the welding procedure.

K328.5 Welding Requirements

K328.5.1 General. The requirements of paras. 328.5.1(b), 328.5.1(d), 328.5.1(e), and 328.5.1(f) apply in addition to the requirements specified below.

(a) All welds, including tack welds, repair welds, and the addition of weld metal for alignment [paras. K328.4.2(b)(3) and K328.4.3(c)(1)], shall be made by qualified welders or welding operators, in accordance with a qualified procedure.

(b) Tack welds at the root of the joint shall be made with filler metal equivalent to that used for the root pass. Tack welds shall be fused with the root pass weld, except that those that have cracked shall be removed. Bridge tacks (above the root) shall be removed.

K328.5.2 Fillet Welds. Fillet welds, where permitted (see para. K311.2.2), shall be fused with and shall merge smoothly into the component surfaces.

K328.5.3 Seal Welds. Seal welds are not permitted.

K328.5.4 Welded Branch Connections. Branch connection fittings (see para. 300.2), attached by smoothly contoured full penetration groove welds of a design that permits 100% interpretable radiographic examination, are the only types acceptable.

Figure K328.5.4-1 shows acceptable details of welded branch connections. The illustrations are typical and are not intended to exclude acceptable details not shown.

K328.5.5 Fabricated Laps. Fabricated laps are not permitted.

K328.6 Weld Repair

Paragraph 328.6 applies, except that procedures and performance shall be qualified as required by para. K328.2.1. See also para. K341.3.3.

K328.7 Attachment Welds

(24)

Paragraph 328.7 applies, except refer to paras. K341.4.1(d) and K341.3.2 instead of paras. 341.4 and 341.3.2, respectively.

K330 PREHEATING

K330.1 General

The preheat requirements herein apply to all types of welding, including tack welds and repair welds.

K330.1.1 Requirements. Paragraph 330.1.1 applies.

K330.1.2 Unlisted Materials. Paragraph 330.1.2 applies.

K330.1.3 Preheat Temperature Verification. Paragraph 330.1.3(a) applies. Temperature-indicating materials and techniques shall not be detrimental to the base metals. (24)

K330.1.4 Preheat Zone. Paragraph 330.1.4 applies.

K330.2 Specific Requirements

Paragraph 330.2 applies in its entirety.

K331 HEAT TREATMENT

K331.1 General

K331.1.1 Postweld Heat Treatment Requirements. The provisions of para. 331.1.1 and Tables 331.1.1-1 through 331.1.3-1 apply, except as specified below.

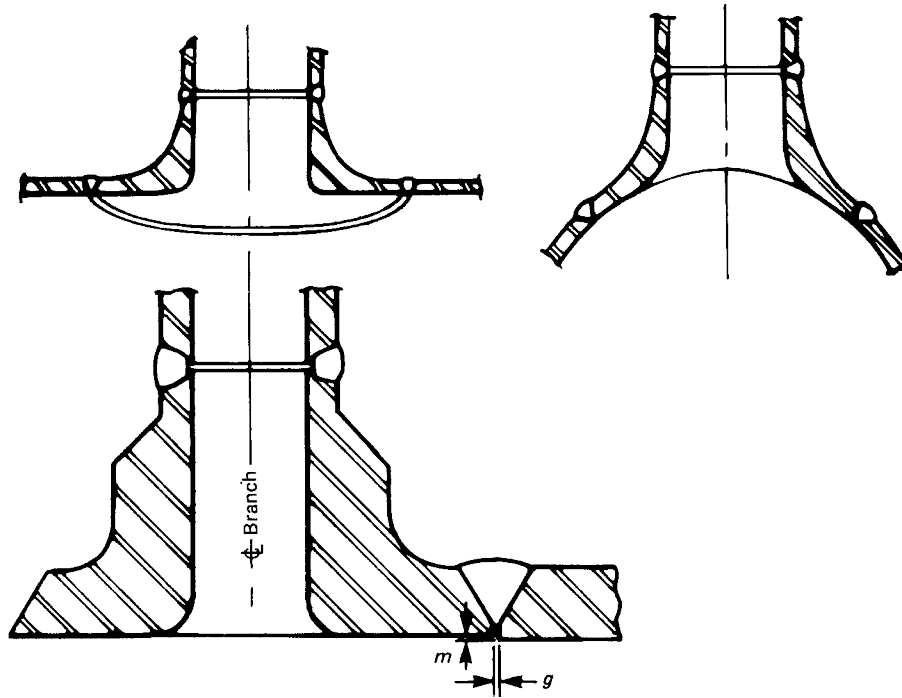
(a) The exemptions to mandatory postweld heat treatment in Table 331.1.3-1 for P-No. 1, all Group Nos., are not permitted when the control thickness exceeds 19 mm ($3/4 \text{ in.}$).

(b) The exemptions to mandatory postweld heat treatment in Table 331.1.3-1 for P-No. 4, Group No. 1 and P-No. 5A, Group No. 1 materials are not permitted.

(c) For quenched and tempered materials, any heat treatment of other than longitudinal welds shall be performed at a temperature no warmer than 28°C (50°F) below the tempering temperature of the material. When this heat treatment temperature is lower than the range specified in Table 331.1.1-1, the decrease in heat treatment temperature shall be accompanied by an increase in holding time in accordance with Table 331.1.2-1.

(d) Longitudinal welds in quenched and tempered material shall be heat treated in accordance with the applicable material specification.

Figure K328.5.4-1
Some Acceptable Welded Branch Connections Suitable for 100% Radiography



- (e) Refer to [para. K331.1.3](#) instead of [para. 331.1.3](#).
- (f) Refer to [Appendix K](#) instead of [Appendix A](#).
- (g) ASME B31P and [para. 328.7](#) do not apply.

K331.1.2 Other Heat Treatments. Heat treatment for bending and forming shall be in accordance with [para. K332.4](#).

K331.1.3 Definition of Thicknesses Governing Post-weld Heat Treatment. The provisions of [para. 331.1.3](#) apply, except for [para. 331.1.3\(b\)\(5\)](#).

K331.1.4 Heating and Cooling. [Paragraph 331.1.4](#) applies.

K331.1.6 Temperature Verification. Heat treatment temperature shall be checked by thermocouple pyrometers or other suitable methods to ensure that the WPS requirements are met. Temperature-indicating materials and techniques shall not be detrimental to the base metals.

K331.2 Specific Requirements

[Paragraph 331.2](#) applies, but refer to [para. K331.1](#) instead of [para 331.1](#).

K332 BENDING AND FORMING

K332.1 General

Pipe shall be hot or cold bent in accordance with a written procedure to any radius that will result in surfaces free of cracks and free of buckles. The procedure shall address at least the following, as applicable:

- (a) material specification and range of size and thickness
- (b) range of bend radii and fiber elongation
- (c) minimum and maximum metal temperature during bending
- (d) method of heating and maximum hold time
- (e) description of bending apparatus and procedure to be used
- (f) mandrels or material and procedure used to fill the bore
- (g) method for protection of thread and machined surfaces
- (h) examination to be performed
- (i) required heat treatment
- (j) postheat treatment dimensional adjustment technique

K332.2 Bending

K332.2.1 Bend Ovality. [Paragraph 332.2.1](#) applies. (24)

K332.2.2 Bending Temperature. Paragraph 332.2.2 applies, except that in cold bending of quenched and tempered ferritic materials, the temperature shall be at least 28°C (50°F) below the tempering temperature.

K332.3 Forming

Piping components shall be formed in accordance with a written procedure. The temperature range shall be consistent with material characteristics, end use, and specified heat treatment. The thickness after forming shall be not less than required by design. The procedure shall address at least the following, as applicable:

- (a) material specification and range of size and thickness
- (b) maximum fiber elongation expected during forming
- (c) minimum and maximum metal temperature during bending
- (d) method of heating and maximum hold time
- (e) description of forming apparatus and procedure to be used
- (f) materials and procedures used to provide internal support during forming
- (g) examination to be performed
- (h) required heat treatment

K332.4 Required Heat Treatment

K332.4.1 Hot Bending and Forming. After hot bending and forming, heat treatment is required for all thicknesses of P-Nos. 3, 4, 5A, 5B, 6, 10A, 10B, and 15E materials that are not quenched and tempered. Times and temperatures shall be in accordance with para. 331. Quenched and tempered materials shall be reheat treated to the original material specification.

K332.4.2 Cold Bending and Forming

(a) After cold bending and forming, heat treatment in accordance with (b) below is required, regardless of thickness, when specified in the engineering design or when the maximum calculated fiber elongation exceeds 5% strain or 50% of the basic minimum specified longitudinal elongation for the applicable specification, grade, and thickness for P-Nos. 1, 3, 4, 5A, 5B, 6, 10A, 10B, and 15E materials (unless it has been demonstrated that the selection of the pipe and the procedure for making the components provide assurance that the most severely formed portion of the material has retained an elongation of not less than 10%).

(b) Heat treatment is required regardless of thickness and shall conform to the temperatures and durations given in Table 331.1.1-1, except that for quenched and tempered materials, the stress relieving temperature shall not exceed a temperature 28°C (50°F) below the tempering temperature of the material.

K333 BRAZING AND SOLDERING

Brazing shall conform to the requirements of this Part and the applicable requirements of para. K317.2. Braze welding and soldering are not permitted.

K333.1 Brazing Responsibility

Each employer is responsible for

- (a) the brazing performed by personnel of its organization
- (b) conducting the qualification tests required to qualify the brazing procedure specifications used by personnel in its organization
- (c) conducting the qualification tests required to qualify its brazers and brazing operators

K333.2 Brazing Qualifications

K333.2.1 Qualification Requirements. The qualification of brazing procedures, brazers, and brazing operators shall be in accordance with the requirements of ASME BPVC, Section IX.

K333.2.2 Procedure Qualification by Others. Brazing procedure specifications qualified by a technically competent group or agency may be used, provided the following requirements are met:

- (a) The owner shall approve the use of brazing procedure specifications qualified by others.
- (b) The brazing procedure specifications shall meet the requirements of para. K333.2.1.
- (c) The employer's business name shall be shown on each brazing procedure specification and on each brazing procedure qualification record.
- (d) The employer shall accept responsibility for each brazing procedure qualification performed by others by signing and dating each brazing procedure qualification record.
- (e) The employer shall conduct the qualification tests required to qualify its brazers and brazing operators.

K333.2.3 Performance Qualification by Others. An employer may accept the performance qualification of a brazer or brazing operator made by a technically competent group or agency, provided the following requirements are met:

- (a) The owner shall approve the use of brazing performance qualifications qualified by others.
- (b) The brazer or brazing operator performance qualifications were made on pipe test coupons.
- (c) The employer shall have a brazing procedure specification such that the performance qualification is within the limits of the essential variables set forth in ASME BPVC, Section IX.
- (d) The employer shall have a copy of the performance qualification test record. The record shall show the name of the technically competent group or agency by whom the brazer or brazing operator was qualified and the date of

that qualification. Evidence shall also be provided that the brazer or brazing operator has maintained qualification in accordance with ASME BPVC, Section IX, Article XIII, QB-322.

(e) The employer's business name shall be shown on the brazing performance qualification record.

(f) The employer shall accept responsibility for the brazing performance qualifications performed by others by signing and dating the qualification.

K333.2.4 Qualification Records. The employer shall maintain copies of the brazing procedure specification and performance qualification records specified by ASME BPVC, Section IX. These records shall be made available to the Inspector upon request.

K333.3 Materials

Brazing filler metals and flux shall be in accordance with [para. 333.2.1](#).

K333.4 Preparation and Cleaning

[Paragraph 333.3](#) applies, except that soldering is not permitted.

(24) K335 ASSEMBLY AND ERECTION

[Paragraphs 335.1, 335.2, 335.9, and 335.10](#) apply.

K335.3 Threaded Joints

[Paragraph 335.3](#) applies, except that threaded joints shall not be seal welded.

K335.6 Special Joints

Special joints (as defined in [para. K318](#)) shall be installed and assembled in accordance with the manufacturer's instructions, as modified by the engineering design. Care shall be taken to ensure full engagement of joint members.

PART 10 INSPECTION, EXAMINATION, AND TESTING

K340 INSPECTION

[Paragraphs 340.1 through 340.4](#) apply.

K341 EXAMINATION

[Paragraphs 341.1 and 341.2](#) apply.

K341.3 Examination Requirements

K341.3.1 General. Prior to initial operation, each piping installation, including components and workmanship, shall be examined in accordance with [para. K341.4](#) and the engineering design. If heat treatment is

performed, examination shall be conducted after its completion.

K341.3.2 Acceptance Criteria. Acceptance criteria (24) shall be as stated in the engineering design and shall at least meet the applicable requirements stated in (a) through (c) below, and elsewhere in this Chapter.

(a) [Table K341.3.2-1](#) states acceptance criteria (limits on imperfections) for welds. See [Figure 341.3.2-1](#) for typical weld imperfections.

(b) Acceptance criteria for castings are specified in [para. K302.3.3](#).

(c) Acceptance criteria for surfaces on pressure-containing components where attachment welds have been removed are specified in [para. 341.3.2\(c\)](#).

K341.3.3 Defective Components and Workmanship. Defects (imperfections of a type or magnitude not acceptable by the criteria specified in [para. K341.3.2](#)) shall be repaired, or the defective item or work shall be replaced.

Examination shall be as follows:

(a) When the defective item or work is repaired, the repaired portion of the item or work shall be examined after the completion of any required heat treatment. The examination shall use the same methods and acceptance criteria employed for the original examination.

(b) When the defective item or work is replaced, the new item or work used to replace the defective item or work shall be examined after the completion of any required heat treatment. The examination shall use any method and applicable acceptance criteria that meet the requirements for the original examination.

K341.4 Extent of Required Examination

(24)

Piping shall be examined to the extent specified herein or to any greater extent specified in the engineering design.

K341.4.1 Visual Examination

(a) The following shall be 100% examined in accordance with [para. 344.2](#):

(1) all materials and components, to verify that they conform to specifications and are free from defects

(2) all fabrication, as defined in [para. 300.2](#)

(3) all completed welds, except those in components made in accordance with a listed standard

(4) all threaded, bolted, and other mechanical joints, to verify that they conform to the applicable requirements of [para. K335](#)

(5) all piping erection, to verify dimensions and alignment, including that of cold spring in [para. 335.1\(b\)](#) prior to assembly

(6) after assembly, all supports, guides, and points of cold spring, to ensure that movement of the piping under all conditions of startup, operation, and shutdown will be accommodated without undue binding or unanticipated constraint

Table K341.3.2-1
Acceptance Criteria for Welds

(24)

Imperfection	Examination Methods		Criteria (A to F) for Type of Welds [Note (1)]			
	Visual	Ultrasonics or Radiography	Girth Groove	Longitudinal Groove [Note (2)]	Fillet [Note (3)]	Branch Connection [Note (4)]
Crack	✓	✓	A	A	A	A
Incomplete fusion	✓	✓	A	A	A	A
Incomplete penetration	✓	✓	A	A	A	A
Rounded indications	...	✓	B	B	N/A	B
Linear indication	...	✓	C	C	N/A	C
Undercutting	✓	✓	A	A	A	A
Surface porosity or exposed slag inclusion	✓	...	A	A	A	A
Concave surface, concave root surface, or burn-through	✓	✓	D	D	N/A	D
Surface finish	✓	...	E	E	E	E
Face reinforcement or root reinforcement	✓	...	F	F	F	F

GENERAL NOTES:

- (a) Imperfections are evaluated by one or more of the examination methods given, as specified in [paras. K341.4.1](#) and [K341.4.2](#).
- (b) "N/A" indicates this Chapter does not establish acceptance criteria or does not require evaluation of this kind of imperfection for this type of weld.
- (c) Check (✓) indicates examination method generally used for evaluating this kind of imperfection.
- (d) Ellipsis (...) indicates examination method not generally used for evaluating this kind of imperfection.
- (e) Symbols A through F are explained in the table on the next page.

NOTES:

- (1) Criteria given are for required examination. More-stringent criteria may be specified in the engineering design.
- (2) Longitudinal welds include only those permitted in [paras. K302.3.4](#) and [K305](#). The criteria shall be met by all welds, including those made in accordance with a standard listed in [Table K326.1.1-1](#) or in [Appendix K](#).
- (3) Fillet welds include only those permitted in [para. K311.2.2](#).
- (4) Branch connection welds include only those permitted in [para. K328.5.4](#).

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Criterion Value Notes for Table K341.3.2-1

Criterion										
Symbol	Measure	Acceptable Value Limits [Note (1)]								
A	Extent of imperfection	Zero (no evident imperfection)								
B	Size and distribution of rounded indications	See ASME BPVC, Section VIII, Division 1, Appendix 4 [Note (2)]								
C	Linear indications characterized as neither incomplete penetration nor incomplete fusion. Linear indications are unacceptable if the amplitude exceeds the reference level, or linear indications have lengths that exceed Individual length [Note (3)] Cumulative length	6 mm ($\frac{1}{4}$ in.) for $\overline{T}_w \leq 19$ mm ($\frac{3}{4}$ in.) $\overline{T}_w/3$ for 19 mm ($\frac{3}{4}$ in.) $< \overline{T}_w \leq 57$ mm ($2\frac{1}{4}$ in.) 19 mm ($\frac{3}{4}$ in.) for $\overline{T}_w > 57$ mm ($2\frac{1}{4}$ in.) $\leq \overline{T}_w$ in any 12 $\overline{T}_w$ weld length [Note (2)]								
D	Depth of surface concavity, root surface concavity, or burn-through	<table><tr><th>$\overline{T}_w$, mm (in.)</th><th>Depth, mm (in.)</th></tr><tr><td>≤ 13 ($\leq \frac{1}{2}$)</td><td>≤ 1.5 ($\leq \frac{1}{16}$)</td></tr><tr><td>> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)</td><td>≤ 3 ($\leq \frac{1}{8}$)</td></tr><tr><td>> 51 (> 2.0)</td><td>≤ 4 ($\leq \frac{5}{32}$)</td></tr></table> and total joint thickness including weld reinforcement $\geq \overline{T}_w$	$\overline{T}_w$, mm (in.)	Depth, mm (in.)	≤ 13 ($\leq \frac{1}{2}$)	≤ 1.5 ($\leq \frac{1}{16}$)	> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)	≤ 3 ($\leq \frac{1}{8}$)	> 51 (> 2.0)	≤ 4 ($\leq \frac{5}{32}$)
$\overline{T}_w$, mm (in.)	Depth, mm (in.)									
≤ 13 ($\leq \frac{1}{2}$)	≤ 1.5 ($\leq \frac{1}{16}$)									
> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)	≤ 3 ($\leq \frac{1}{8}$)									
> 51 (> 2.0)	≤ 4 ($\leq \frac{5}{32}$)									
E	Surface roughness	$\leq 12.5\text{ }\mu\text{m}$ ($\leq 500\text{ }\mu\text{in.}$) R_a (see ASME B46.1 for definition of roughness average, R_a)								
F	Height of face reinforcement or root reinforcement [Note (4)] in any plane through the weld shall be within the limits of the applicable height value in the tabulation at the right. Weld metal shall be fused with and merge smoothly into the component surfaces.	<table><tr><th>$\overline{T}_w$, mm (in.)</th><th>Height, mm (in.)</th></tr><tr><td>≤ 13 ($\leq \frac{1}{2}$)</td><td>≤ 1.5 ($\leq \frac{1}{16}$)</td></tr><tr><td>> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)</td><td>≤ 3 ($\leq \frac{1}{8}$)</td></tr><tr><td>> 51 (> 2.0)</td><td>≤ 4 ($\leq \frac{5}{32}$)</td></tr></table>	$\overline{T}_w$, mm (in.)	Height, mm (in.)	≤ 13 ($\leq \frac{1}{2}$)	≤ 1.5 ($\leq \frac{1}{16}$)	> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)	≤ 3 ($\leq \frac{1}{8}$)	> 51 (> 2.0)	≤ 4 ($\leq \frac{5}{32}$)
$\overline{T}_w$, mm (in.)	Height, mm (in.)									
≤ 13 ($\leq \frac{1}{2}$)	≤ 1.5 ($\leq \frac{1}{16}$)									
> 13 ($> \frac{1}{2}$) and ≤ 51 (≤ 2.0)	≤ 3 ($\leq \frac{1}{8}$)									
> 51 (> 2.0)	≤ 4 ($\leq \frac{5}{32}$)									

NOTES:

- (1) Where two limiting values are given, the lesser measured value governs acceptance. $\bar{T}_w$ is the nominal wall thickness of the thinner of two components joined by a butt weld.
- (2) Porosity and inclusions such as slag or tungsten are defined as rounded indications where the maximum length is three times the width or less. These indications may be circular, elliptical, or irregular in shape; may have tails; and may vary in density. Indications where the length is greater than three times the width are defined as linear indications and may also be slag, porosity, or tungsten.
- (3) For ultrasonic examination, refer to para. K344.6.3 for acceptable value limits.
- (4) For groove welds, height is the lesser of the measurements made from the surfaces of the adjacent components. For fillet welds, height is measured from the theoretical throat; root reinforcement does not apply. Required thickness t_m shall not include face reinforcement or root reinforcement.

(7) all erected piping, for evidence of defects that would require repair or replacement, and for other evident deviations from the intent of the design

(b) *Pressure-Containing Threads.* 100% examination for finish and fit is required. Items with visible imperfections in thread finish or any of the following defects shall be rejected:

(1) *Tapered Threads.* Failure to meet gaging requirements in API Spec 5B or ASME B1.20.1, as applicable.

(2) *Straight Threads.* Excessively loose or tight fit when gaged for light interference fit.

(c) *Brazed Joints.* 100% of all brazed joints shall be examined by in-process examination in accordance with para. K344.7.

(d) *Attachment Welds.* Surfaces on pressure-containing components where attachment welds have been removed shall be subjected to visual examination in accordance with para. 344.2.

K341.4.2 Radiographic and Ultrasonic Examination

(a) All girth, longitudinal, and branch connection welds shall be 100% radiographically examined, except as permitted in (b) below.

(b) When specified in the engineering design and with the owner's approval, ultrasonic examination of welds may be substituted for radiographic examination where $T_w \geq 13 \text{ mm } (\frac{1}{2} \text{ in.})$.

(c) In-process examination (see para. 344.7) shall not be substituted for radiographic or ultrasonic examination of welds.

K341.4.3 Magnetic Particle and Liquid Penetrant Examination. Surfaces on pressure-containing components where attachment welds have been removed shall be subjected to magnetic particle or liquid penetrant examination in accordance with para. K344.3 or para. K344.4, respectively.

K341.4.4 Certifications and Records. Paragraph 341.4.1(c) applies.

K341.5 Supplementary Examination

Any of the examination methods described in para. K344 may be specified by the engineering design to supplement the examination required by para. K341.4. The extent of supplementary examination to be performed and any acceptance criteria that differ from those specified in para. K341.3.2 shall be specified in the engineering design.

K341.5.1 Hardness Tests. Paragraph 341.5.2 applies.

K341.5.2 Examinations to Resolve Uncertainty. Paragraph 341.5.3 applies.

K342 EXAMINATION PERSONNEL

Paragraph 342 applies in its entirety.

K343 EXAMINATION PROCEDURES

Paragraph 343 applies, except that the examination methods shall comply with para. K344.

K344 EXAMINATION METHODS

K344.1 General

(24)

Paragraph 344.1 applies in its entirety.

K344.2 Visual Examination

Paragraph 344.2 applies in its entirety.

K344.3 Magnetic Particle Examination

The method for magnetic particle examination shall be as specified in

(a) paragraph K302.3.3(b) for castings

(b) ASME BPVC, Section V, Article 7 for welds and other components

K344.4 Liquid Penetrant Examination

The method for liquid penetrant examination shall be as specified in

(a) paragraph K302.3.3(b) for castings

(b) ASME BPVC, Section V, Article 6 for welds and other components

K344.5 Radiographic Examination

The method for radiographic examination shall be as specified in

(a) paragraph K302.3.3(c) for castings

(b) ASME BPVC, Section V, Article 2 for welds and other components

K344.6 Ultrasonic Examination

K344.6.1 Castings. The method for ultrasonic examination of castings shall be as specified in para. K302.3.3(c).

K344.6.2 Pipe and Tubing

(24)

(a) *Method.* Pipe and tubing, required or selected in accordance with Table K305.1.2-1 to undergo ultrasonic examination, shall pass a 100% examination for longitudinal defects in accordance with ASTM E213.⁹ Longitudinal (axial) reference notches shall be introduced on the outer and inner surfaces of the calibration (reference) standard in accordance with Figure 3(c) of ASTM E213 to a depth not greater than the larger of 0.10 mm (0.004 in.) or 4% of specimen thickness and a length not more than 10 times the notch depth.

(b) *Acceptance Criteria.* Any indication greater than that produced by the calibration notch represents a defect; defective pipe and tubing shall be rejected.

⁹ ASTM E213. Standard Practice for Ultrasonic Testing of Metal Pipe and Tubing. ASTM International.

(c) *Records.* For pipe and tubing that passes this examination, a report shall be prepared that contains at least the information specified in 14.2.1 through 14.2.6 of ASTM E213.⁹

(24) K344.6.3 Welds

(a) *Method.* The method for ultrasonic examination of welds shall be as specified in ASME BPVC, Section V, Article 4 for nominal thickness, $\bar{T}_w$, greater than or equal to 13 mm ($\frac{1}{2}$ in.) but less than 25 mm (1.0 in.), and ASME BPVC, Section VIII, Division 3, Article KE-3, KE-301 and KE-302 for nominal wall thickness, $\bar{T}_w$, greater than or equal to 25 mm (1.0 in.).

(b) *Acceptance Criteria.* Cracks, incomplete fusion, incomplete penetration, or undercutting are unacceptable regardless of size or length (see Table K341.3.2-1). In addition, for a linear indication

(1) for nominal wall thickness, $\bar{T}_w$, greater than or equal to 13 mm ($\frac{1}{2}$ in.) but less than 25 mm (1.0 in.), the acceptance criterion for the thickness to be examined specified in para. 344.6.2 applies.

(2) for nominal wall thickness, $\bar{T}_w$, greater than or equal to 25 mm (1.0 in.), the acceptance criteria specified in ASME BPVC, Section VIII, Division 3, Article KE-3, KE-333 for the thickness to be examined apply.

K344.7 In-Process Examination

Paragraph 344.7 applies in its entirety.

(24) K344.8 Eddy Current Examination

K344.8.1 Method. The method for eddy current examination of pipe and tubing shall follow the general guidelines of ASME BPVC, Section V, Article 8, subject to the following specific requirements:

(a) Cold drawn austenitic stainless steel pipe and tubing, selected in accordance with Table K305.1.2-1 for eddy current examination, shall pass a 100% examination for longitudinal defects.

(b) A calibration (reference) standard shall be prepared from a representative sample. A longitudinal (axial) reference notch shall be introduced on the inner surface of the standard to a depth not greater than the larger of 0.10 mm (0.004 in.) or 5% of specimen thickness and a length not more than 6.4 mm (0.25 in.).

K344.8.2 Acceptance Criteria. Any indication greater than that produced by the calibration notch represents a defect; defective pipe or tubing shall be rejected.

K344.8.3 Records. For pipe and tubing that passes this examination, a report shall be prepared that includes at least the following information:

- (a) material identification by type, size, lot, heat, etc.
- (b) listing of examination equipment and accessories
- (c) details of examination technique (including examination speed and frequency) and end effects, if any

(d) description of the calibration standard, including dimensions of the notch, as measured

(e) examination results

K345 LEAK TESTING

K345.1 Required Leak Test

(24)

Prior to initial operation and after completion of the examinations required by para. K341, each piping system shall be leak tested.

(a) Each weld and each piping component, except bolting and individual gaskets to be used during final system assembly and pressure relief devices to be used during operation, shall be hydrostatically or pneumatically leak tested in accordance with para. K345.4 or para. K345.5, respectively. The organization conducting the test shall ensure that during the required leak testing of components and welds, adequate protection is provided to prevent injury to people and damage to property from missile fragments, shock waves, or other consequences of any failure that might occur in the pressurized system.

(b) In addition to the requirements of (a) above, a leak test of the installed piping system, excluding pressure relief devices to be used during operation, shall be conducted at a pressure not less than 110% of the design pressure to ensure tightness, except as provided in (c) or (d) below.

(c) If the leak test required in (a) above is conducted on the installed piping system, the additional test in (b) above is not required.

(d) With the owner's approval, pressure relief devices to be used during operation may be included in the leak test required in (b) above. The leak test pressure may be reduced to prevent the operation of, or damage to, the pressure relief devices, but shall not be less than 90% of the lowest set pressure of the pressure relief devices in the system.

(e) For closure welds, examination in accordance with para. K345.2.3(c) may be substituted for the leak test required in (a) above.

(f) None of the following leak tests may be used in lieu of the leak tests required in para. K345.1:

- (1) initial service leak test (see para. 345.7)
- (2) sensitive leak test (see para. 345.8)
- (3) alternative leak test (see para. 345.9)

K345.2 General Requirements for Leak Tests

Paragraphs 345.2.4 through 345.2.7 apply. See below for paras. K345.2.1, K345.2.2, and K345.2.3.

K345.2.1 Limitations on Pressure

(a) *Through-Thickness Yielding.* If the test pressure would produce stress (exclusive of stress intensification) in excess of S_{yt} at the outside surface of a component at test temperature, as determined by calculation or by testing in

accordance with [para. K304.7.2\(b\)](#), the test pressure may be reduced to the maximum pressure that will result in a stress (exclusive of stress intensification) at the outside surface that will not exceed S_{yt} .

(b) The provisions of [paras. 345.2.1\(b\)](#) and [345.2.1\(c\)](#) apply.

- (24) **K345.2.2 Other Test Requirements.** Paragraph [345.2.2](#) applies. In addition, except for those piping components exempted from impact testing in [para. K323.3.1\(a\)](#), the minimum metal temperature during leak testing shall be not less than the impact test temperature (see [para. K323.3.4](#)).

- (24) **K345.2.3 Special Provisions for Leak Testing.** Paragraphs (a), (b), and (c) below apply only to the leak test specified in [para. K345.1\(a\)](#). They are not applicable to the installed piping system leak test specified in [para. K345.1\(b\)](#).

(a) *Piping Components and Subassemblies.* Piping components and subassemblies may be leak tested either separately or as assembled piping.

(b) *Reassembled Mechanical Joints.* The following mechanical joints need not be subjected to an additional leak test in accordance with [para. K345.1\(a\)](#) if disconnected and reassembled after leak testing:

(1) mechanical joints used to connect piping components and subassemblies that have previously been leak tested

(2) mechanical joints used to connect temporary piping components for the purpose of isolating equipment, instruments, or other piping during the leak test

(c) *Closure Welds.* Leak testing of the final weld connecting piping systems or components that have been successfully leak tested is not required, provided the weld is examined in-process in accordance with [para. 344.7](#) and passes the required 100% radiographic examination in accordance with [para. K341.4.2](#).

(d) See also [Appendix F, para. F345.2.3](#).

K345.3 Preparation for Leak Test

[Paragraph 345.3](#) applies in its entirety.

K345.4 Hydrostatic Leak Test

[Paragraph 345.4.1](#) applies. See [paras. K345.4.2](#) and [K345.4.3](#).

K345.4.2 Test Pressure for Components and Welds. Except as provided in [para. K345.4.3](#), the hydrostatic test pressure at every point in a metallic piping system shall be as follows:

- (a) not less than 1.25 times the design pressure.
 (b) when the design temperature is greater than the test temperature, the minimum test pressure, at the point under consideration, shall be calculated by [eq. \(38\)](#)

$$P_T = 1.25PS_T/S \quad (38)$$

where

P = internal design gage pressure

P_T = minimum test gage pressure

S = allowable stress at component design temperature for the prevalent pipe material; see [Table K-1 \(Table K-1C\)](#)

S_T = allowable stress at test temperature for the prevalent pipe material; see [Table K-1 \(Table K-1C\)](#)

(c) in those cases where the piping system may not include pipe itself, any other component in the piping system, other than pipe-supporting elements and bolting, may be used to determine the S_T/S ratio based on the applicable allowable stresses obtained from [Table K-1 \(Table K-1C\)](#). In those cases where the piping system may be made up of equivalent lengths of more than one material, the S_T/S ratio shall be based on the minimum calculated ratio of the included materials.

K345.4.3 Hydrostatic Test of Piping With Vessels as a System. [Paragraph 345.4.3\(a\)](#) applies.

K345.5 Pneumatic Leak Test

[Paragraph 345.5](#) applies, except for [para. 345.5.4](#). See [para. K345.5.4](#).

K345.5.4 Test Pressure. The pneumatic test pressure for components and welds shall be identical to that required for the hydrostatic test in accordance with [para. K345.4.2](#).

K345.6 Hydrostatic-Pneumatic Leak Test for Components and Welds

If a combination hydrostatic-pneumatic leak test is used, the requirements of [para. K345.5](#) shall be met, and the pressure in the liquid-filled part of the piping shall not exceed the limits stated in [para. K345.4.2](#).

K346 RECORDS

K346.1 Responsibility

It is the responsibility of the designer, the manufacturer, the fabricator, and the erector, as applicable, to prepare the records required by this Chapter and by the engineering design.

K346.2 Required Records

At least the following records, as applicable, shall be provided to the owner or the Inspector by the person responsible for their preparation:

- (a) the engineering design
 (b) the written report required in [para. K300\(b\)\(2\)](#)
 (c) material certifications

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(d) procedures used for fabrication, welding, brazing, heat treatment, examination, and testing

(e) repair records of materials and piping components, including the welding procedure used for each, and location of repairs

(f) performance qualifications for welders, brazers, and welding and brazing operators

(g) qualifications of examination personnel

(h) records of examination of pipe and tubing for longitudinal defects as specified in [paras. K344.6.2\(c\)](#) and [K344.8.3](#), as applicable

K346.3 Retention of Records

The owner shall retain one set of the required records for at least 5 yr after they are received.

Chapter X High Purity Piping

U300 GENERAL STATEMENTS

(a) [Chapter X](#) pertains to piping designated by the owner as being in High Purity Fluid Service. See also [Appendix M](#).

(b) The organization, content, and paragraph designations of this Chapter correspond to those of the base Code ([Chapters I](#) through [VI](#)), [Chapter VII](#), and [Chapter VIII](#). The prefix U is used to designate [Chapter X](#) requirements.

(c) Provisions and requirements of the base Code, [Chapter VII](#), and [Chapter VIII](#) apply only as stated in this Chapter.

(d) For piping not in High Purity Fluid Service, Code requirements are found in [Chapters I](#) through [IX](#).

(e) *High Purity Piping*. [Chapter X](#) provides alternative rules for design and construction of piping designated by the owner as being High Purity Fluid Service.

(1) These rules apply only when specified by the owner, and only as a whole, not in part.

(2) [Chapter X](#) rules do not provide for High Pressure Fluid Service.

(3) [Chapter VII](#) applies to nonmetallic piping and piping lined with nonmetals in High Purity Fluid Service.

(f) [Chapter I](#) applies.

PART 1 CONDITIONS AND CRITERIA

[Chapter II, Part 1](#) applies. See [para. U301.3.2\(b\)\(5\)](#).

U301 DESIGN CONDITIONS

U301.3 Design Temperature

U301.3.2 Uninsulated Piping Components

(b)

(5) compression, face seal, and hygienic clamped fittings and joints — 100% of the fluid temperature

PART 2 PRESSURE DESIGN OF PIPING COMPONENTS

[Chapter II, Part 2](#) applies.

U304 PRESSURE DESIGN OF COMPONENTS

U304.5 Pressure Design of Flanges and Blanks

U304.5.3 Blanks. See [Figure U304.5.3-1](#) for representative configuration for metal face seal blanks.

PART 3 FLUID SERVICE REQUIREMENTS FOR PIPING COMPONENTS

[Chapter II, Part 3](#) applies. See [paras. U306.6, U307.3, and U308](#).

U306 FITTINGS, BENDS, MITERS, LAPS, AND BRANCH CONNECTIONS

U306.6 Tube Fittings

(a) Tube fittings not listed in [Table 326.1.1-1](#) or [Appendix A](#) shall meet the pressure design requirements described in [para. 302.2.3](#) and the mechanical strength requirements described in [para. 303](#).

(b) Compression-type tube fittings may be used in accordance with [para. U315.2](#) provided that the type of fitting selected complies with the following:

(1) The gripping action of the fitting shall provide vibration resistance as demonstrated by exhibiting a stress intensity factor equal to or less than 1.5.

(2) Intermixing of components from different manufacturers is permitted only when specified in the engineering design.

(c) Face seal or hygienic clamp-type fittings in which the tightness of the joint is provided by a seating surface other than the threads (e.g., a metal face-seal fitting comprising internal and external threaded components, glands, and gasket or other designs shown typically in [Figure U335.7.1-1](#)) may be used.

U307 VALVES AND SPECIALTY COMPONENTS

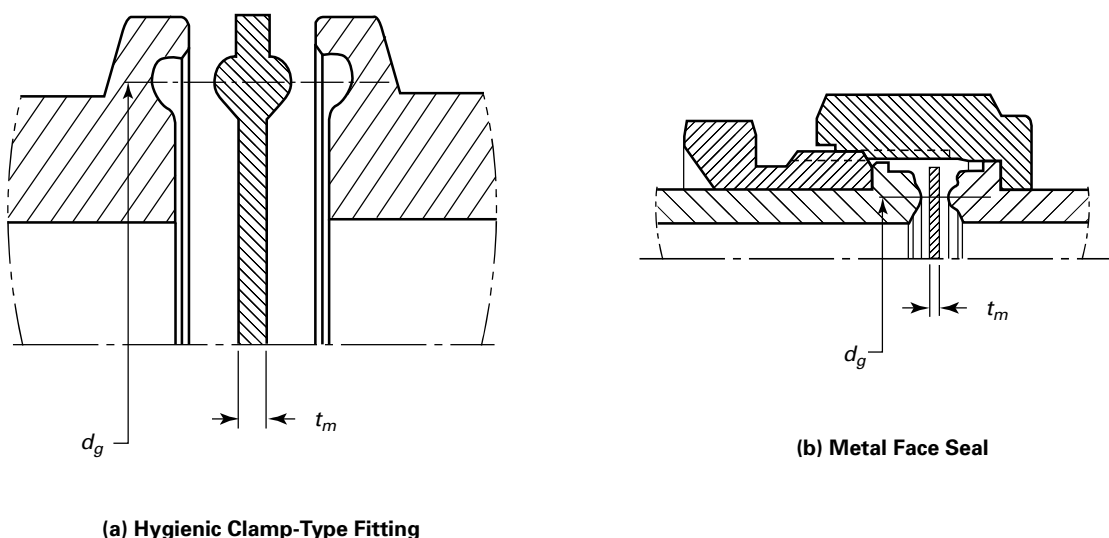
U307.3 High Purity Fluid Service Valves

Valves such as ball, bellows, and diaphragm valves designed for High Purity Fluid Service that are not listed in [Table 326.1.1-1](#) shall meet the pressure design requirements described in [para. 302.2.2](#) and the mechanical strength requirements described in [para. 303](#).

(24)

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Figure U304.5.3-1
Blanks



U308 FLANGES, BLANKS, FLANGE FACINGS, AND GASKETS

Flanges should be avoided whenever possible. When flanges are utilized, [para. 308](#) applies, except expanded joint flanges described in [para. 308.2.2](#) are not permitted.

PART 4 FLUID SERVICE REQUIREMENTS FOR PIPING JOINTS

[Chapter II, Part 4](#) applies, except expanded joints, flared tube fittings, and caulked joints, described in [paras. 313, 315, and 316](#), respectively, are not permitted. See [paras. U311, U311.1\(c\), U314, and U315](#).

U311 WELDED JOINTS

[Paragraph 311](#) applies, except for [para. 311.1\(c\)](#). See [para. U311.1\(c\)](#).

U311.1 General

(c) Examination shall be in accordance with [para. U341.4.1](#).

U314 THREADED JOINTS

Threaded joints should be avoided whenever possible. When threaded joints are utilized, [para. 314](#) applies.

U315 TUBING JOINTS

[Paragraph 315](#) applies. See [paras. U315.1, U315.2\(c\), and U315.3](#).

U315.1 General

In selecting and applying compression, face seal, and hygienic clamp-type tube fittings, the designer shall consider the possible adverse effects on the joints of such factors as assembly and disassembly, cyclic loading, vibration, shock, and thermal expansion and contraction. See [para. FU315](#).

U315.2 Joints Conforming to Listed Standards

(c) Joints using compression, face seal, hygienic clamp, and automatic welding tube fittings covered by listed standards may be used.

U315.3 Joints Not Conforming to Listed Standards

(a) Compression-type tube fitting joints shall be fully gageable on initial installation to ensure sufficient tightening.

(b) Safeguarding is required for face seal or hygienic clamp-type joints used under severe cyclic conditions.

PART 5 FLEXIBILITY AND SUPPORT

[Chapter II, Part 5](#) applies. See [para. U319.3.6](#).

U319 PIPING FLEXIBILITY

U319.3 Properties for Flexibility Analysis

U319.3.6 Flexibility and Stress Intensification Factors. [Paragraph 319.3.6](#) applies; however, piping components used in high-purity applications, e.g.,

multiport block valves, hygienic unions, crosses, and point-of-use and adaptor fittings, often do not have geometries similar to those in ASME B31J.

PART 6 SYSTEMS

Chapter II, Part 6 applies.

PART 7 METALLIC MATERIALS

The provisions and requirements in Chapter III for materials apply. Materials commonly used in high purity process piping systems include austenitic, ferritic, and duplex stainless steels, and nickel and nickel alloys.

PART 8 STANDARDS FOR PIPING COMPONENTS

Chapter IV applies.

PART 9 FABRICATION, ASSEMBLY, AND ERECTION

U327 GENERAL

Metallic piping materials and components are prepared for assembly and erection by one or more of the fabrication processes covered in paras. U328, U330, U331, and U332. When any of these processes is used in assembly or erection, requirements are the same as for fabrication.

U328 WELDING

Paragraph 328 applies, except for paras. 328.3.2, 328.5.4, and 328.5.5. See paras. U328.2(a), U328.2(b), U328.2(c), U328.4, U328.4.2, U328.4.4, and U328.5.1(g) for additional requirements.

U328.2 Welding and Brazing Qualification

The welding process shall be orbital GTAW, except for the following:

- (a) Tack welds made prior to orbital welding may be manual GTAW.
- (b) Manual GTAW may be used, with the owner's approval, on those welds where the orbital welding equipment cannot be used.
- (c) A change in the type or nominal composition of the backing (purge) gas shall require requalification.

U328.4 Preparation for Welding

Paragraph 328.4.1 applies. Additionally, when weld coupon examination is specified in the engineering design or in the referencing code or standard (e.g., ASME BPE or SEMI), primary weld coupons shall be made in accordance with para. U328.4.4(b)(1) and examined in accordance with para. U344.8 prior to the start of production welding. This will demonstrate that the orbital welding equipment is set up properly and the weld program is sufficient to make repeatable production welds in accordance with the qualified welding procedure specification (WPS).

U328.4.2 End Preparations

(24)

(a) Paragraph 328.4.2(a)(1) applies, with the exception that discoloration requirements shall be based on the referencing code or standard (e.g., ASME BPE or SEMI).

(b) End preparations for GTAW are as follows:

(1) Components having nominal wall thickness of 3.18 mm (0.125 in.) and less shall have ends prepared in accordance with the referencing code or standard (e.g., ASME BPE or SEMI).

(2) Pipe from 4.76 mm to 22.22 mm (0.188 in. to 0.875 in.) nominal wall thickness shall be in accordance with ASME B16.25. "J" or bevel groove angles suitable for orbital welding are shown in Figure U328.4.2-1.

(3) For pipe with nominal wall thickness 4.76 mm to 22.22 mm (0.188 in. to 0.875 in.), a modified "J" end preparation with an extended land as shown in Figure U328.4.2-1, illustration (a) may be used.

(4) A pipe with a modified "J" end preparation may be welded to a pipe fitting with a 37½-deg bevel end as shown in Figure U328.4.2-1, illustration (b).

U328.4.4 Preparation of Weld Coupons

(a) Weld coupons shall be made by qualified welding operators using the same qualified WPS and the same variables used for production welds.

(b) Methods

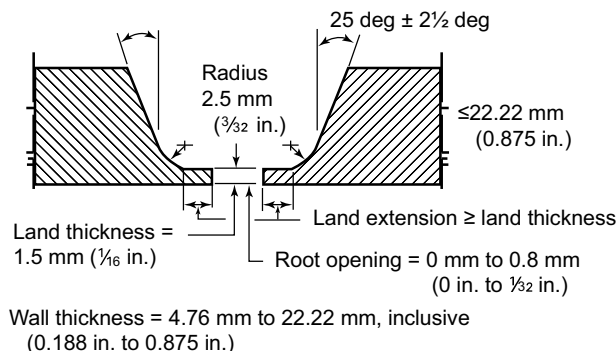
(1) Primary weld coupons shall be made from two short sections of tubing selected from the same diameter, wall thickness, and alloy as the material used for production. Sections shall be of sufficient length for fit up in the weld head allowing for attachment of inside diameter purge apparatus outside of the weld head. The sections shall be welded together in a square groove weld on a butt joint.

(2) Production weld coupons may be made in accordance with (1) or, at the owner's discretion, may be cut from actual production welds. The weld coupons shall be selected to ensure that the work product of each welding operator doing the production welding is represented.

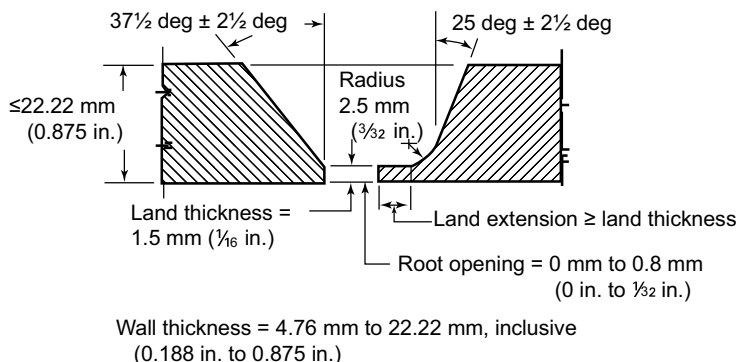
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(24)

Figure U328.4.2-1
Modified Pipe End Preparations



(a) Modified "J" Preparation



(b) Modified Butt Weld Preparation

U328.5 Welding Requirements

U328.5.1 General

(g) Tack welds shall be fully consumed after completion of the weld. Tack welds shall be made by a qualified welder or welding operator.

U330 PREHEATING

Paragraph 330 applies.

U331 HEAT TREATMENT

Paragraph 331 applies.

U332 BENDING AND FORMING

Paragraph 332 applies in its entirety.

U333 BRAZING AND SOLDERING

Brazing and soldering are not permitted.

U335 ASSEMBLY AND ERECTION

Paragraph 335 applies, except for paras. 335.4.1, 335.5, and 335.6. See paras. U335.7 and U335.8.

U335.7 Face Seal Joints

U335.7.1 Metal Face Seal. Metal face seal joints shall be installed and assembled in accordance with manufacturer's instructions. See Figure U335.7.1-1, illustration (a).

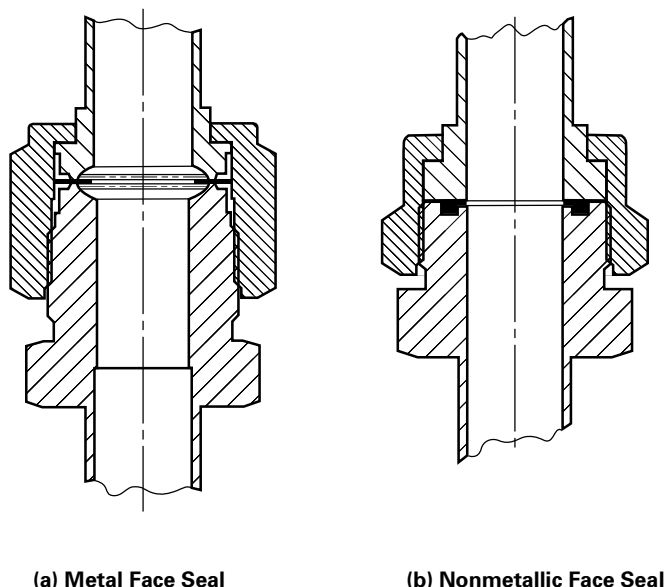
U335.7.2 Nonmetallic Face Seal. Nonmetallic face seal joints shall be installed and assembled in accordance with manufacturer's instructions. Care shall be taken to avoid distorting the seal when incorporating such joints into piping assemblies by welding. See Figure U335.7.1-1, illustration (b).

U335.8 Hygienic Clamp Joint Assembly

Hygienic clamp joint assembly components, e.g., those shown in Figures U335.8-1 through U335.8-3, shall be installed and assembled in accordance with the manufacturer's instructions. Care shall be taken to avoid distorting

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**Figure U335.7.1-1
Face Seal Joints**



(a) Metal Face Seal

(b) Nonmetallic Face Seal

the seal when incorporating such joints into piping assemblies by welding.

PART 10 INSPECTION, EXAMINATION, AND TESTING

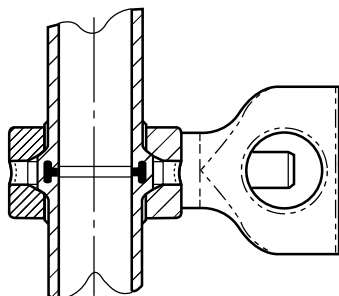
U340 INSPECTION

Paragraph 340 applies in its entirety.

U341 EXAMINATION

Paragraph 341 applies. See paras. U341.3.2 and U341.4.1.

**Figure U335.8-1
Hygienic Clamp Joint Assembly**



U341.3 Examination Requirements

U341.3.2 Acceptance Criteria. Acceptance criteria for all coupon and production welds shall be as stated in the engineering design or in the referencing code or standard (e.g., ASME BPE or SEMI) and shall at least meet the applicable requirements in para. 341.3.2.

U341.4 Extent of Required Examination

U341.4.1 Examination. Production weld coupon examination in accordance with para. U344.8 may be used in lieu of the 5% random radiography/ultrasonic examination required in para. 341.4.1(b)(1) when any of the following are employed in fabrication:

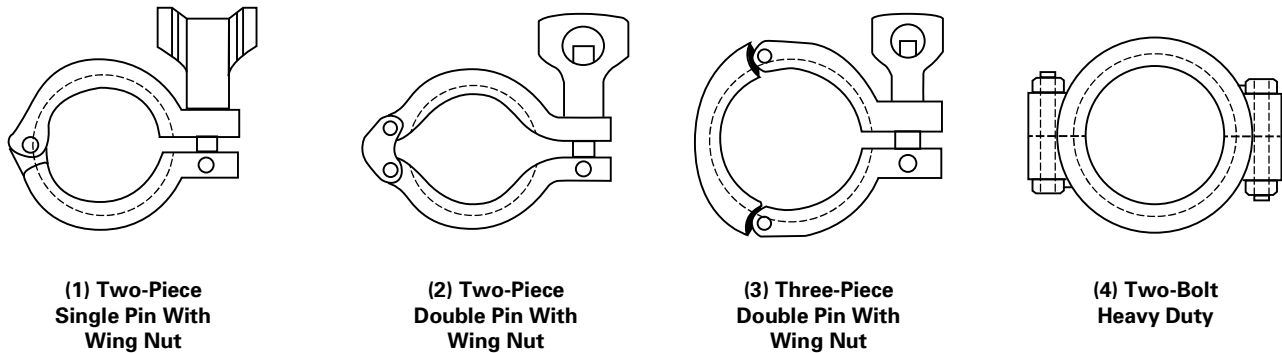
- (a) autogenous automatic or machine orbital welding
- (b) automatic or machine orbital welding with the use of consumable inserts
- (c) automatic or machine orbital welding with the use of filler wire

U341.4.5 Production Weld Coupon Examination. Production weld coupons shall be made and examined in accordance with para. U344.8 at the beginning of each shift and after changes in any of the following:

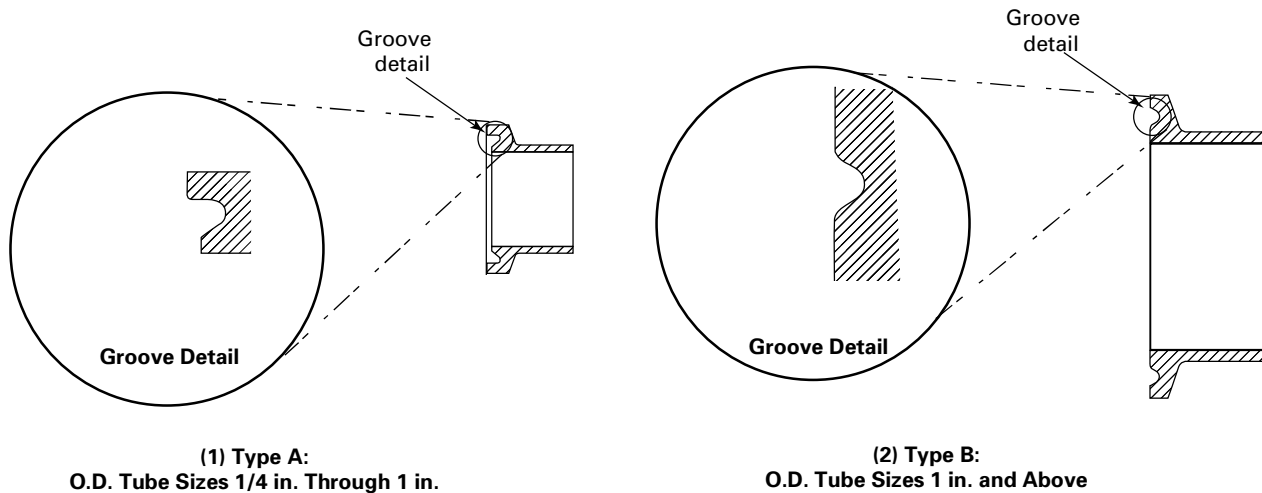
- (a) purge gas source
- (b) welding power supply
- (c) welding equipment, e.g., weld head, weld-head extensions
- (d) tungsten electrode
- (e) O.D. tube size or nominal pipe diameter
- (f) nominal wall thickness
- (g) P-Number combination of base metals

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**Figure U335.8-2
Hygienic Clamp Types**



**Figure U335.8-3
Hygienic Ferrules**



(h) welding operator

U342 EXAMINATION PERSONNEL

Paragraph 342 applies in its entirety. See para. U342.2(a).

U342.2 Specific Requirement

(a) For weld coupon examination

- (1) the examinations shall be performed by personnel other than those performing the production work or
- (2) with the owner's approval, the personnel performing the production work shall be permitted to perform the examination, provided the personnel meet the personnel qualification and certification requirements in para. 342.1

U343 EXAMINATION PROCEDURES

Paragraph 343 applies.

U344 EXAMINATION METHODS

Paragraph 344 applies. See paras. U344.2 and U344.8.

U344.2 Visual Examination

Paragraph 344.2 applies, except that in addition to the method described in para. 344.2.2, borescopic examination shall be acceptable.

U344.8 Weld Coupon Examination

U344.8.1 Requirements. Weld coupon examination comprises examination of weld coupons for the following, as applicable:

(a) prior to welding of coupons made in accordance with [para. U328.4.4\(b\)\(1\)](#)

(1) joint preparation and cleanliness

(2) fit-up, collet or clamp grip, and alignment in the weld head

(3) variables in the orbital welding machine specified in the WPS

(b) after welding of coupons made in accordance with [para. U328.4.4\(b\)\(1\)](#), and for weld coupons made in accordance with [para. U328.4.4\(b\)\(2\)](#), for compliance with [para. U341.3.2](#)

(1) alignment

(2) weld penetration

(3) weld bead width variation

(4) weld bead meander

(5) discoloration

(6) weld defects, e.g., cracks, porosity, or sulfur stringers

To allow direct visual examination of the inside surfaces, the weld coupon may be cut or a suitable indirect visual examination method (e.g., borescopic examination) may be used.

U344.8.2 Method. A weld coupon shall be made to allow visual examination in accordance with [para. U344.2](#), unless otherwise specified in the engineering design.

U345 TESTING

[Paragraph 345](#) applies except for [paras. 345.1](#), [345.8](#), and [345.9](#). See [paras. U345.1](#), [U345.8](#), and [U345.9](#).

U345.1 Required Leak Test

[Paragraph 345.1](#) applies, except that, at the owner's option, a helium mass spectrometer test in accordance with [para. U345.8.1](#) may be substituted for the hydrostatic leak test.

U345.8 Sensitive Leak Test

[Paragraph 345.8](#) applies, except that the helium mass spectrometer test described in [para. U345.8.1](#) is also an acceptable method.

U345.8.1 Helium Mass Spectrometer Test. The test shall be one of the following methods and performed in accordance with the following:

(a) For pressurized systems, the test shall be in accordance with ASME BPVC, Section V, Article 10, Appendix IV (Helium Mass Spectrometer — Detector Probe Technique).

(1) The test pressure shall be the lesser of 105 kPa (15 psig) gage or 25% of the design pressure.

(2) Prior to testing, the test pressure shall be held a minimum of 30 min.

(3) Unless otherwise specified in the engineering design, the system tested is acceptable when no leakage is detected that exceeds the allowable leakage rate of 1×10^{-4} std cc/s.

(b) For evacuated systems, the test shall be in accordance with ASME BPVC, Section V, Article 10, Appendix V (Helium Mass Spectrometer Test — Tracer Probe Technique).

(1) The piping system shall be evacuated to an absolute pressure sufficient for connection of the helium mass spectrometer to the system.

(2) Unless otherwise specified in the engineering design, the system tested is acceptable when no leakage is detected that exceeds the allowable leakage rate of 1×10^{-5} std cc/s.

U345.9 Alternative Leak Test

[Paragraph 345.9](#) applies, except that welds may be examined by weld coupon examination method in accordance with [para. U341.4.5](#) and the test method may be helium mass spectrometer test in accordance with [para. U345.8.1](#).

U346 RECORDS

U346.2 Responsibility

(24)

It is the responsibility of the designer, the manufacturer, the fabricator, and the erector, as applicable, to prepare the records required by this Code, ASME BPE, SEMI, or other industry standard as specified in the engineering design.

U346.3 Retention of Records

[Paragraph 346.3](#) applies.

PART 11

HIGH PURITY PIPING IN CATEGORY M FLUID SERVICE

UM300 GENERAL STATEMENTS

(a) [Chapter X](#), [Part 11](#) pertains to piping designated by the owner as being high purity piping in Category M Fluid Service. See also [Appendix M](#).

(b) The organization, content, and paragraph designations of these Parts correspond to those of [Chapter VIII](#). The prefix UM is used.

(c) [Paragraphs M300\(d\)](#), [M300\(e\)](#), and [M300\(f\)](#) apply.

(d) Provisions and requirements of [Chapter VIII](#) apply with the additional requirements in [paras. UM307](#), [UM307.2](#), [UM322](#), [UM322.3](#), [UM328](#), [UM335](#), [UM335.3.3](#), [UM341](#), [UM341.4\(b\)\(1\)](#) and [UM341.4\(b\)\(2\)](#), and [UM345\(b\)](#).

UM307 METALLIC VALVES AND SPECIALTY COMPONENTS

Paragraph M307 applies in its entirety. See also para. UM307.2(a).

UM307.2 Specific Requirements

(a) For bellows or diaphragm sealed type valves, the bonnet or cover plate closure shall be secured by a straight thread sufficient for mechanical strength, have a metal-to-metal seat, and include a secondary stem seal.

UM322 SPECIFIC PIPING SYSTEMS

Paragraph M322 applies, except for para. M322.3(c). See para. UM322.3(c).

UM322.3 Instrument Piping

(c) joining methods shall conform to the requirements of para. U315

UM328 WELDING OF MATERIALS

Welding shall be in accordance with paras. M311.1 and U328, except examination shall be in accordance with para. UM341.

UM335 ASSEMBLY AND ERECTION OF METALLIC PIPING

Paragraph M335 applies, except for para. M335.3.3. See para. UM335.3.3.

UM335.3.3 Straight-Threaded Joints. The requirements of para. M335.3.3 are subject to the limitations in para. UM322.

UM341 EXAMINATION

Paragraph M341 applies. See UM341.4(b)(1) and UM341.4(b)(2).

UM341.4 Extent of Required Examination

(b) *Other Examination*

(1) The 100% radiography/ultrasonic examination required in para. M341.4(b) applies.

(2) The in-process examination alternative permitted in para. 341.4.1(b)(1) applies, except a weld coupon examination in accordance with para. U344.8 is also an acceptable substitute when specified in the engineering design or by the Inspector.

UM345 TESTING

Paragraph M345(a) applies. See (b).

(b) A sensitive leak test in accordance with para. U345.8 shall be included in the required leak test (see para. U345.1).

APPENDIX A

ALLOWABLE STRESSES AND QUALITY FACTORS FOR METALLIC PIPING AND BOLTING MATERIALS

(24)

The [Specification Index for Appendix A](#) begins on the next page. [Tables A-1 \(Table A-1C\)](#) through [A-4 \(Table A-4C\)](#) follow the Index.

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Specification Index for **Appendix A**

Spec. No.	Title	Spec. No.	Title
ASTM		ASTM (Cont'd)	
A36	Carbon Structural Steel	A285	Pressure Vessel Plates, Carbon Steel, Low- and Intermediate-Tensile Strength
A47	Ferritic Malleable Iron Castings	A299	Pressure Vessel Plates, Carbon Steel, Manganese-Silicon
A48	Gray Iron Castings	A302	Pressure Vessel Plates, Alloy Steel, Manganese-Molybdenum and Manganese-Molybdenum-Nickel
A53	Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless	A307	Carbon Steel Bolts and Studs, 60 000 PSI Tensile Strength
A105	Carbon Steel Forgings for Piping Applications	A312	Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
A106	Seamless Carbon Steel Pipe for High-Temperature Service	A320	Alloy-Steel and Stainless Steel Bolting for Low-Temperature Service
A126	Gray Iron Castings for Valves, Flanges, and Pipe Fittings	A333	Seamless and Welded Steel Pipe for Low-Temperature Service and Other Applications with Required Notch Toughness
A134	Pipe, Steel, Electric-Fusion (Arc)-Welded (Sizes NPS 16 and Over)	A334	Seamless and Welded Carbon and Alloy-Steel Tubes for Low-Temperature Service
A135	Electric-Resistance-Welded Steel Pipe	A335	Seamless Ferritic Alloy-Steel Pipe for High-Temperature Service
A139	Electric-Fusion (Arc)-Welded Steel Pipe (NPS 4 and Over)	A350	Carbon and Low-Alloy Steel Forgings, Requiring Notch Toughness Testing for Piping Components
A179	Seamless Cold-Drawn Low-Carbon Steel Heat-Exchanger and Condenser Tubes	A351	Castings, Austenitic, for Pressure-Containing Parts
A181	Carbon Steel Forgings, for General-Purpose Piping	A352	Steel Castings, Ferritic and Martensitic, for Pressure-Containing Parts, Suitable for Low-Temperature Service
A182	Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service	A353	Pressure Vessel Plates, Alloy Steel, Double-Normalized and Tempered 9% Nickel
A193	Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications	A354	Quenched and Tempered Alloy Steel Bolts, Studs, and Other Externally Threaded Fasteners
A194	Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both	A358	Electric-Fusion-Welded Austenitic Chromium-Nickel Stainless Steel Pipe for High-Temperature Service and General Applications
A197	Cupola Malleable Iron	A369	Carbon and Ferritic Alloy Steel Forged and Bored Pipe for High-Temperature Service
A203	Pressure Vessel Plates, Alloy Steel, Nickel	A376	Seamless Austenitic Steel Pipe for High-Temperature Service
A204	Pressure Vessel Plates, Alloy Steel, Molybdenum	A381	Metal-Arc-Welded Carbon or High-Strength Low-Alloy Steel Pipe for Use With High-Pressure Transmission Systems
A213	Seamless Ferritic and Austenitic Alloy-Steel Boiler, Superheater, and Heat-Exchanger Tubes	A387	Pressure Vessel Plates, Alloy Steel, Chromium-Molybdenum
A216	Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service	A395	Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures
A217	Steel Castings, Martensitic Stainless and Alloy, for Pressure-Containing Parts, Suitable for High-Temperature Service	A403	Wrought Austenitic Stainless Steel Piping Fittings
A234	Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service	A409	Welded Large Diameter Austenitic Steel Pipe for Corrosive or High-Temperature Service
A240	Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications	A420	Piping Fittings of Wrought Carbon Steel and Alloy Steel for Low-Temperature Service
A268	Seamless and Welded Ferritic and Martensitic Stainless Steel Tubing for General Service	A426	Centrifugally Cast Ferritic Alloy Steel Pipe for High-Temperature Service
A269	Seamless and Welded Austenitic Stainless Steel Tubing for General Service	A437	Stainless and Alloy-Steel Turbine-Type Bolting Material Specially Heat Treated for High-Temperature Service
A270	Seamless and Welded Austenitic and Ferritic/Austenitic Stainless Steel Sanitary Tubing		
A276	Stainless Steel Bars and Shapes		
A278	Gray Iron Castings for Pressure-Containing Parts for Temperatures Up to 650°F (350°C)		
A283	Low and Intermediate Tensile Strength Carbon Steel Plates		

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Specification Index for **Appendix A (Cont'd)**

Spec. No.	Title	Spec. No.	Title
ASTM (Cont'd)		ASTM (Cont'd)	
A451	Centrifugally Cast Austenitic Steel Pipe for High-Temperature Service	A815	Wrought Ferritic, Ferritic/Austenitic, and Martensitic Stainless Steel Piping Fittings
A453	High-Temperature Bolting, with Expansion Coefficients Comparable to Austenitic Stainless Steels	A860	Wrought High-Strength Ferritic Steel Butt-Welding Fittings
A479	Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels	A928	Ferritic/Austenitic (Duplex) Stainless Steel Pipe Electric Fusion Welded with Addition of Filler Metal
A487	Steel Castings Suitable for Pressure Service	A992	Structural Steel Shapes
A494	Castings, Nickel and Nickel Alloy	A995	Castings, Austenitic-Ferritic (Duplex) Stainless Steel, for Pressure-Containing Parts
A515	Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher-Temperature Service	A1010	Higher-Strength Martensitic Stainless Steel Plate, Sheet, and Strip
A516	Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service	A1011	Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
A524	Seamless Carbon Steel Pipe for Atmospheric and Lower Temperatures	A1053	Welded Ferritic-Martensitic Stainless Steel Pipe
A536	Ductile Iron Castings	B21	Naval Brass Rod, Bar, and Shapes
A537	Pressure Vessel Plates, Heat-Treated, Carbon-Manganese-Silicon Steel	B26	Aluminum-Alloy Sand Castings
A553	Pressure Vessel Plates, Alloy Steel, Quenched and Tempered 7, 8, and 9 % Nickel	B42	Seamless Copper Pipe, Standard Sizes
A563	Carbon and Alloy Steel Nuts (Inch and Metric)	B43	Seamless Red Brass Pipe, Standard Sizes
A571	Austenitic Ductile Iron Castings for Pressure-Containing Parts Suitable for Low-Temperature Service	B61	Steam or Valve Bronze Castings
A587	Electric-Resistance-Welded Low-Carbon Steel Pipe for the Chemical Industry	B62	Composition Bronze or Ounce Metal Castings
A645	Pressure Vessel Plates, 5 % and 5 ½ % Nickel Alloy Steels, Specially Heat Treated	B68	Seamless Copper Tube, Bright Annealed
A671	Electric-Fusion-Welded Steel Pipe for Atmospheric and Lower Temperatures	B75	Seamless Copper Tube
A672	Electric-Fusion-Welded Steel Pipe for High-Pressure Service at Moderate Temperatures	B88	Seamless Copper Water Tube
A675	Steel Bars, Carbon, Hot-Wrought, Special Quality, Mechanical Properties	B96	Copper-Silicon Alloy Plate, Sheet, Strip, and Rolled Bar for General Purposes and Pressure Vessels
A691	Carbon and Alloy Steel Pipe, Electric-Fusion-Welded for High-Pressure Service at High Temperatures	B98	Copper-Silicon Alloy Rod, Bar and Shapes
A694	Carbon and Alloy Steel Forgings for Pipe Flanges, Fittings, Valves, and Parts for High-Pressure Transmission Service	B127	Nickel-Copper Alloy (UNS N04400) Plate, Sheet, and Strip
A696	Steel Bars, Carbon, Hot-Wrought or Cold-Finished, Special Quality, for Pressure Piping Components	B148	Aluminum-Bronze Sand Castings
A707	Forged Carbon and Alloy Steel Flanges for Low-Temperature Service	B150	Aluminum Bronze Rod, Bar and Shapes
A789	Seamless and Welded Ferritic/Austenitic Stainless Steel Tubing for General Service	B152	Copper Sheet, Strip, Plate and Rolled Bar
A790	Seamless and Welded Ferritic/Austenitic Stainless Steel Pipe	B160	Nickel Rod and Bar
A813	Single- or Double-Welded Austenitic Stainless Steel Pipe	B161	Nickel Seamless Pipe and Tube
A814	Cold-Worked Welded Austenitic Stainless Steel Pipe	B162	Nickel Plate, Sheet and Strip
		B163	Seamless Nickel and Nickel Alloy Condenser and Heat Exchanger Tubes
		B164	Nickel-Copper Alloy Rod, Bar, and Wire
		B165	Nickel-Copper Alloy (UNS N04400) Seamless Pipe and Tube
		B166	Nickel-Chromium-Aluminum Alloy, Nickel-Chromium-Iron Alloys, Nickel-Chromium-Cobalt-Molybdenum Alloy, Nickel-Iron-Chromium-Tungsten Alloy, and Nickel-Chromium-Molybdenum-Copper Alloy Rod, Bar, and Wire

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Specification Index for **Appendix A (Cont'd)**

Spec. No.	Title	Spec. No.	Title
ASTM (Cont'd)		ASTM (Cont'd)	
B167	Nickel-Chromium-Iron Alloys (UNS N06600, N06601, N06603, N06690, N06693, N06025, N06045, and N06696), Nickel-Chromium-Cobalt-Molybdenum Alloy (UNS N06617), and Nickel-Iron-Chromium-Tungsten Alloy (UNS N06674) Seamless Pipe and Tube	B424	Ni-Fe-Cr-Mo-Cu Alloy (UNS N08825, UNS N08221, and UNS N06845) Plate, Sheet, and Strip
B168	Nickel-Chromium-Iron Alloys (UNS N06600, N06601, N06603, N06690, N06693, N06025, N06045, and N06696), Nickel-Chromium-Cobalt-Molybdenum Alloy (UNS N06617), and Nickel-Iron-Chromium-Tungsten Alloy (UNS N06674) Plate, Sheet and Strip	B425	Ni-Fe-Cr-Mo-Cu Alloy (UNS N08825, UNS N08221, and UNS N06845) Rod and Bar
B169	Aluminum Bronze Sheet, Strip, and Rolled Bar	B435	UNS N06002, UNS N06230, UNS N12160, and UNS R30556 Plate, Sheet, and Strip
B171	Copper-Alloy Plate and Sheet for Pressure Vessels, Condensers, and Heat Exchangers	B443	Nickel-Chromium-Molybdenum-Columbium Alloy (UNS N06625) and Nickel-Chromium-Molybdenum-Silicon Alloy (UNS N06219) Plate, Sheet, and Strip
B187	Copper, Bus Bar, Rod, and Shapes and General Purpose Rod, Bar, and Shapes	B444	Nickel-Chromium-Molybdenum-Columbium Alloys (UNS N06625 and UNS N06852) and Nickel-Chromium-Molybdenum-Silicon Alloy (UNS N06219) Pipe and Tube
B209	Aluminum and Aluminum-Alloy Sheet and Plate	B446	Nickel-Chromium-Molybdenum-Columbium Alloy (UNS N06625), Nickel-Chromium-Molybdenum-Silicon Alloy (UNS N06219), and Nickel-Chromium-Molybdenum-Tungsten Alloy (UNS N06650) Rod and Bar
B210	Aluminum and Aluminum-Alloy Drawn Seamless Tubes	B462	Forged or Rolled Nickel Alloy Pipe Flanges, Forged Fittings, and Valves and Parts for Corrosive High-Temperature Service
B211	Aluminum and Aluminum-Alloy Rolled or Cold Finished Bar, Rod, and Wire	B463	UNS N08020 Alloy Plate, Sheet, and Strip
B221	Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes	B464	Welded UNS N08020 Alloy Pipe
B241	Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube	B466	Seamless Copper-Nickel Pipe and Tube
B247	Aluminum and Aluminum-Alloy Die Forgings, Hand Forgings, and Rolled Ring Forgings	B467	Welded Copper-Nickel Pipe
B265	Titanium and Titanium Alloy Strip, Sheet, and Plate	B474	Electric Fusion Welded Nickel and Nickel Alloy Pipe
B280	Seamless Copper Tube for Air Conditioning and Refrigeration Field Service	B491	Aluminum and Aluminum-Alloy Extruded Round Tubes for General-Purpose Applications
B283	Copper and Copper-Alloy Die Forgings (Hot-Pressed)	B493	Zirconium and Zirconium Alloy Forgings
B333	Nickel-Molybdenum Alloy Plate, Sheet, and Strip	B514	Welded Nickel-Iron-Chromium Alloy Pipe
B335	Nickel-Molybdenum Alloy Rod	B515	Welded UNS N08120, UNS N08800, UNS N08810, and UNS N08811 Alloy Tubes
B345	Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube for Gas and Oil Transmission and Distribution Piping Systems	B517	Welded Nickel-Chromium-Iron-Alloy (UNS N06600, UNS N06603, UNS N06025, and UNS N06045) Pipe
B348	Titanium and Titanium Alloy Bars and Billets	B523	Seamless and Welded Zirconium and Zirconium Alloy Tubes
B361	Factory-Made Wrought Aluminum and Aluminum-Alloy Welding Fittings	B550	Zirconium and Zirconium Alloy Bar and Wire
B363	Seamless and Welded Unalloyed Titanium and Titanium Alloy Welding Fittings	B551	Zirconium and Zirconium Alloy Strip, Sheet, and Plate
B366	Factory-Made Wrought Nickel and Nickel Alloy Fittings	B564	Nickel Alloy Forgings
B367	Titanium and Titanium Alloy Castings	B572	UNS N06002, UNS N06230, UNS N12160, and UNS R30556 Rod
B371	Copper-Zinc-Silicon Alloy Rod	B574	Low-Carbon Nickel-Chromium-Molybdenum, Low-Carbon Nickel-Molybdenum-Chromium, Low-Carbon Nickel-Molybdenum-Chromium-Tantalum, Low-Carbon Nickel-Chromium-Molybdenum-Copper, and Low-Carbon Nickel-Chromium-Molybdenum-Tungsten Alloy Rod
B381	Titanium and Titanium Alloy Forgings	B575	Low-Carbon Nickel-Chromium-Molybdenum, Low-Carbon Nickel-Chromium-Molybdenum-Copper, Low-Carbon Nickel-Chromium-Molybdenum-Tantalum, Low-Carbon Nickel-Chromium-Molybdenum-Tungsten, and Low-Carbon Nickel-Molybdenum-Chromium Alloy Plate, Sheet and Strip
B407	Nickel-Iron-Chromium Alloy Seamless Pipe and Tube		
B408	Nickel-Iron-Chromium Alloy Rod and Bar		
B409	Nickel-Iron-Chromium Alloy Plate, Sheet, and Strip		
B423	Nickel-Iron-Chromium-Molybdenum-Copper Alloy Seamless Pipe and Tube		

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Specification Index for **Appendix A (Cont'd)**

Spec. No.	Title	Spec. No.	Title
ASTM (Cont'd)		ASTM (Cont'd)	
B581	Nickel-Chromium-Iron-Molybdenum-Copper Alloy Rod	B690	Iron-Nickel-Chromium-Molybdenum Alloy (UNS N08367) Seamless Pipe and Tube
B582	Nickel-Chromium-Iron-Molybdenum-Copper Alloy Plate, Sheet, and Strip	B704	Welded UNS N06625, UNS N06219 and UNS N08825 Alloy Tubes
B584	Copper Alloy Sand Castings for General Applications	B705	Nickel-Alloy (UNS N06625, N06219 and N08825) Welded Pipe
B619	Welded Nickel and Nickel-Cobalt Alloy Pipe	B709	Iron-Nickel-Chromium-Molybdenum Alloy (UNS N08028) Plate, Sheet, and Strip
B620	Nickel-Iron-Chromium-Molybdenum Alloy (UNS N08320) Plate, Sheet, and Strip	B725	Welded Nickel and Nickel Copper Alloy Pipe
B621	Nickel-Iron-Chromium-Molybdenum Alloy (UNS N08320) Rod	B729	Seamless UNS N08020, UNS N08026, and UNS N08024 Nickel-Alloy Pipe and Tube
B622	Seamless Nickel and Nickel-Cobalt Alloy Pipe and Tube	B804	UNS N08367 and UNS N08926 Welded Pipe
B625	UNS N08925, UNS N08031, UNS N08932, UNS N08926, UNS N08354, UNS N08830, and UNS R20033 Plate, Sheet, and Strip	B861	Titanium and Titanium Alloy Seamless Pipe
B626	Welded Nickel and Nickel-Cobalt Alloy Tube	B862	Titanium and Titanium Alloy Welded Pipe
B637	Precipitation-Hardening and Cold Worked Nickel Alloy Bars, Forgings, and Forging Stock for Moderate or High Temperature Service	E112	Standard Test Methods for Determining Average Grain Size
B649	Ni-Fe-Cr-Mo-Cu-N Low-Carbon Alloys (UNS N08925, UNS N08031, UNS N08034, UNS N08354, and UNS N08926), and Cr-Ni-Fe-N Low-Carbon Alloy (UNS R20033) Bar and Wire, and Ni-Cr-Fe-Mo-N Alloy (UNS N08936) Wire	F3125	High Strength Structural Bolts, Steel and Alloy Steel, Heat Treated, 120 ksi (830 MPa) and 150 ksi (1040 MPa) Minimum Tensile Strength
B658	Seamless and Welded Zirconium and Zirconium Alloy Pipe	API	
B668	UNS N08028 Seamless Pipe and Tube	5L	Line Pipe
B675	UNS N08367 Welded Pipe	CSA	
B688	Chromium-Nickel-Molybdenum-Iron (UNS N08367) Plate, Sheet, and Strip	Z245.1	Steel Pipe

GENERAL NOTE: It is not practical to refer to a specific edition of each standard throughout the Code text. Instead, the approved edition references, along with the names and websites of the sponsoring organizations, are shown in [Appendix E](#).

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NOTES FOR TABLES A-1 and A-1C

GENERAL NOTES:

- (a) The allowable stress values, P-Number assignments, and minimum temperatures, together with the referenced Notes in the stress tables, are requirements of this Code.
- (b) Notes (1) through (7) are referenced in column headings and in body headings for material type and product form; Note (8) and all subsequent Notes are referenced in the Notes column for specific materials. Notes marked with an asterisk (*) restate requirements found in the text of the Code.
- (c) The stress values given in MPa as shown in Table A-1, and given in ksi as shown in Table A-1C, may be used. The values stated in ksi are not exact equivalents to the values stated in MPa. Therefore, for any given material, the user of the Code should use only the MPa or the ksi values.
- (d) For copper and copper alloys, the following symbols are used in the Temper column: H = drawn; H01 = quarter hard; H02 = half hard; H06 = extra hard; H55 = light drawn; H58 = drawn, general purpose; H80 = hard drawn; HR50 = drawn, stress relieved; M20 = hot rolled; O25 = hot rolled, annealed; O50 = light annealed; O60 = soft annealed; O61 = annealed; W050 = welded, annealed; and W061 = welded, fully finished, annealed.
- (e) For nickel and nickel alloys, the following abbreviations are used in the Class column: ann., annealed; C.D., cold worked; forg., forged; H.F., hot finished; H.R., hot rolled; H.W., hot worked; plt., plate; R., rolled; rel., relieved; sol., solution; str., stress; and tr., treated.
- (f) In Table A-1, the following abbreviations are used in the Product Form column: CW, continuous welded; EW, electric welded; forg., forgings; ftg., fittings; pl., plate; shps., shapes; sht., sheet; smls., seamless; struct., structural; and wld., welded.

NOTES:

- (1) *The stress values are basic allowable stresses in tension in accordance with para. 302.3.1(a). For pressure design, the stress values are multiplied by the appropriate quality factor E (E_c from Table A-2 or E_t from Table A-3). Stress values in shear and bearing are stated in para. 302.3.1(b); those in compression in para. 302.3.1(c).
- (2a) *In Table A-1C, stress values printed in *italics* exceed two-thirds of the expected yield strength at temperature. Stress values in **boldface** are equal to 90% of expected yield strength at temperature. See paras. 302.3.2(d)(3) and 302.3.2(e).
- (2b) *In Table A-1, stress values printed in *italics* are tensile-controlled values. Yield-controlled stress values are in normal font and time-dependent stress values are in **boldface**.
- (3) The stress values for austenitic stainless steels in these Tables may not be applicable if the material has been given a final heat treatment other than that required by

the material specification or by reference to Note (30) or Note (31).

- (4) Reserved
- (5) *See ASME BPVC, Section IX, QW-200.3 for a description of P-Number groupings. P-Numbers are indicated by number or by a number followed by a letter (e.g., 8, 5B, or 11A).
- (6) *The minimum temperature shown is that design minimum temperature for which the material is normally suitable without impact testing other than that required by the material specification. However, the use of a material at a design minimum temperature colder than -29°C (-20°F) is established by rules elsewhere in this Code, including para. 323.2.2 and other impact test requirements. For carbon steels with a letter designation in the Min. Temp. column, see para. 323.2.2(e) and the applicable curve and Notes in Figure 323.2.2-1.
- (7) The letter "a" indicates alloys that are not recommended for welding and that, if welded, shall be individually qualified. The letter "b" indicates copper base alloys that shall be individually qualified.
- (8) *There are restrictions on the use of this material in the text of the Code as follows:
 - (a) See para. 305.2.1; temperature limits are -29°C to 186°C (-20°F to 366°F).
 - (b) See para. 305.2.2; pipe shall be safeguarded when used outside the temperature limits in Note (8a).
 - (c) See Table 323.2.2-1, box B-2.
 - (d) See para. 323.4.2(a).
 - (e) See para. 323.4.2(b).
- (9) *For pressure-temperature ratings of components made in accordance with standards listed in Table 326.1.1-1, see para. 326.2.1. Stress values may be used to calculate ratings for unlisted components, and special ratings for listed components, as permitted by para. 303.
- (10) Component standards listed in Table 326.1.1-1 impose the following restrictions on this material when used as a forging: composition, properties, heat treatment, and grain size shall conform to this specification; manufacturing procedures, tolerances, tests, certification, and markings shall be in accordance with ASTM B564.
- (11) *For use under this Code, radiography shall be performed after heat treatment.
- (12) *Certain forms of this material, as stated in Table 323.2.2-1, shall be impact tested to qualify for service below -29°C (-20°F). Alternatively, if provisions for impact testing are included in the material specification as supplementary requirements and are invoked, the material may be used down to the temperature at which the test was conducted in accordance with the specification.
- (13) Properties of this material vary with thickness or size. Stress values are based on minimum properties for the thickness listed.

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- (14) For use in Code piping at the stated stress values, the required minimum tensile and yield properties must be verified by tensile test. If such tests are not required by the material specification, they shall be specified in the purchase order.
- (15) For service temperature $>454^{\circ}\text{C}$ ($>850^{\circ}\text{F}$), weld metal shall have a carbon content $>0.05\%$.
- (16) Heat treatment is required after welding for all products of zirconium Grade R60705. See [Table 331.1.1-1](#).
- (17) Mechanical properties of fittings made from forging stock shall meet the minimum tensile requirements of one of the bar, forging, or rod specifications listed in Table 2 of B366 for which tensile testing is required.
- (18) Stress values shown are for materials in the normalized and tempered condition, or when the heat treatment is unknown. If material is annealed, use the following values above 510°C (950°F):

Temp., $^{\circ}\text{C}$	538	566	593	621	649
S, MPa	55.1	39.3	26.2	16.5	9.6

Temp., $^{\circ}\text{F}$	1,000	1,050	1,100	1,150	1,200
S, ksi	8.0	5.7	3.8	2.4	1.4

- (19) The minimum tensile strength of the reduced section tensile specimen in accordance with ASME BPVC, Section IX, QW-462.1, shall not be less than 758 MPa (110.0 ksi).
- (20) For pipe sizes $\geq \text{DN } 200$ ($\geq \text{NPS } 8$) with wall thicknesses $\geq \text{Sch } 140$, the specified minimum tensile strength is 483 MPa (70 ksi).
- (21) For material thickness $>130 \text{ mm}$ ($>5 \text{ in.}$), the specified minimum tensile strength is 485 MPa (70 ksi).
- (22) The minimum tensile strength for weld (qualification) and stress values shown shall be multiplied by 0.90 for pipe having an outside diameter less than 51 mm (2.0 in.) and a D/t value less than 15. This requirement may be waived if it can be shown that the welding procedure to be used will consistently produce welds that meet the listed minimum tensile strength of 165 MPa (24 ksi).
- (23) For material thickness $>130 \text{ mm}$ ($>5 \text{ in.}$), the specified minimum tensile strength is 450 MPa (65 ksi).
- (24) Yield strength is not stated in the material specification. The value shown is based on yield strengths of materials with similar characteristics.
- (25) This steel may develop embrittlement after service at approximately 316°C (600°F) and higher temperature.
- (26) This unstabilized grade of stainless steel increasingly tends to precipitate intergranular carbides as the carbon content increases above 0.03%. See also [para. F323.4\(c\)\(2\)](#).
- (27) For temperatures above 427°C (800°F), these stress values apply only when the carbon content is 0.04% or higher.
- (28) For temperatures above 538°C ($1,000^{\circ}\text{F}$), these stress values apply only when the carbon content is 0.04% or higher.
- (29) The stress values above 538°C ($1,000^{\circ}\text{F}$) listed here shall be used only when the steel's austenitic micrograin size, as defined in ASTM E112, is No. 6 or less (coarser grain). Otherwise, the lower stress values listed for the same material, specification, and grade shall be used.
- (30) For temperatures above 538°C ($1,000^{\circ}\text{F}$), these stress values may be used only if the material has been heat treated by heating to a minimum temperature of 1093°C ($2,000^{\circ}\text{F}$) and quenching in water or rapidly cooling by other means.
- (31) For temperatures above 538°C ($1,000^{\circ}\text{F}$), these stress values may be used only if the material has been heat treated by heating to a minimum temperature of 1038°C ($1,900^{\circ}\text{F}$) and quenching in water or rapidly cooling by other means.
- (32) Stress values shown are for the lowest strength base material permitted by the specification to be used in the manufacture of this grade of fitting. If a higher strength base material is used, the higher stress values for that material may be used in design.
- (33) For welded construction with work hardened grades, use the stress values for annealed material; for welded construction with precipitation hardened grades, use the special stress values for welded construction given in the tables.
- (34) If material is welded, brazed, or soldered, the allowable stress values for the annealed condition shall be used.
- (35) This steel is intended for use at high temperatures; it may have low ductility and/or low impact properties at room temperature after being used above the temperature indicated by [para. F323.4\(c\)\(4\)](#).
- (36) The specification permits this material to be furnished without solution heat treatment or with other than a solution heat treatment. When the material has not been solution heat treated, the minimum temperature shall be -29°C (-20°F) unless the material is impact tested in accordance with [para. 323.3](#).
- (37) Impact requirements for seamless fittings shall be governed by those listed in this table for the particular base material specification in the grades permitted (A312, A240, and A182). When A276 materials are used in the manufacture of these fittings, the Notes, minimum temperatures, and allowable stresses for comparable grades of A240 materials shall apply.
- (38) For use under this Code, impact testing shall be performed in accordance with [para. 323.3](#) at the design minimum temperature but not warmer than -29°C (-20°F).

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- (39) This material when used below -29°C (-20°F) shall be impact tested if the carbon content is above 0.10%.
- (40) Stress values shown are 90% of those for the corresponding core material.
- (41) The tension test specimen from plate 12.7 mm ($\frac{1}{2}$ in.) and thicker is machined from the core and does not include the cladding alloy; therefore, the stress values listed are those for materials less than 12.7 mm.
- (42) This material may be used only in nonpressure applications.
- (43) *The stress values given for this material are not applicable when either welding or thermal cutting is employed [see [para. 323.4.2\(c\)](#)].
- (44) For stress relieved tempers (T351, T3510, T3511, T451, T4510, T4511, T651, T6510, T6511), stress values for material in the listed temper shall be used.
- (45) Stress values shown are applicable for "die" forgings only.
- (46) The minimum temperature shown is for the heaviest wall meeting the specified mechanical property requirements in the specification. The minimum temperature for lighter walls shall be as shown in the following tabulation:

Impact Test Temperature ($^{\circ}\text{C}$) for Plate Thicknesses Shown		
Spec. No. and Grade	Max. 51 mm	Over 51 mm to 76 mm
A203 A	-68	-59
A203 B	-68	-59
A203 D	-101	-87
A203 E	-101	-87

Impact Test Temperature ($^{\circ}\text{F}$) for Plate Thicknesses Shown		
Spec. No. and Grade	Max. 2.0 in.	Over 2.0 in. to 3.0 in.
A203 A	-90	-75
A203 B	-90	-75
A203 D	-150	-125
A203 E	-150	-125

- (47) If no welding is employed in fabrication of piping from these materials, the stress values may be increased to 230 MPa (33.3 ksi).
- (48) The stress value to be used for this gray iron material at its upper temperature limit of 232°C (450°F) is the same as that shown in the 204°C (400°F) column.
- (49) If the chemical composition of this Grade is such as to render it hardenable, qualification under P-No. 6 is required.
- (50) This material is grouped in P-No. 7 because its hardenability is low.
- (51) This material may require special consideration for welding qualification. See ASME BPVC, Section IX, Table QW/QB-422. For use in this Code, a qualified WPS is required for each strength level of material.
- (52) These materials are normally microalloyed with Cb, V, and/or Ti. Supplemental specifications agreed to by manufacturer and purchaser commonly establish chemistry more restrictive than the base specification, as well as plate rolling specifications and requirements for weldability (i.e., C-equivalent) and toughness.

- (53) Stress relief heat treatment is required for service above 232°C (450°F).
- (54) The maximum operating temperature is arbitrarily set at 260°C (500°F) because hard temper adversely affects design stress in the creep rupture temperature ranges.
- (55) Pipe produced to this specification is not intended for high temperature service. The stress values apply to any heat treatment condition, delivery condition, and service condition allowed for the applicable pipe strength level. Thermomechanical rolled or thermomechanical formed pipe shall not be heat treated at a temperature warmer than 580°C ($1,075^{\circ}\text{F}$).
- (56) Because of thermal instability, this material is not recommended for service above 427°C (800°F).
- (57) Conversion of carbides to graphite may occur after prolonged exposure to temperatures over 427°C (800°F). See [para. F323.4\(b\)\(2\)](#).
- (58) Conversion of carbides to graphite may occur after prolonged exposure to temperatures over 468°C (875°F). See [para. F323.4\(b\)\(3\)](#).
- (59) For temperatures above 482°C (900°F), consider the advantages of killed steel. See [para. F323.4\(b\)\(4\)](#).
- (60) Alloy 625 (UNS N06625) in the annealed condition is subject to severe loss of impact strength at room temperature after exposure in the range of 538°C to 760°C ($1,000^{\circ}\text{F}$ to $1,400^{\circ}\text{F}$).
- (61) Annealed at approximately 982°C ($1,800^{\circ}\text{F}$).
- (62) Annealed at approximately $1\,121^{\circ}\text{C}$ ($2,050^{\circ}\text{F}$).
- (63) The pipe grades listed below, marked as either or both grades in accordance with API Specification 5L, shall be considered as equivalents and treated as listed materials.

API 5L Grade Equivalents	
X-Grade	L-Grade
A25	L175
A	L210
B	L245
X42	L290
X46	L320
X52	L360
X56	L390
X60	L415
X65	L450
X70	L485
X80	L555

- (64) For use under this Code, the heat treatment requirements for pipe manufactured to A671, A672, and A691 shall be as required by [para. 331](#) for the particular material being used.
- (65) For use under this Code, the designer should consider additional material requirements for flanges when the design minimum temperature is colder than -18°C (0°F). See [Appendix F, para. F323.2.2\(b\)](#).

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Table A-1

Basic Allowable Stresses in Tension for Metals — SI Units

(24)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated											
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Notes	Min. Temp., °C (6)	Min. Tensile Strgth., MPa	Min. Yield Strgth., MPa	Max. Temp., °C
1	Fe	Castings	A126	A	F11501	...	(8e) (9) (48)	−29	145	...	204
2	Fe	Castings	A48	150	F11401	...	(8e) (48)	−29	150	...	204
3	Fe	Castings	A278	150	F11401	...	(8e) (48)	−29	150	...	204
4	Fe	Castings	A48	175	F11701	...	(8e) (48)	−29	175	...	204
5	Fe	Castings	A278	175	F11701	...	(8e) (48)	−29	175	...	204
6	Fe	Castings	A48	200	F12101	...	(8e) (48)	−29	200	...	204
7	Fe	Castings	A278	200	F12101	...	(8e) (48)	−29	200	...	204
8	Fe	Castings	A126	B	F12102	...	(8e) (9) (48)	−29	214	...	204
9	Fe	Castings	A48	225	...	...	(8e) (48)	−29	225	...	204
10	Fe	Castings	A278	225	...	...	(8e) (48)	−29	225	...	204
11	Fe	Castings	A48	250	F12401	...	(8e) (9) (48)	−29	250	...	204
12	Fe	Castings	A278	250	F12401	...	(8e) (53)	−29	250	...	343
13	Fe	Castings	A48	275	F12801	...	(8e) (48)	−29	275	...	204
14	Fe	Castings	A278	275	F12803	...	(8e) (53)	−29	275	...	204
15	Fe	Castings	A48	375	...	...	(8e) (48)	−29	375	...	204
16	Fe	Castings	A278	380	F13801	...	(8e) (53)	−29	380	...	204
17	Fe	Castings	A197	...	F22000	...	(8e) (9)	−29	275	200	343
18	Fe	Castings	A126	C	F12802	...	(8e) (9) (48)	−29	283	...	204
19	Fe	Castings	A48	300	F13101	...	(8e) (48)	−29	300	...	204
20	Fe	Castings	A278	300	F13101	...	(8e) (53)	−29	300	...	343
21	Fe	Castings	A48	325	...	...	(8e) (48)	−29	325	...	204
22	Fe	Castings	A278	325	...	...	(8e) (53)	−29	325	...	204
23	Fe	Castings	A47	22010	F22200	...	(8e) (9)	−29	340	220	343
24	Fe	Castings	A48	350	F13501	...	(8e) (48)	−29	350	...	204
25	Fe	Castings	A278	350	F13502	...	(8e) (53)	−29	350	...	343
26	Fe	Castings	A48	400	F14101	...	(8e) (48)	−29	400	...	204
27	Fe	Castings	A278	415	F14102	...	(8e) (53)	−29	415	...	204
28	Fe	Castings	A395	60-40-18	F32800	...	(8d) (9)	−29	415	275	343
29	Fe	Castings	A536	65-45-12	F33100	...	(8d) (9)	−29	448	310	260
30	Fe	Castings	A571	D-2M	F43010	3	(8d)	−29	450	205	40

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]												
	Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325	350
1	13.8	13.8	13.8	13.8	13.8	13.8	13.8	13.8	...	...	...	...	...
2	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	...	...	...	...	...
3	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	...	...	...	...	...
4	17.5	17.5	17.5	17.5	17.5	17.5	17.5	17.5	...	...	...	...	...
5	17.5	17.5	17.5	17.5	17.5	17.5	17.5	17.5	...	...	...	...	...
6	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	...	...	...	...	...
7	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	...	...	...	...	...
8	20.7	20.7	20.7	20.7	20.7	20.7	20.7	20.7	...	...	...	...	...
9	22.5	22.5	22.5	22.5	22.5	22.5	22.5	22.5	...	...	...	...	...
10	22.5	22.5	22.5	22.5	22.5	22.5	22.5	22.5	...	...	...	...	...
11	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	...	...	...	...	...
12	25.0	25.0	25.0	25.0	25.0	25.0	25.0	25.0	27.6	27.6	27.6	27.6	27.6
13	27.5	27.5	27.5	27.5	27.5	27.5	27.5	27.5	...	...	...	...	...
14	27.5	27.5	27.5	27.5	27.5	27.5	27.5	27.5	...	...	...	...	...
15	37.5	37.5	37.5	37.5	37.5	37.5	37.5	37.5	...	...	...	...	...
16	38.0	38.0	38.0	38.0	38.0	38.0	38.0	38.0	...	...	...	...	...
17	55.2	55.2	55.2	55.2	55.2	55.2	55.2	55.2	55.2	55.2	55.2	55.2	55.2
18	27.6	27.6	27.6	27.6	27.6	27.6	27.6	27.6	...	...	...	...	...
19	30.0	30.0	30.0	30.0	30.0	30.0	30.0	30.0	...	...	...	...	...
20	30.0	30.0	30.0	30.0	30.0	30.0	30.0	30.0	34.5	34.5	34.5	34.5	34.5
21	32.5	32.5	32.5	32.5	32.5	32.5	32.5	32.5	...	...	...	...	...
22	32.5	32.5	32.5	32.5	32.5	32.5	32.5	32.5	...	...	...	...	...
23	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0	68.0
24	35.0	35.0	35.0	35.0	35.0	35.0	35.0	35.0	...	...	...	...	...
25	35.0	35.0	35.0	35.0	35.0	35.0	35.0	35.0	41.4	41.4	41.4	41.4	41.4
26	37.5	37.5	40.0	40.0	40.0	40.0	40.0	40.0	...	...	...	...	...
27	41.5	41.5	41.5	41.5	41.5	41.5	41.5	41.5	...	...	...	...	...
28	137	133	128	125	122	119	116	112	109	106	103	98.0	93.5
29	149	149	149	149	149	149	148	148	148	147	...	...	...
30	137	...	...	...	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Cond./ Temper	Size, mm	P- No. (5)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C
1	Carbon steel	Pipe & tube	A134	A285A	K01700	...	...	1	(8b) (57)	B	310	165	482
2	Carbon steel	Pipe & tube	A672	A45	K01700	...	...	1	(57) (59) (64)	B	310	165	593
3	Carbon steel	CW pipe	API 5L	A25	...	PSL 1	...	1	(8a) (8c) (63)	-29	310	175	204
4	Carbon steel	Smls. & EW pipe	API 5L	A25	...	PSL 1	...	1	(57) (59) (63)	B	310	175	204
5	Carbon steel	Pipe & tube	A179	...	K01200	...	...	1	(57) (59)	-29	325	180	593
6	Carbon steel	CW Pipe	A53	F/A	K02504	...	...	1	(8a) (8c)	-29	330	205	204
7	Carbon steel	Pipe & tube	A139	A	...	...	...	1	(8b)	A	330	205	149
8	Carbon steel	Pipe & tube	A587	...	K11500	...	...	1	(57) (59)	-29	330	205	454
9	Carbon steel	Wld. pipe	A53	E/A	K02504	...	...	1	(57) (59)	B	330	205	593
10	Carbon steel	Smls. pipe	A53	S/A	K02504	...	...	1	(57) (59)	B	330	205	593
11	Carbon steel	Pipe & tube	A106	A	K02501	...	...	1	(57)	B	330	205	593
12	Carbon steel	Pipe & tube	A135	A	...	...	...	1	(57) (59)	B	330	205	593
13	Carbon steel	Pipe & tube	A369	FPA	K02501	...	...	1	(57)	B	330	210	593
14	Carbon steel	Pipe	API 5L	A	...	PSL 1	...	1	(57) (59) (63)	B	335	210	593
15	Carbon steel	Pipe & tube	A134	A285B	K02200	...	...	1	(8b) (57)	B	345	185	482
16	Carbon steel	Pipe & tube	A672	A50	K02200	...	...	1	(57) (59) (64)	B	345	185	593
17	Carbon steel	Pipe & tube	A134	A285C	K02801	...	...	1	(8b) (57)	A	380	205	482
18	Carbon steel	Pipe & tube	A524	II	K02104	...	...	1	(57)	-29	380	205	538
19	Carbon steel	Pipe & tube	A333	1	K03008	...	...	1	(57) (59)	-46	380	205	593
20	Carbon steel	Pipe & tube	A334	1	K03008	...	...	1	(57) (59)	-46	380	205	593
21	Carbon steel	Pipe & tube	A671	CA55	K02801	...	...	1	(59) (64)	A	380	205	593
22	Carbon steel	Pipe & tube	A672	A55	K02801	...	...	1	(57) (59) (64)	A	380	205	593
23	Carbon steel	Pipe & tube	A672	C55	K01800	...	...	1	(57) (64)	C	380	205	593
24	Carbon steel	Pipe & tube	A671	CC60	K02100	...	...	1	(57) (64)	C	415	220	538
25	Carbon steel	Pipe & tube	A671	CB60	K02401	...	...	1	(57) (64)	B	415	220	593
26	Carbon steel	Pipe & tube	A672	B60	K02401	...	...	1	(57) (64)	B	415	220	593
27	Carbon steel	Pipe & tube	A672	C60	K02100	...	...	1	(57) (64)	C	415	220	593
28	Carbon steel	Pipe & tube	A139	B	K03003	...	...	1	(8b)	A	415	240	149
29	Carbon steel	Pipe & tube	A135	B	K03018	...	...	1	(57) (59)	B	415	240	538
30	Carbon steel	Pipe & tube	A524	I	K02104	...	...	1	(57)	-29	415	240	538
31	Carbon steel	Wld. pipe	A53	E/B	K03005	...	...	1	(57) (59)	B	415	240	593
32	Carbon steel	Smls. pipe	A53	S/B	K03005	...	...	1	(57) (59)	B	415	240	593
33	Carbon steel	Pipe & tube	A106	B	K03006	...	...	1	(57)	B	415	240	593
34	Carbon steel	Pipe & tube	A333	6	K03006	...	...	1	(57)	-46	415	240	593

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated																			
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]																		
	Min. Temp. to 40	65	100	150	200	250	300	325	350	375	400	425	450	475	500	525	550	575	600
1	103	103	101	97.5	94.6	90.8	86.1	83.6	81.1	78.6	73.3	64.0	55.8	43.9	40.7	...	...	...	...
2	103	103	101	97.5	94.6	90.8	86.1	83.6	81.1	78.6	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
3	103	103	103	102	98.5	91.0	...	...	...	...	...	...	...	...	...	...	...	...	...
4	103	103	103	102	98.5	91.0	...	...	...	...	...	...	...	...	...	...	...	...	...
5	108	108	108	106	102	98.3	93.3	90.6	87.8	84.3	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
6	110	110	110	110	110	...	...	...	...	...	...	...	...	...	...	...	...	...	...
7	110	110	110	110	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
8	110	110	110	110	110	110	108	105	97.0	84.3	73.3	64.0	55.8	54.5	...	...	...	...	...
9	110	110	110	110	110	110	108	105	97.0	84.3	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
10	110	110	110	110	110	110	108	105	97.0	84.3	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
11	110	110	110	110	110	110	108	105	97.0	84.3	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
12	110	110	110	110	110	110	108	105	97.0	84.3	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
13	110	110	110	110	110	110	108	105	97.0	84.3	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
14	110	110	110	110	110	110	108	105	97.0	84.3	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
15	115	115	113	110	106	102	96.9	94.1	91.2	84.3	73.3	64.0	55.8	43.9	40.7	...	...	...	...
16	115	115	113	110	106	102	96.9	94.1	91.2	84.3	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
17	126	126	126	122	118	113	108	105	101	98.3	89.1	75.4	62.6	45.5	31.6	...	...	...	...
18	126	126	126	122	118	113	108	105	101	98.3	89.1	75.4	62.6	45.5	31.6	21.9	12.7	...	...
19	126	126	126	122	118	113	108	105	101	98.3	89.1	75.4	62.6	45.5	31.6	21.9	12.7	9.40	6.89
20	126	126	126	122	118	113	108	105	101	98.3	89.1	75.4	62.6	45.5	31.6	21.9	12.7	9.40	6.89
21	126	126	126	122	118	113	108	105	101	98.3	89.1	75.4	62.6	45.5	31.6	21.9	12.7	9.40	6.89
22	126	126	126	122	118	113	108	105	101	98.3	89.1	75.4	62.6	45.5	31.6	21.9	12.7	9.40	6.89
23	126	126	126	122	118	113	108	105	101	98.3	89.1	75.4	62.6	45.5	31.6	21.9	12.7	9.40	6.89
24	138	138	134	130	126	121	115	111	108	105	88.9	75.3	62.7	45.5	31.6	21.9	12.7	...	...
25	138	138	134	130	126	121	115	111	108	105	88.9	75.3	62.7	45.5	31.6	21.9	12.7	9.40	6.89
26	138	138	134	130	126	121	115	111	108	105	88.9	75.3	62.7	45.5	31.6	21.9	12.7	9.40	6.89
27	138	138	134	130	126	121	115	111	108	105	88.9	75.3	62.7	45.5	31.6	21.9	12.7	9.40	6.89
28	138	138	138	138	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
29	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	17.2	...	...
30	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	17.2	...	...
31	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
32	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
33	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
34	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Cond./ Temper	Size, mm	P- No. (5)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C
35	Carbon steel	Pipe & tube	A334	6	K03006	...	...	1	(57)	-46	415	240	593
36	Carbon steel	Pipe & tube	A369	FPB	K03006	...	...	1	(57)	-29	415	240	593
37	Carbon steel	Pipe	A381	Y35	...	...	...	1	...	A	415	240	593
38	Carbon steel	Pipe	CSA Z245.1	241	...	...	...	1	(57) (59)	B	414	241	593
39	Carbon steel	Pipe	API 5L	B	...	PSL 1	...	1	(57) (59) (63)	B	415	245	593
40	Carbon steel	Pipe	API 5L	L245	...	PSL 2	...	1	(57) (59) (63)	0	415	245	593
41	Carbon steel	Pipe & tube	A139	C	K03004	...	...	1	(8b)	A	415	290	149
42	Carbon steel	Pipe & tube	A139	D	K03010	...	...	1	(8b)	A	415	315	149
43	Carbon steel	Pipe	A381	Y42	...	...	...	1	...	A	415	290	204
44	Carbon steel	Pipe	CSA Z245.1	290	...	...	...	1	(55)	A	414	290	260
45	Carbon steel	Pipe	API 5L	L290	...	PSL 1	...	1	(55) (63)	A	415	290	260
46	Carbon steel	Pipe	API 5L	L290	...	PSL 2	...	1	(55) (63)	0	415	290	260
47	Carbon steel	Pipe	A381	Y48	...	...	...	1	...	A	430	330	343
48	Carbon steel	Pipe	A381	Y46	...	...	...	1	...	A	435	316	204
49	Carbon steel	Pipe	API 5L	L320	...	PSL 1	...	1	(55) (63)	A	435	320	260
50	Carbon steel	Pipe	API 5L	L320	...	PSL 2	...	1	(55) (63)	0	435	320	260
51	Carbon steel	Pipe	A381	Y50	...	...	...	1	...	A	440	345	343
52	Carbon steel	Pipe & tube	A671	CC65	K02403	...	...	1	(57) (64)	B	450	240	538
53	Carbon steel	Pipe & tube	A671	CB65	K02800	...	...	1	(57) (64)	A	450	240	593
54	Carbon steel	Pipe & tube	A672	B65	K02800	...	...	1	(57) (64)	A	450	240	593
55	Carbon steel	Pipe & tube	A672	C65	K02403	...	...	1	(57) (64)	B	450	240	593
56	Carbon steel	Pipe & tube	A139	E	K03012	...	...	1	(8b)	A	455	360	149
57	Carbon steel	Pipe	A381	Y52	...	...	...	1	...	A	455	360	204
58	Carbon steel	Pipe	CSA Z245.1	359	...	...	...	1	(55)	A	455	359	260
59	Carbon steel	Pipe	API 5L	L360	...	PSL 1	...	1	(55) (63)	A	460	360	260
60	Carbon steel	Pipe	API 5L	L360	...	PSL 2	...	1	(55) (63)	0	460	360	260
61	Carbon steel	Pipe & tube	A671	CC70	K02700	...	...	1	(57) (64)	B	485	260	538
62	Carbon steel	Pipe & tube	A671	CB70	K03101	...	...	1	(57) (64)	A	485	260	593
63	Carbon steel	Pipe & tube	A672	B70	K03101	...	...	1	(57) (64)	A	485	260	593
64	Carbon steel	Pipe & tube	A672	C70	K02700	...	...	1	(57) (64)	B	485	260	593
65	Carbon steel	Pipe & tube	A106	C	K03501	...	...	1	(57)	B	485	275	427
66	Carbon steel	Pipe & tube	A671	CD70	K12437	...	≤65	1	(64)	D	485	345	371
67	Carbon steel	Pipe & tube	A672	D70	K12437	...	≤65	1	(64)	D	485	345	371
68	Carbon steel	Pipe & tube	A691	CMSH-70	K12437	...	≤65	1	(64)	D	485	345	371
69	Carbon steel	Pipe & tube	A381	Y56	...	...	...	1	(52)	A	490	385	204

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated																			
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]																		
	Min. Temp. to 40	65	100	150	200	250	300	325	350	375	400	425	450	475	500	525	550	575	600
35	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
36	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
37	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
38	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
39	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
40	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
41	138	138	138	138	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
42	138	138	138	138	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
43	138	138	138	138	138	138	...	...	...	...	...	...	...	...	...	...	...	...	...
44	138	138	138	138	138	138	132	...	...	...	...	...	...	...	...	...	...	...	...
45	138	138	138	138	138	138	132	...	...	...	...	...	...	...	...	...	...	...	...
46	138	138	138	138	138	138	132	...	...	...	...	...	...	...	...	...	...	...	...
47	142	142	142	142	142	142	142	142	129	...	...	...	...	...	...	...	...	...	...
48	145	145	145	145	145	145	...	...	...	...	...	...	...	...	...	...	...	...	...
49	145	145	145	145	145	145	145	...	...	...	...	...	...	...	...	...	...	...	...
50	145	145	145	145	145	145	145	...	...	...	...	...	...	...	...	...	...	...	...
51	147	147	147	147	147	147	147	147	129	...	...	...	...	...	...	...	...	...	...
52	149	149	147	142	138	132	126	122	118	113	95.0	79.6	63.2	45.3	31.7	21.9	12.7	...	...
53	149	149	147	142	138	132	126	122	118	113	95.0	79.6	63.2	45.3	31.7	21.9	12.7	9.40	6.89
54	149	149	147	142	138	132	126	122	118	113	95.0	79.6	63.2	45.3	31.7	21.9	12.7	9.40	6.89
55	149	149	147	142	138	132	126	122	118	113	95.0	79.6	63.2	45.3	31.7	21.9	12.7	9.40	6.89
56	152	152	152	152	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
57	152	152	152	152	152	152	...	...	...	...	...	...	...	...	...	...	...	...	...
58	152	152	152	152	152	152	152	...	...	...	...	...	...	...	...	...	...	...	...
59	153	153	153	153	153	153	153	...	...	...	...	...	...	...	...	...	...	...	...
60	153	153	153	153	153	153	153	...	...	...	...	...	...	...	...	...	...	...	...
61	161	161	159	154	150	144	136	132	128	122	101	83.8	67.1	51.0	33.6	21.3	12.9	...	...
62	161	161	159	154	150	144	136	132	128	122	101	83.8	67.1	51.0	33.6	21.3	12.9	9.40	6.89
63	161	161	159	154	150	144	136	132	128	122	101	83.8	67.1	51.0	33.6	21.3	12.9	9.40	6.89
64	161	161	159	154	150	144	136	132	128	122	101	83.8	67.1	51.0	33.6	21.3	12.9	9.40	6.89
65	161	161	161	161	158	151	144	139	135	122	101	83.8	82.7	...	...	...	...	...	...
66	161	161	161	157	156	156	156	154	148	126	...	...	...	...	...	...	...	...	...
67	161	161	161	157	156	156	156	154	148	126	...	...	...	...	...	...	...	...	...
68	161	161	161	157	156	156	156	154	148	126	...	...	...	...	...	...	...	...	...
69	163	163	163	163	163	163	...	...	...	...	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Cond./ Temper	Size, mm	P- No. (5)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C
70	Carbon steel	Pipe	CSA Z245.1	386	...	...	...	1	(52) (55)	A	490	386	260
71	Carbon steel	Pipe	API 5L	L390	...	PSL 1	...	1	(52) (55) (63)	A	490	390	260
72	Carbon steel	Pipe	API 5L	L390	...	PSL 2	...	1	(52) (55) (63)	0	490	390	260
73	Carbon steel	Pipe & tube	A671	CK75	K02803	...	>25	1	(57) (64)	A	515	275	593
74	Carbon steel	Pipe & tube	A672	N75	K02803	...	>25	1	(57) (64)	A	515	275	593
75	Carbon steel	Pipe & tube	A691	CMS-75	K02803	...	>25	1	(57) (64)	A	515	275	593
76	Carbon steel	Pipe & tube	A671	CK75	K02803	...	≤25	1	(57) (64)	A	515	290	371
77	Carbon steel	Pipe & tube	A672	N75	K02803	...	≤25	1	(57) (64)	A	515	290	371
78	Carbon steel	Pipe & tube	A691	CMS-75	K02803	...	≤25	1	(57) (64)	A	515	290	371
79	Carbon steel	Pipe	A381	Y60	...	...	...	1	(52)	A	515	415	204
80	Carbon steel	Pipe	CSA Z245.1	414	...	...	...	1	(52) (55)	A	517	414	260
81	Carbon steel	Pipe	API 5L	L415	...	PSL 1	...	1	(52) (55) (63)	A	520	415	260
82	Carbon steel	Pipe	API 5L	L415	...	PSL 2	...	1	(52) (55) (63)	0	520	415	260
83	Carbon steel	Pipe	CSA Z245.1	448	...	...	...	1	(52) (55)	A	531	448	260
84	Carbon steel	Pipe	API 5L	L450	...	PSL 1	...	1	(52) (55) (63)	A	535	450	260
85	Carbon steel	Pipe	API 5L	L450	...	PSL 2	...	1	(52) (55) (63)	0	535	450	260
86	Carbon steel	Pipe	CSA Z245.1	483	...	...	...	1	(51) (52) (55)	A	565	483	204
87	Carbon steel	Pipe	API 5L	L485	...	PSL 1	...	1	(51) (52) (55) (63)	A	570	485	204
88	Carbon steel	Pipe	API 5L	L485	...	PSL 2	...	1	(51) (52) (55) (63)	0	570	485	204
89	Carbon steel	Pipe	CSA Z245.1	550	...	...	...	1	(51) (52) (55)	A	620	550	204
90	Carbon steel	Pipe	API 5L	L555	...	PSL 2	...	1	(51) (52) (55) (63)	0	625	555	204
91	Carbon steel	Structural pipe	A134	A1011SS30	K02502	...	...	1	(8a) (8c)	-29	340	205	204
92	Carbon steel	Structural pipe	A134	A1011SS33	K02502	...	...	1	(8a) (8c)	-29	360	230	204
93	Carbon steel	Structural pipe	A134	A1011SS36-T1	K02502	...	...	1	(8a) (8c)	-29	365	250	204
94	Carbon steel	Structural pipe	A134	A1011SS40	K02502	...	...	1	(8a) (8c)	-29	380	275	204
95	Carbon steel	Structural pipe	A134	A36	K02600	...	...	1	(8a) (8c)	-29	400	250	204
96	Carbon steel	Structural pipe	A134	A283D	K02702	...	...	1	(8a) (8c)	-29	415	230	149
97	Carbon steel	Structural pipe	A134	A1011SS45-T1	K02507	...	...	1	(8a) (8c)	-29	410	310	204

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated																			
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]																		
	Min. Temp. to 40	65	100	150	200	250	300	325	350	375	400	425	450	475	500	525	550	575	600
70	163	163	163	163	163	163	163	...	...	...	...	...	...	...	...	...	...	...	...
71	163	163	163	163	163	163	163	...	...	...	...	...	...	...	...	...	...	...	...
72	163	163	163	163	163	163	163	...	...	...	...	...	...	...	...	...	...	...	...
73	172	172	168	163	158	151	144	139	135	131	107	88.0	67.3	50.3	33.2	21.4	14.2	9.40	6.89
74	172	172	168	163	158	151	144	139	135	131	107	88.0	67.3	50.3	33.2	21.4	14.2	9.40	6.89
75	172	172	168	163	158	151	144	139	135	131	107	88.0	67.3	50.3	33.2	21.4	14.2	9.40	6.89
76	172	172	172	171	165	159	151	146	142	131	...	...	...	...	...	...	...	...	...
77	172	172	172	171	165	159	151	146	142	131	...	...	...	...	...	...	...	...	...
78	172	172	172	171	165	159	151	146	142	131	...	...	...	...	...	...	...	...	...
79	172	172	172	172	172	172	...	...	...	...	...	...	...	...	...	...	...	...	...
80	172	172	172	172	172	172	172	...	...	...	...	...	...	...	...	...	...	...	...
81	173	173	173	173	173	173	173	...	...	...	...	...	...	...	...	...	...	...	...
82	173	173	173	173	173	173	173	...	...	...	...	...	...	...	...	...	...	...	...
83	177	177	177	177	177	177	177	...	...	...	...	...	...	...	...	...	...	...	...
84	178	178	178	178	178	178	178	...	...	...	...	...	...	...	...	...	...	...	...
85	178	178	178	178	178	178	178	...	...	...	...	...	...	...	...	...	...	...	...
86	188	188	188	188	188	188	...	...	...	...	...	...	...	...	...	...	...	...	...
87	190	190	190	190	190	190	...	...	...	...	...	...	...	...	...	...	...	...	...
88	190	190	190	190	190	190	...	...	...	...	...	...	...	...	...	...	...	...	...
89	207	207	207	207	207	207	...	...	...	...	...	...	...	...	...	...	...	...	...
90	208	208	208	208	208	208	...	...	...	...	...	...	...	...	...	...	...	...	...
91	104	104	104	104	104	...	...	...	...	...	...	...	...	...	...	...	...	...	...
92	110	110	110	110	110	...	...	...	...	...	...	...	...	...	...	...	...	...	...
93	112	112	112	112	112	...	...	...	...	...	...	...	...	...	...	...	...	...	...
94	116	116	116	116	116	...	...	...	...	...	...	...	...	...	...	...	...	...	...
95	123	123	123	123	123	...	...	...	...	...	...	...	...	...	...	...	...	...	...
96	127	127	127	123	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
97	127	127	127	127	127	...	...	...	...	...	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Cond./ Temper	Size, mm	P- No. (5)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C
98	Carbon steel	Structural pipe	A134	A1011SS45-T2	K02507	...	...	1	(8a) (8c)	−29	410	310	204
99	Carbon steel	Structural pipe	A134	A1011SS50	K02507	...	...	1	(8a) (8c)	−29	450	340	204
100	Carbon steel	Plate, bar, shps., sheet	A285	A	K01700	...	...	1	(57) (59)	B	310	165	593
101	Carbon steel	Plate, bar, shps., sheet	A285	B	K02200	...	...	1	(57) (59)	B	345	185	593
102	Carbon steel	Plate, bar, shps., sheet	A516	55	K01800	...	...	1	(57)	C	380	205	454
103	Carbon steel	Plate, bar, shps., sheet	A285	C	K02801	...	...	1	(57) (59)	A	380	205	593
104	Carbon steel	Plate, bar, shps., sheet	A516	60	K02100	...	...	1	(57)	C	415	220	454
105	Carbon steel	Plate, bar, shps., sheet	A515	60	K02401	...	...	1	(57)	B	415	220	538
106	Carbon steel	Plate, bar, shps., sheet	A696	B	K03200	...	...	1	(57)	A	415	240	371
107	Carbon steel	Plate, bar, shps., sheet	A516	65	K02403	...	...	1	(57)	B	450	240	454
108	Carbon steel	Plate, bar, shps., sheet	A515	65	K02800	...	...	1	(57)	A	450	240	538
109	Carbon steel	Plate, bar, shps., sheet	A516	70	K02700	...	...	1	(57)	B	485	260	454
110	Carbon steel	Plate, bar, shps., sheet	A515	70	K03101	...	...	1	(57)	A	485	260	538
111	Carbon steel	Plate, bar, shps., sheet	A696	C	K03200	...	...	1	(57)	A	485	275	371
112	Carbon steel	Plate, bar, shps., sheet	A537	...	K12437	1	≤65	1	...	D	485	345	371
113	Carbon steel	Plate, bar, shps., sheet	A299	A	K02803	...	>25	1	(57)	A	515	275	593
114	Carbon steel	Plate, bar, shps., sheet	A299	A	K02803	...	≤25	1	(57)	A	515	290	593
115	Carbon steel	Struct. sht., strip	A1011	SS30	K02502	...	...	1	(8c) (57)	A	340	205	399
116	Carbon steel	Struct. sht., strip	A1011	SS33	K02502	...	...	1	(8c) (57)	A	360	230	399
117	Carbon steel	Struct. sht., strip	A1011	SS36-T1	K02502	...	...	1	(8c) (57)	A	365	250	399
118	Carbon steel	Structural plate	A283	C	K02401	...	...	1	(8c) (57)	A	380	205	399
119	Carbon steel	Struct. sht., strip	A1011	SS40	K02502	...	...	1	(8c) (57)	A	380	275	399
120	Carbon steel	Structural steel	A36	...	K02600	...	...	1	(8c)	A	400	250	371
121	Carbon steel	Structural plate	A283	D	K02702	...	...	1	(8c) (57)	A	415	230	399
122	Carbon steel	Struct. sht., strip	A1011	SS45-T1	K02507	...	...	1	(8c) (57)	A	410	310	399
123	Carbon steel	Struct. sht., strip	A1011	SS45-T2	K02507	...	...	1	(8c) (57)	A	410	310	399
124	Carbon steel	Struct. sht., strip	A1011	SS50	K02507	...	...	1	(8c) (57)	A	450	340	399
125	Carbon steel	Structural shapes	A992	...	...	...	...	1	(8c) (57)	A	450	345	427
126	Carbon steel	Forgings & fittings	A350	LF1	K03009	...	...	1	(9) (57) (59)	−29	415	205	538
127	Carbon steel	Forgings & fittings	A181	...	K03502	60	...	1	(9) (57) (59)	A	415	205	593
128	Carbon steel	Forgings & fittings	A420	WPL6	K03006	...	...	1	(57)	−46	415	240	538

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated																			
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]																		
	Min. Temp. to 40	65	100	150	200	250	300	325	350	375	400	425	450	475	500	525	550	575	600
98	127	127	127	127	127	...	...	...	...	...	...	...	...	...	...	...	...	...	...
99	137	137	137	137	137	...	...	...	...	...	...	...	...	...	...	...	...	...	...
100	103	103	101	97.5	94.6	90.8	86.1	83.6	81.1	78.6	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
101	115	115	113	110	106	102	96.9	94.1	91.2	84.3	73.3	64.0	55.8	43.9	31.7	21.4	14.2	9.40	6.89
102	126	126	126	122	118	113	108	105	101	98.3	89.1	75.4	62.6	45.5	31.6	21.9	12.7	...	...
103	126	126	126	122	118	113	108	105	101	98.3	89.0	75.3	62.1	45.0	31.7	21.4	14.2	9.40	6.89
104	138	138	134	130	126	121	115	111	108	105	88.9	75.3	62.7	45.5	31.6	21.9	12.7	...	...
105	138	138	134	130	126	121	115	111	108	105	88.9	75.3	62.7	45.5	31.6	21.9	12.7	...	...
106	138	138	138	138	138	132	125	122	118	115	...	...	...	...	...	...	...	...	...
107	149	149	147	142	138	132	126	122	118	113	95.0	79.6	63.2	45.3	31.7	21.9	12.7	...	...
108	149	149	147	142	138	132	126	122	118	113	95.0	79.6	63.2	45.3	31.7	21.9	12.7	...	...
109	161	161	159	154	150	144	136	132	128	122	101	83.8	67.1	51.0	33.6	21.3	12.9	...	...
110	161	161	159	154	150	144	136	132	128	122	101	83.8	67.1	51.0	33.6	21.3	12.9	...	...
111	161	161	161	161	158	151	144	139	135	131	...	...	...	...	...	...	...	...	...
112	161	161	161	157	156	156	156	154	148	126	...	...	...	...	...	...	...	...	...
113	172	172	168	163	158	151	144	139	135	131	107	88.0	67.3	50.3	33.2	21.4	14.2	9.40	6.89
114	172	172	172	171	165	159	151	146	142	131	107	88.0	67.3	50.3	33.2	21.4	14.2	9.40	6.89
115	104	104	104	104	104	104	99.4	96.6	89.2	77.6	67.9	...	...	...	...	...	...	...	...
116	110	110	110	110	110	110	109	106	89.2	77.6	67.9	...	...	...	...	...	...	...	...
117	112	112	112	112	112	112	112	112	89.2	77.6	67.9	...	...	...	...	...	...	...	...
118	116	116	116	112	109	104	99.4	96.6	92.9	90.4	82.4	...	...	...	...	...	...	...	...
119	116	116	116	116	116	116	116	116	114	96.6	82.4	...	...	...	...	...	...	...	...
120	123	123	123	123	123	123	119	115	112	99.4	...	...	...	...	...	...	...	...	...
121	127	127	127	123	120	115	109	106	102	99.4	88.1	...	...	...	...	...	...	...	...
122	127	127	127	127	127	127	127	127	124	104	88.1	...	...	...	...	...	...	...	...
123	127	127	127	127	127	127	127	127	124	104	88.1	...	...	...	...	...	...	...	...
124	137	137	137	137	137	137	137	137	124	104	88.1	...	...	...	...	...	...	...	...
125	137	137	137	137	137	137	137	137	117	105	87.5	73.2	58.1	...	...	...	...	...	...
126	138	130	126	122	118	113	108	105	101	98.3	95.1	79.5	62.6	45.0	31.7	21.4	17.2	...	...
127	138	130	126	122	118	113	108	105	101	98.3	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
128	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	17.2	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Cond./ Temper	Size, mm	P- No. (5)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C
129	Carbon steel	Forgings & fittings	A234	WPB	K03006	...	...	1	(57) (59)	B	415	240	593
130	Carbon steel	Forgings & fittings	A694	F42	K03014	...	...	1	(9)	−29	415	290	260
131	Carbon steel	Forgings & fittings	A707	L1	K02302	1	...	1	(9)	−29	415	290	260
132	Carbon steel	Forgings & fittings	A707	L2	K03301	1	...	1	(9)	−46	415	290	260
133	Carbon steel	Forgings & fittings	A707	L3	K12510	1	...	1	(9)	−46	415	290	260
134	Carbon steel	Forgings & fittings	A860	WPHY 42	...	...	...	1	...	−46	415	290	260
135	Carbon steel	Forgings & fittings	A694	F46	K03014	...	...	1	(9)	−29	435	315	260
136	Carbon steel	Forgings & fittings	A860	WPHY 46	...	...	...	1	...	−46	435	315	260
137	Carbon steel	Forgings & fittings	A694	F52	K03014	...	...	1	(9)	−29	455	360	260
138	Carbon steel	Forgings & fittings	A707	L1	K02302	2	...	1	(9)	−29	455	360	260
139	Carbon steel	Forgings & fittings	A707	L2	K03301	2	...	1	(9)	−46	455	360	260
140	Carbon steel	Forgings & fittings	A707	L3	K12510	2	...	1	(9)	−46	455	360	260
141	Carbon steel	Forgings & fittings	A860	WPHY 52	...	...	...	1	...	−46	455	360	260
142	Carbon steel	Forgings & fittings	A350	LF2	K03011	1	...	1	(9) (57)	−46	485	250	538
143	Carbon steel	Forgings & fittings	A350	LF2	K03011	2	...	1	(9) (57)	−18	485	250	538
144	Carbon steel	Forgings & fittings	A105	...	K03504	...	...	1	(9) (57) (59) (65)	−29	485	250	593
145	Carbon steel	Forgings & fittings	A181	...	K03502	70	...	1	(9) (57) (59)	A	485	250	593
146	Carbon steel	Forgings & fittings	A234	WPC	K03501	...	...	1	(57) (59)	B	485	275	427
147	Carbon steel	Forgings & fittings	A694	F56	K03014	...	...	1	(9)	−29	490	385	260
148	Carbon steel	Forgings & fittings	A694	F60	K03014	...	...	1	(9)	−29	515	415	260
149	Carbon steel	Forgings & fittings	A707	L2	K03301	3	...	1	(9)	−46	515	415	260
150	Carbon steel	Forgings & fittings	A707	L3	K12510	3	...	1	(9)	−46	515	415	260
151	Carbon steel	Forgings & fittings	A860	WPHY 60	...	...	...	1	...	−46	515	415	260
152	Carbon steel	Forgings & fittings	A694	F65	K03014	...	...	1	(9)	−29	530	450	260
153	Carbon steel	Forgings & fittings	A860	WPHY 65	...	...	...	1	...	−46	530	450	260
154	Carbon steel	Forgings & fittings	A694	F70	K03014	...	...	1	(9) (38) (51)	...	565	485	204
155	Carbon steel	Forgings & fittings	A860	WPHY 70	...	...	...	1	(51)	−46	565	485	204
156	Carbon steel	Castings	A216	WCA	J02502	...	...	1	(57)	−29	415	205	593
157	Carbon steel	Castings	A352	LCB	J03003	...	...	1	(9) (57)	−46	450	240	593
158	Carbon steel	Castings	A352	LCC	J02505	...	...	1	(9)	−46	485	275	371
159	Carbon steel	Castings	A216	WCB	J03002	...	...	1	(9) (57)	−29	485	250	593
160	Carbon steel	Castings	A216	WCC	J02503	...	...	1	(9) (57)	−29	485	275	538

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated																			
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]																		
	Min. Temp. to 40	65	100	150	200	250	300	325	350	375	400	425	450	475	500	525	550	575	600
129	138	138	138	138	138	132	126	122	118	113	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
130	138	138	138	138	138	137	132	...	...	...	...	...	...	...	...	...	...	...	...
131	138	138	138	138	138	137	132	...	...	...	...	...	...	...	...	...	...	...	...
132	138	138	138	138	138	137	132	...	...	...	...	...	...	...	...	...	...	...	...
133	138	138	138	138	138	137	132	...	...	...	...	...	...	...	...	...	...	...	...
134	138	138	138	138	138	137	132	...	...	...	...	...	...	...	...	...	...	...	...
135	145	145	145	145	145	145	144	...	...	...	...	...	...	...	...	...	...	...	...
136	145	145	145	145	145	145	144	...	...	...	...	...	...	...	...	...	...	...	...
137	150	150	150	150	150	150	150	...	...	...	...	...	...	...	...	...	...	...	...
138	150	150	150	150	150	150	150	...	...	...	...	...	...	...	...	...	...	...	...
139	150	150	150	150	150	150	150	...	...	...	...	...	...	...	...	...	...	...	...
140	150	150	150	150	150	150	150	...	...	...	...	...	...	...	...	...	...	...	...
141	150	150	150	150	150	150	150	...	...	...	...	...	...	...	...	...	...	...	...
142	161	156	151	146	142	136	129	125	122	118	101	83.8	66.8	50.3	33.2	21.4	17.2	...	...
143	161	156	151	146	142	136	129	125	122	118	101	83.8	66.8	50.3	33.2	21.4	17.2	...	...
144	161	156	151	146	142	136	129	125	122	118	101	83.8	66.8	50.3	33.2	21.4	14.2	9.40	6.89
145	161	156	151	146	142	136	129	125	122	118	101	83.8	66.8	50.3	33.2	21.4	14.2	9.40	6.89
146	161	161	161	161	158	151	144	139	135	122	101	83.8	82.7	...	...	...	...	...	...
147	163	163	163	163	163	163	163	...	...	...	...	...	...	...	...	...	...	...	...
148	172	172	172	172	172	172	172	...	...	...	...	...	...	...	...	...	...	...	...
149	172	172	172	172	172	172	172	...	...	...	...	...	...	...	...	...	...	...	...
150	172	172	172	172	172	172	172	...	...	...	...	...	...	...	...	...	...	...	...
151	172	172	172	172	172	172	172	...	...	...	...	...	...	...	...	...	...	...	...
152	177	177	177	177	177	177	177	...	...	...	...	...	...	...	...	...	...	...	...
153	177	177	177	177	177	177	177	...	...	...	...	...	...	...	...	...	...	...	...
154	188	188	188	188	188	188	...	...	...	...	...	...	...	...	...	...	...	...	...
155	188	188	188	188	188	188	...	...	...	...	...	...	...	...	...	...	...	...	...
156	138	130	126	122	118	113	108	105	101	98.3	95.1	79.5	62.6	45.0	31.7	21.4	14.2	9.40	6.89
157	149	149	147	142	138	132	126	122	118	113	95.1	79.5	64.4	47.7	32.5	21.4	14.2	9.40	6.89
158	161	161	161	161	158	151	139	137	136	132	...	...	...	...	...	...	...	...	...
159	161	156	151	146	142	136	129	125	122	118	101	83.8	66.8	50.3	33.2	21.4	14.2	9.40	6.89
160	161	161	161	161	158	151	144	139	135	122	101	83.8	66.8	50.3	33.2	21.4	17.2	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C
1	½Cr–½Mo	Pipe	A335	P2	K11547	...	...	3	...	–29	380	205	538
2	½Cr–½Mo	Pipe	A691	½CR	K12143	...	...	3	(11) (64)	–29	380	230	538
3	C–½Mo	Pipe	A335	P1	K11522	...	...	3	(58)	–29	380	205	593
4	C–½Mo	Pipe	A369	FP1	K11522	...	...	3	(58)	–29	380	210	593
5	½Cr–½Mo	Pipe	A369	FP2	K11547	...	...	3	...	–29	380	210	593
6	1Cr–½Mo	Pipe	A691	1CR	K11757	...	...	4	(11) (64)	–29	380	230	649
7	½Cr–½Mo	Pipe	A426	CP2	J11547	...	...	3	...	–29	415	205	593
8	1½Si–½Mo	Pipe	A335	P15	K11578	...	...	3	...	–29	415	205	593
9	C–½Mo–Si	Pipe	A426	CP15	J11522	...	...	3	...	–29	415	205	593
10	1Cr–½Mo	Pipe	A426	CP12	J11562	...	...	4	...	–29	415	205	649
11	5Cr–1½Si–½Mo	Pipe	A426	CP5b	J51545	...	...	5B	...	–29	415	205	649
12	3Cr–Mo	Pipe	A426	CP21	J31545	...	...	5A	...	–29	415	205	649
13	¾Cr–¾Ni–Cu–Al	Pipe	A333	4	K11267	...	...	4	...	–101	415	240	40
14	2Cr–½Mo	Pipe	A369	FP3b	K21509	...	...	4	...	–29	415	210	649
15	1Cr–½Mo	Pipe	A335	P12	K11562	...	...	4	...	–29	415	220	649
16	1Cr–½Mo	Pipe	A369	FP12	K11562	...	...	4	...	–29	415	220	649
17	1¼Cr–½Mo–Si	Pipe	A335	P11	K11597	...	...	4	...	–29	415	205	649
18	1¼Cr–½Mo–Si	Pipe	A369	FP11	K11597	...	...	4	...	–29	415	210	649
19	1¼Cr–½Mo–Si A387 Gr. 11 Cl. 1	Pipe	A691	1¼CR	K11789	...	...	4	(11) (64)	–29	415	240	649
20	5Cr–½Mo	Pipe	A691	5CR	K41545	...	...	5B	(11) (64)	–29	415	205	649
21	5Cr–½Mo	Pipe	A335	P5	K41545	...	...	5B	...	–29	415	205	649
22	5Cr–½Mo–Si	Pipe	A335	P5b	K51545	...	...	5B	...	–29	415	205	649
23	5Cr–½Mo–Ti	Pipe	A335	P5c	K41245	...	...	5B	...	–29	415	205	649
24	5Cr–½Mo	Pipe	A369	FP5	K41545	...	...	5B	...	–29	415	210	649
25	9Cr–1Mo	Pipe	A335	P9	K90941	...	...	5B	...	–29	415	205	649
26	9Cr–1Mo	Pipe	A369	FP9	K90941	...	...	5B	...	–29	415	210	649
27	9Cr–1Mo	Pipe	A691	9CR	K90941	...	...	5B	(11) (64)	–29	415	205	649
28	3Cr–1Mo	Pipe	A335	P21	K31545	...	...	5A	...	–29	415	205	649
29	3Cr–1Mo	Pipe	A369	FP21	K31545	...	...	5A	...	–29	415	210	649
30	3Cr–1Mo	Pipe	A691	3CR	K31545	...	...	5A	(11) (64)	–29	415	205	649
31	2¼Cr–1Mo	Pipe	A691	2¼CR	K21590	...	...	5A	(11) (15) (18) (64)	–29	415	205	649
32	2¼Cr–1Mo	Pipe	A369	FP22	K21590	...	...	5A	(15) (18)	–29	415	210	649
33	2¼Cr–1Mo	Pipe	A335	P22	K21590	...	...	5A	(15) (18)	–29	415	205	649
34	2Ni–1Cu	Pipe	A333	9	K22035	...	...	9A	...	–73	435	315	40
35	2Ni–1Cu	Pipe	A334	9	K22035	...	...	9A	...	–73	435	315	40

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated																											
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]																										
	Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325	350	375	400	425	450	475	500	525	550	575	600	625	650		
1	126	126	126	126	124	122	120	119	117	115	114	112	110	108	106	103	100	97.1	74.4	49.9	40.7	...	...	...	...		
2	126	126	126	126	126	126	126	126	126	126	125	123	121	119	116	114	110	107	74.4	49.9	40.7	...	...	...	...		
3	126	126	126	126	124	122	120	119	117	115	114	112	110	108	106	103	100	97.1	68.0	42.3	30.5	23.2	16.5	...	...		
4	126	126	126	126	124	122	120	119	117	115	114	112	110	108	106	103	100	97.1	68.0	42.3	30.5	23.2	16.5	...	...		
5	126	126	126	126	124	122	120	119	117	115	114	112	110	108	106	103	100	97.1	93.5	49.9	34.7	23.9	17.2	...	...		
6	126	126	123	122	122	122	122	121	120	118	116	115	114	112	110	109	106	104	92.1	61.1	40.4	26.4	17.4	11.6	7.58		
7	138	133	129	126	124	122	120	119	117	115	114	112	110	108	106	103	100	97.1	74.4	49.9	34.3	23.2	16.5	...	...		
8	138	133	129	127	125	124	122	121	120	118	117	115	114	112	110	107	103	88.4	74.7	53.7	35.6	23.2	16.5	...	...		
9	138	133	129	127	125	124	122	121	120	118	117	115	114	112	110	107	103	88.4	74.7	53.7	35.6	23.2	16.5	...	...		
10	138	129	124	120	117	115	112	110	109	107	106	105	103	102	100	98.7	96.8	94.6	92.0	61.1	40.4	26.4	17.4	11.6	7.58		
11	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
12	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	89.2	68.8	54.2	43.4	34.0	25.1	17.1	10.3		
13	138	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
14	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	118	74.7	53.3	36.0	24.6	15.5	9.21	6.89		
15	138	138	132	128	125	122	120	118	116	114	113	112	110	109	107	105	103	101	92.1	61.1	40.4	26.4	17.4	11.6	7.58		
16	138	138	132	128	125	122	120	118	116	114	113	112	110	109	107	105	103	101	92.1	61.1	40.4	26.4	17.4	11.6	7.58		
17	138	131	126	124	121	119	116	115	113	111	109	107	106	104	102	99.6	97.2	94.5	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
18	138	131	126	124	121	119	116	115	113	111	109	107	106	104	102	99.6	97.2	94.5	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
19	138	138	138	138	138	138	136	134	131	129	127	125	123	121	119	116	113	104	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
20	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
21	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
22	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
23	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
24	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
25	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	98.3	83.2	60.2	42.9	29.9	20.6	14.4	10.3		
26	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	98.3	83.2	60.2	42.9	29.9	20.6	14.4	10.3		
27	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	98.3	83.2	60.2	42.9	29.9	20.6	14.4	10.3		
28	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	89.2	68.8	54.2	43.4	34.0	25.1	17.1	10.3		
29	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	89.2	68.8	54.2	43.4	34.0	25.1	17.1	10.3		
30	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	89.2	68.8	54.2	43.4	34.0	25.1	17.1	10.3		
31	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	99.6	80.9	63.3	47.5	34.2	23.5	15.3	9.65		
32	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	99.6	80.9	63.3	47.5	34.2	23.5	15.3	9.65		
33	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	99.6	80.9	63.3	47.5	34.2	23.5	15.3	9.65		
34	145	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
35	145	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C
36	2¼Ni	Pipe	A333	7	K21903	...	...	9A	...	-73	450	240	593
37	2¼Ni	Pipe	A334	7	K21903	...	...	9A	...	-73	450	240	593
38	3½Ni	Pipe	A333	3	K31918	...	...	9B	...	-101	450	240	593
39	3½Ni	Pipe	A334	3	K31918	...	...	9B	...	-101	450	240	593
40	C-½Mo	Pipe	A426	CP1	J12521	...	...	3	(58)	-29	450	240	593
41	C-½Mo	Pipe	A672	L65	K11820	...	...	3	(11) (58) (64)	-29	450	255	593
42	C-½Mo	Pipe	A691	CM-65	K11820	...	...	3	(11) (58) (64)	-29	450	255	593
43	2¼Ni	Pipe	A671	CFB70	K22103	...	...	9A	(11) (46) (64)	-29	485	275	40
44	3½Ni	Pipe	A671	CFE70	K32018	...	...	9B	(11) (46) (64)	-29	485	275	40
45	C-½Mo	Pipe	A672	L70	K12020	...	...	3	(11) (58) (64)	-29	485	275	593
46	C-½Mo	Pipe	A691	CM-70	K12020	...	...	3	(11) (58) (64)	-29	485	275	593
47	1¼Cr-½Mo	Pipe	A426	CP11	J12072	...	...	4	...	-29	485	275	649
48	2¼Cr-1Mo	Pipe	A426	CP22	J21890	...	...	5A	(15)	-29	485	275	649
49	C-½Mo	Pipe	A672	L75	K12320	...	...	3	(11) (58) (64)	-29	515	295	593
50	C-½Mo	Pipe	A691	CM-75	K12320	...	...	3	(11) (58) (64)	-29	515	295	593
51	1¼Cr-½Mo-Si A387 Gr. 11. Cl. 2	Pipe	A691	1¼Cr	K11789	...	...	4	(11) (64)	-29	515	310	649
52	9Cr-1Mo-V	Pipe	A335	P91	K90901	...	≤75	15E	...	-29	585	415	649
53	9Cr-1Mo-V	Pipe	A691	91	K90901	...	≤75	15E	(11) (64)	-29	585	415	649
54	5Cr-½Mo	Pipe	A426	CP5	J42045	...	...	5B	...	-29	620	415	649
55	9Cr-1Mo	Pipe	A426	CP9	J82090	...	...	5B	...	-29	620	415	649
56	9Ni	Pipe	A333	8	K81340	...	...	11A	(47)	-196	690	515	93
57	9Ni	Pipe	A334	8	K81340	...	...	11A	...	-196	690	520	93
58	½Cr-½Mo	Plate	A387	2	K12143	1	...	3	...	-29	380	230	538
59	1Cr-½Mo	Plate	A387	12	K11757	1	...	4	...	-29	380	230	649
60	9Cr-1Mo	Plate	A387	9	K90941	1	...	5B	...	-29	415	205	649
61	1¼Cr-½Mo-Si	Plate	A387	11	K11789	1	...	4	...	-29	415	240	649
62	5Cr-½Mo	Plate	A387	5	K41545	1	...	5B	...	-29	415	205	649
63	3Cr-1Mo	Plate	A387	21	K31545	1	...	5A	...	-29	415	205	649
64	2¼Cr-1Mo	Plate	A387	22	K21590	1	...	5A	(15)	-29	415	205	649
65	2¼Ni	Plate	A203	A	K21703	...	...	9A	(12) (46)	-29	450	255	538
66	3½Ni	Plate	A203	D	K31718	...	...	9B	(12) (46)	-29	450	255	538
67	C-½Mo	Plate	A204	A	K11820	...	...	3	(58)	-29	450	255	593
68	1Cr-½Mo	Plate	A387	12	K11757	2	...	4	...	-29	450	275	649

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated																											
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]																										
	Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325	350	375	400	425	450	475	500	525	550	575	600	625	650		
36	149	149	147	144	142	140	138	135	132	128	124	119	113	107	95.1	79.5	64.4	48.8	35.4	22.6	14.2	9.40	6.89	...	...		
37	149	149	147	144	142	140	138	135	132	128	124	119	113	107	95.1	79.5	64.4	48.8	35.4	22.6	14.2	9.40	6.89	...	...		
38	149	149	147	144	142	140	138	135	132	128	124	119	113	107	95.1	79.5	64.4	48.8	35.4	22.6	14.2	9.48	7.06	...	...		
39	149	149	147	144	142	140	138	135	132	128	124	119	113	107	95.1	79.5	64.4	48.8	35.4	22.6	14.2	9.48	7.06	...	...		
40	149	149	149	148	145	143	140	138	137	135	133	131	129	126	123	120	117	109	68.0	42.3	30.5	23.2	16.5	...	...		
41	149	149	149	149	149	149	148	146	144	142	140	138	136	133	131	127	124	109	68.0	42.3	30.5	23.2	16.5	...	...		
42	149	149	149	149	149	149	148	146	144	142	140	138	136	133	131	127	124	109	68.0	42.3	30.5	23.2	16.5	...	...		
43	161	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
44	161	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
45	161	161	161	161	161	161	161	158	156	154	152	149	147	144	141	138	134	109	68.0	42.3	30.5	23.2	16.5	...	...		
46	161	161	161	161	161	161	161	158	156	154	152	149	147	144	141	138	134	109	68.0	42.3	30.5	23.2	16.5	...	...		
47	161	161	161	161	161	158	155	153	150	148	146	143	141	138	136	133	130	104	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
48	161	161	160	157	156	156	156	156	156	156	156	156	156	156	156	156	156	119	88.4	64.0	44.6	30.0	19.7	12.8	8.27		
49	172	172	172	172	172	172	172	170	168	165	163	161	158	155	152	148	144	109	68.0	42.3	30.5	23.2	16.5	...	...		
50	172	172	172	172	172	172	172	170	168	165	163	161	158	155	152	148	144	109	68.0	42.3	30.5	23.2	16.5	...	...		
51	172	172	172	172	172	172	172	172	169	166	164	161	159	156	153	149	146	104	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
52	195	195	195	195	195	195	195	195	195	194	193	192	190	187	183	178	172	165	156	147	137	115	87.0	64.7	45.1	29.6	
53	195	195	195	195	195	195	195	195	195	194	193	192	190	187	183	178	172	165	156	147	137	115	87.0	64.7	45.1	29.6	
54	207	207	205	202	200	199	199	199	198	198	196	194	191	187	182	176	169	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
55	207	207	205	202	200	199	199	199	198	198	196	194	191	187	182	176	169	160	87.5	61.2	42.9	29.9	20.6	14.4	10.3		
56	230	230	230	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
57	230	230	230	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
58	126	126	126	126	126	126	126	126	126	126	125	123	121	119	116	114	110	107	74.4	49.9	40.7	...	...	...	...		
59	126	126	123	122	122	122	122	121	120	118	116	115	114	112	110	109	106	104	92.1	61.1	40.4	26.4	17.4	11.6	7.58		
60	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	98.3	83.2	60.2	42.9	29.9	20.6	14.4	10.3		
61	138	138	138	138	138	138	136	134	131	129	127	125	123	121	119	116	113	104	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
62	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
63	138	130	126	123	121	119	117	116	115	114	112	111	110	109	107	105	103	89.2	68.8	54.2	43.4	34.0	25.1	17.1	10.3		
64	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	99.6	80.9	63.3	47.5	34.2	23.5	15.3	9.65		
65	149	149	149	149	149	148	146	143	140	136	131	126	120	113	95.1	79.5	64.4	48.8	35.4	22.6	17.2	...	...	...	...		
66	149	149	149	149	149	148	146	143	140	136	131	126	120	113	95.1	79.5	64.4	48.8	35.4	22.6	17.2	...	...	...	...		
67	149	149	149	149	149	149	148	146	144	142	140	138	136	133	131	127	124	109	68.0	42.3	30.5	23.2	16.5	...	...		
68	149	149	146	144	144	144	144	144	144	143	141	139	138	136	134	132	129	126	92.1	61.1	40.4	26.4	17.4	11.6	7.58		

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C
69	2¼Ni	Plate	A203	B	K22103	...	...	9A	(12) (46)	-29	485	275	538
70	3½Ni	Plate	A203	E	K32018	...	...	9B	(12) (46)	-29	485	275	538
71	½Cr-½Mo	Plate	A387	2	K12143	2	...	3	...	-29	485	310	538
72	C-½Mo	Plate	A204	B	K12020	...	...	3	(58)	-29	485	275	593
73	Mn-½Mo	Plate	A302	A	K12021	...	...	3	...	-29	515	310	538
74	C-½Mo	Plate	A204	C	K12320	...	...	3	(58)	-29	515	295	593
75	1¼Cr-½Mo-Si	Plate	A387	11	K11789	2	...	4	...	-29	515	310	649
76	5Cr-½Mo	Plate	A387	5	K41545	2	...	5B	...	-29	515	310	649
77	3Cr-1Mo	Plate	A387	21	K31545	2	...	5A	...	-29	515	310	649
78	2¼Cr-1Mo	Plate	A387	22	K21590	2	...	5A	(15)	-29	515	310	649
79	Mn-½Mo	Plate	A302	B	K12022	...	...	3	...	-29	550	345	538
80	Mn-½Mo-½Ni	Plate	A302	C	K12039	...	...	3	...	-29	550	345	538
81	Mn-½Mo-¾Ni	Plate	A302	D	K12054	...	...	3	...	-29	550	345	538
82	9Cr-1Mo-V	Plate	A387	91	K90901	2	≤75	15E	...	-29	585	415	649
83	8Ni	Plate	A553	II	K71340	...	...	11A	(47)	-171	690	585	40
84	5Ni-¼Mo	Plate	A645	A	K41583	...	...	11A	...	-171	655	450	93
85	9Ni	Plate	A553	I	K81340	...	...	11A	(47)	-196	690	585	93
86	9Ni	Plate	A353	...	K81340	...	...	11A	(47)	-196	690	515	93
87	C-½Mo	Forg. & ftg.	A234	WP1	K12821	...	...	3	(58)	-29	380	205	593
88	1Cr-½Mo	Forg. & ftg.	A182	F12	K11562	1	...	4	(9)	-29	415	220	649
89	1Cr-½Mo	Forg. & ftg.	A234	WP12	K12062	1	...	4	...	-29	415	220	649
90	1¼Cr-½Mo-Si	Forg. & ftg.	A182	F11	K11597	1	...	4	(9)	-29	415	205	649
91	1¼Cr-½Mo-Si	Forg. & ftg.	A234	WP11	K11597	1	...	4	...	-29	415	205	649
92	2¼Cr-1Mo	Forg. & ftg.	A182	F22	K21590	1	...	5A	(9) (15) (18)	-29	415	205	649
93	2¼Cr-1Mo	Forg. & ftg.	A234	WP22	K21590	1	...	5A	(15)	-29	415	205	649
94	5Cr-½Mo	Forg. & ftg.	A234	WP5	K41545	...	...	5B	...	-29	415	205	649
95	9Cr-1Mo	Forg. & ftg.	A234	WP9	K90941	...	...	5B	...	-29	415	205	649
96	3½Ni	Forg. & ftg.	A420	WPL3	K31918	...	...	9B	...	-101	450	240	343
97	3½Ni	Forg. & ftg.	A350	LF3	K32025	...	...	9B	(9)	-101	485	260	343
98	½Cr-½Mo	Forg. & ftg.	A182	F2	K12122	...	...	3	(9)	-29	485	275	538
99	C-½Mo	Forg. & ftg.	A182	F1	K12822	...	...	3	(9) (58)	-29	485	275	593

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated																											
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]																										
	Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325	350	375	400	425	450	475	500	525	550	575	600	625	650		
69	161	161	161	161	161	160	158	155	151	147	142	136	130	113	95.1	79.5	64.4	48.8	35.4	22.6	17.2	...	...	...	...		
70	161	161	161	161	161	160	158	155	151	147	142	136	130	122	101	83.8	66.8	49.2	35.4	22.6	17.2	...	...	...	...		
71	161	161	161	161	161	161	161	161	161	161	161	161	161	161	159	155	151	146	93.5	49.9	40.7	...	...	...	...		
72	161	161	161	161	161	161	161	158	156	154	152	149	147	144	141	138	134	109	68.0	42.3	30.5	23.2	16.5	...	...		
73	172	172	172	172	172	172	172	172	172	172	172	172	171	168	165	160	154	104	68.0	42.3	33.1	...	...	...	...		
74	172	172	172	172	172	172	172	170	168	165	163	161	158	155	152	148	144	109	68.0	42.3	30.5	23.2	16.5	...	...		
75	172	172	172	172	172	172	172	172	169	166	164	161	159	156	153	149	146	104	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
76	172	172	171	169	167	166	165	165	165	164	164	162	159	156	130	126	104	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
77	172	172	171	168	167	167	167	167	167	167	167	167	167	167	167	167	167	98.2	73.5	54.7	40.6	29.2	20.6	15.2	8.96		
78	172	172	171	168	167	167	167	167	167	167	167	167	167	167	167	167	167	119	88.4	64.0	44.6	30.0	19.7	12.8	8.27		
79	184	184	184	184	184	184	184	184	184	184	184	184	184	184	183	178	172	104	68.0	42.3	33.1	...	...	...	...		
80	184	184	184	184	184	184	184	184	184	184	184	184	184	184	184	183	178	172	104	68.0	42.3	33.1	...	...	...	...	
81	184	184	184	184	184	184	184	184	184	184	184	184	184	184	184	183	178	172	104	68.0	42.3	33.1	...	...	...	...	
82	195	195	195	195	195	195	195	195	194	193	192	190	187	183	178	172	165	156	147	137	115	87.0	64.7	45.1	29.6		
83	230	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
84	218	218	218	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
85	230	230	230	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
86	230	230	230	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
87	126	126	126	126	124	122	120	119	117	115	114	112	110	108	106	103	100	97.1	68.0	42.3	30.5	23.2	16.5	...	...		
88	138	129	124	120	117	115	112	110	109	107	106	105	103	102	100	98.7	96.8	94.6	92.0	61.1	40.4	26.4	17.4	11.6	7.58		
89	138	138	132	128	125	122	120	118	116	114	113	112	110	109	107	105	103	101	92.1	61.1	40.4	26.4	17.4	11.6	7.58		
90	138	131	126	124	121	119	116	115	113	111	109	107	106	104	102	99.6	97.2	94.5	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
91	138	131	126	124	121	119	116	115	113	111	109	107	106	104	102	99.6	97.2	94.5	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
92	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	99.6	80.9	63.3	47.5	34.2	23.5	15.3	9.65		
93	138	132	128	126	125	124	124	124	124	124	124	124	124	124	123	122	121	99.6	80.9	63.3	47.5	34.2	23.5	15.3	9.65		
94	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
95	138	129	124	122	120	119	119	118	118	117	117	116	114	112	110	106	103	98.3	87.5	61.2	42.9	29.9	20.6	14.4	10.3		
96	149	149	147	144	142	140	138	135	132	128	124	119	113	...	...	...	...	...	...	...	...	...	...	...	...		
97	161	160	157	155	152	150	148	145	142	137	133	128	122	...	...	...	...	...	...	...	...	...	...	...	...		
98	161	161	161	161	161	161	161	158	156	154	152	149	147	144	141	138	134	129	93.5	49.9	40.7	...	...	...	...		
99	161	161	161	161	161	161	161	158	156	154	152	149	147	144	141	138	134	109	68.0	42.3	30.5	23.2	16.5	...	...		

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C
100	1Cr-½Mo	Forg. & ftg.	A182	F12	K11564	2	...	4	(9)	-29	485	275	649
101	1Cr-½Mo	Forg. & ftg.	A234	WP12	K12062	2	...	4	...	-29	485	275	649
102	1¼Cr-½Mo-Si	Forg. & ftg.	A182	F11	K11572	2	...	4	(9)	-29	485	275	649
103	1¼Cr-½Mo-Si	Forg. & ftg.	A234	WP11	K11572	2	...	4	...	-29	485	275	649
104	5Cr-½Mo	Forg. & ftg.	A182	F5	K41545	...	...	5B	(9)	-29	485	275	649
105	3Cr-1Mo	Forg. & ftg.	A182	F21	K31545	...	...	5A	(9)	-29	515	310	649
106	2¼Cr-1Mo	Forg. & ftg.	A182	F22	K21590	3	...	5A	(9) (15)	-29	515	310	649
107	2¼Cr-1Mo	Forg. & ftg.	A234	WP22	K21590	3	...	5A	(15)	-29	520	310	649
108	9Cr-1Mo	Forg. & ftg.	A182	F9	K90941	...	...	5B	(9)	-29	585	380	649
109	9Cr-1Mo-V	Forg. & ftg.	A182	F91	K90901	...	≤75	15E	...	-29	586	415	649
110	9Cr-1Mo-V	Forg. & ftg.	A234	WP91	K90901	...	≤75	15E	...	-29	586	415	649
111	5Cr-½Mo	Forg. & ftg.	A182	F5a	K42544	...	...	5B	(9)	-29	620	450	649
112	9Ni	Forg. & ftg.	A420	WPL8	K81340	...	...	11A	(47)	-196	690	515	93
113	C-½Mo	Castings	A352	LC1	J12522	...	...	3	(9) (58)	-59	450	240	371
114	C-½Mo	Castings	A217	WC1	J12524	...	...	3	(9) (58)	-29	450	240	593
115	2½Ni	Castings	A352	LC2	J22500	...	...	9A	(9)	-73	485	275	343
116	3½Ni	Castings	A352	LC3	J31550	...	...	9B	(9)	-101	485	275	343
117	1Ni-½Cr-½Mo	Castings	A217	WC4	J12082	...	...	4	(9)	-29	485	275	538
118	¾Ni-1Mo-¾Cr	Castings	A217	WC5	J22000	...	...	4	(9)	-29	485	275	593
119	1¼Cr-½Mo	Castings	A217	WC6	J12072	...	...	4	(9)	-29	485	275	649
120	2¼Cr-1Mo	Castings	A217	WC9	J21890	...	...	5A	(9)	-29	485	275	649
121	5Cr-½Mo	Castings	A217	C5	J42045	...	...	5B	(9)	-29	620	415	649
122	9Cr-1Mo	Castings	A217	C12	J82090	...	...	5B	(9)	-29	620	415	649

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated																											
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]																										
	Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325	350	375	400	425	450	475	500	525	550	575	600	625	650		
100	161	161	157	155	155	153	150	147	145	143	141	139	138	136	134	132	129	126	92.1	61.1	40.4	26.4	17.4	11.6	7.58		
101	161	161	157	155	155	153	150	147	145	143	141	139	138	136	134	132	129	126	92.1	61.1	40.4	26.4	17.4	11.6	7.58		
102	161	161	161	161	161	158	155	153	150	148	146	143	141	138	136	133	130	104	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
103	161	161	161	161	161	158	155	153	150	148	146	143	141	138	136	133	130	104	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
104	161	161	160	157	156	155	155	155	154	154	153	151	149	146	142	137	131	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
105	172	172	172	169	168	167	166	166	165	165	164	164	162	161	158	155	152	98.2	73.5	54.7	40.6	29.2	20.6	15.2	8.96		
106	172	172	171	168	167	167	167	167	167	167	167	167	167	167	167	167	167	119	88.4	64.0	44.6	30.0	19.7	12.8	8.27		
107	172	172	171	168	167	167	167	167	167	167	167	167	167	167	167	167	167	119	88.4	64.0	44.6	30.0	19.7	12.8	8.27		
108	195	195	194	191	189	188	188	188	187	187	186	184	181	177	172	166	159	151	83.2	60.2	42.9	29.9	20.6	14.4	10.3		
109	195	195	195	195	195	195	195	195	194	193	192	190	187	183	178	172	165	156	147	137	115	87.0	64.7	45.1	29.6		
110	195	195	195	195	195	195	195	195	194	193	192	190	187	183	178	172	165	156	147	137	115	87.0	64.7	45.1	29.6		
111	207	207	205	202	200	199	199	199	198	198	196	194	191	187	182	176	169	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
112	230	230	230	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...		
113	149	149	149	148	145	143	140	138	137	135	133	131	129	127	...	...	...	...	...	...	...	...	...	...	...		
114	149	149	149	148	145	143	140	138	137	135	133	131	129	126	123	120	117	109	68.0	42.3	30.5	23.2	16.5	...	...		
115	161	161	161	161	161	160	158	155	151	147	142	136	131	...	...	...	...	...	...	...	...	...	...	...	...		
116	161	161	161	161	161	160	158	155	151	147	142	136	131	...	...	...	...	...	...	...	...	...	...	...	...		
117	161	161	161	161	161	161	161	161	159	158	156	154	152	149	146	142	137	131	74.4	49.9	40.7	...	...	...	...		
118	161	161	161	161	161	161	161	161	159	158	156	154	152	149	146	142	137	131	74.4	49.9	34.3	23.2	16.5	...	...		
119	161	161	161	161	161	158	155	153	150	148	146	143	141	138	136	133	130	104	73.7	52.0	36.3	25.2	17.6	12.3	8.27		
120	161	161	160	157	156	156	156	156	156	156	156	156	156	156	156	156	156	119	88.4	64.0	44.6	30.0	19.7	12.8	8.27		
121	207	207	205	202	200	199	199	199	198	198	196	194	191	187	182	176	169	80.6	61.7	46.4	34.7	25.5	17.8	11.4	6.89		
122	207	207	205	202	200	199	199	199	198	198	196	194	191	187	182	176	169	160	83.2	60.2	42.9	29.9	20.6	14.4	10.3		

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
1	18Cr-10Ni-Ti	Smls. pipe	A312	TP321	S32100	...	>9.50	8
2	18Cr-10Ni-Ti	Smls. pipe	A376	TP321	S32100	...	>9.50	8
3	18Cr-8Ni	Tube	A213	TP304L	S30403	...	...	8
4	18Cr-8Ni	Tube	A269	TP304L	S30403	...	...	8
5	18Cr-8Ni	Tube	A270	TP304L	S30403	...	...	8
6	18Cr-8Ni	Smls. & wld. pipe	A312	TP304L	S30403	...	...	8
7	18Cr-8Ni	Wld. pipe	A358	304L	S30403	...	...	8
8	16Cr-12Ni-2Mo	Tube	A213	TP316L	S31603	...	...	8
9	16Cr-12Ni-2Mo	Tube	A269	TP316L	S31603	...	...	8
10	16Cr-12Ni-2Mo	Tube	A270	TP316L	S31603	...	...	8
11	16Cr-12Ni-2Mo	Smls. & wld. pipe	A312	TP316L	S31603	...	...	8
12	16Cr-12Ni-2Mo	Wld. pipe	A358	316L	S31603	...	...	8
13	16Cr-12Ni-2Mo-Ti	Tube	A213	TP316Ti	S31635	...	...	8
14	18Cr-10Ni-Ti	Smls. pipe	A312	TP321	S32100	...	>9.50	8
15	18Cr-10Ni-Ti	Smls. pipe	A376	TP321	S32100	...	>9.50	8
16	18Cr-10Ni-Ti	Smls. pipe	A312	TP321H	S32109	...	>9.50	8
17	18Cr-10Ni-Ti	Smls. pipe	A376	TP321H	S32109	...	>9.50	8
18	25Cr-12Ni	Pipe & tube	A451	CPH8	J93400	...	...	8
19	25Cr-20Ni	Pipe & tube	A451	CPK20	J94202	...	...	8
20	11Cr-Ti	Tube	A268	TP409	S40900	...	...	7
21	18Cr-Ti	Tube	A268	TP430Ti	S43036	...	...	7
22	16Cr-14Ni-2Mo	Pipe & tube	A451	CPF10MC	J92971	...	...	8
23	12Cr-Al	Tube	A268	TP405	S40500	...	...	7
24	13Cr	Tube	A268	TP410	S41000	...	...	6
25	17Cr	Tube	A268	TP430	S43000	...	...	7
26	18Cr-13Ni-3Mo	Smls. & wld. pipe	A312	TP317L	S31703	...	...	8
27	18Cr-13Ni-3Mo	Wld. pipe	A358	317L	S31703	...	...	8
28	25Cr-20Ni	Smls. & wld. pipe	A312	TP310S	S31008	...	...	8
29	25Cr-20Ni	Wld. pipe	A358	310S	S31008	...	...	8
30	25Cr-20Ni	Pipe	A409	TP310S	S31008	...	...	8
31	18Cr-10Ni-Ti	Smls. pipe	A312	TP321	S32100	...	≤9.50	8
32	18Cr-10Ni-Ti	Wld. pipe	A312	TP321	S32100	...	...	8
33	18Cr-10Ni-Ti	Wld. pipe	A358	321	S32100	...	...	8
34	18Cr-10Ni-Ti	Smls. pipe	A376	TP321	S32100	...	≤9.50	8
35	18Cr-10Ni-Ti	Wld. pipe	A409	TP321	S32100	...	...	8

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]						
						Min. Temp. to 40	65	100	125	150	175	200
1	(28)	-254	485	170	816	115	115	115	115	115	115	115
2	(28) (36)	-254	480	170	816	115	115	115	115	115	115	115
3	(14) (36)	-254	485	170	816	115	115	115	115	115	114	110
4	(14) (36)	-254	483	172	816	115	115	115	115	115	114	110
5	(14)	-254	483	172	816	115	115	115	115	115	114	110
6	...	-254	485	170	816	115	115	115	115	115	114	110
7	(36)	-254	485	170	816	115	115	115	115	115	114	110
8	(14) (36)	-254	485	170	816	115	115	115	115	115	113	109
9	(14) (36)	-254	483	172	816	115	115	115	115	115	113	109
10	(14)	-254	483	172	816	115	115	115	115	115	113	109
11	...	-254	485	170	816	115	115	115	115	115	113	109
12	(36)	-254	485	170	816	115	115	115	115	115	113	109
13	(30)	-198	515	205	816	138	138	138	138	138	138	134
14	(28) (30)	-254	485	170	816	115	115	115	115	115	115	115
15	(28) (30) (36)	-254	480	170	816	115	115	115	115	115	115	115
16	(30)	-198	480	170	816	115	115	115	115	115	115	115
17	(30) (36)	-198	480	170	816	115	115	115	115	115	115	115
18	(26) (28) (35)	-198	448	195	816	129	129	129	129	127	125	124
19	(12) (28) (35) (39)	-198	448	195	816	129	129	129	129	127	125	124
20	(35)	-29	380	170	40	115	...	...	...	...	...	...
21	(35) (49)	-29	415	240	40	138	...	...	...	...	...	...
22	(28)	-198	485	205	40	138	...	...	...	...	...	...
23	(35)	-29	415	205	538	138	138	138	137	135	134	133
24	(35)	-29	415	205	649	138	138	138	137	135	134	133
25	(35) (49)	-29	415	240	649	138	138	138	137	135	134	133
26	...	-198	515	205	454	138	138	138	138	138	136	131
27	(36)	-198	515	205	454	138	138	138	138	138	136	131
28	(28) (35)	-198	515	205	816	138	138	138	138	138	138	138
29	(28) (35) (36)	-198	515	205	816	138	138	138	138	138	138	138
30	(28) (31) (35) (36)	-198	515	205	816	138	138	138	138	138	138	138
31	(28)	-254	515	205	816	138	138	138	138	138	138	138
32	(28)	-254	515	205	816	138	138	138	138	138	138	138
33	(28) (36)	-254	515	205	816	138	138	138	138	138	138	138
34	(28) (36)	-254	515	205	816	138	138	138	138	138	138	138
35	(28) (36)	-254	515	205	816	138	138	138	138	138	138	138

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]												
	225	250	275	300	325	350	375	400	425	450	475	500	525
1	115	112	109	106	104	102	100	99.0	97.0	96.3	95.3	94.4	93.6
2	115	112	109	106	104	102	100	99.0	97.0	96.3	95.3	94.4	93.6
3	106	103	99.9	97.7	95.7	94.1	92.6	91.3	90.0	88.7	87.3	85.6	83.7
4	106	103	99.9	97.7	95.7	94.1	92.6	91.3	90.0	88.7	87.3	85.6	83.7
5	106	103	99.9	97.7	95.7	94.1	92.6	91.3	90.0	88.7	87.3	85.6	83.7
6	106	103	99.9	97.7	95.7	94.1	92.6	91.3	90.0	88.7	87.3	85.6	83.7
7	106	103	99.9	97.7	95.7	94.1	92.6	91.3	90.0	88.7	87.3	85.6	83.7
8	106	103	100	98.1	96.1	94.3	92.6	90.9	89.3	87.6	85.9	84.2	82.5
9	106	103	100	98.1	96.1	94.3	92.6	90.9	89.3	87.6	85.9	84.2	82.5
10	106	103	100	98.1	96.1	94.3	92.6	90.9	89.3	87.6	85.9	84.2	82.5
11	106	103	100	98.1	96.1	94.3	92.6	90.9	89.3	87.6	85.9	84.2	82.5
12	106	103	100	98.1	96.1	94.3	92.6	90.9	89.3	87.6	85.9	84.2	82.5
13	129	124	120	117	115	113	112	111	110	109	108	107	106
14	115	112	109	106	104	102	100	99.0	97.0	96.3	95.3	94.4	93.6
15	115	112	109	106	104	102	100	99.0	97.0	96.3	95.3	94.4	93.6
16	115	112	109	106	104	102	100	99.0	97.0	96.3	95.3	94.4	93.6
17	115	112	109	106	104	102	100	99.0	97.0	96.3	95.3	94.4	93.6
18	124	123	121	119	117	115	112	109	106	103	100	96.9	93.7
19	124	123	121	119	117	115	112	109	106	103	100	96.9	93.7
20	...	...	...	...	...	...	...	...	...	...	...	...	...
21	...	...	...	...	...	...	...	...	...	...	...	...	...
22	...	...	...	...	...	...	...	...	...	...	...	...	...
23	132	131	130	129	127	124	121	118	114	109	103	70.1	38.8
24	132	131	130	129	127	124	121	118	114	109	92.5	68.4	51.1
25	132	131	130	129	127	124	121	118	114	109	88.7	69.8	52.6
26	127	123	120	118	115	113	111	109	107	105	103	...	...
27	127	123	120	118	115	113	111	109	107	105	103	...	...
28	137	134	131	129	127	125	123	122	120	119	117	116	84.9
29	137	134	131	129	127	125	123	122	120	119	117	116	84.9
30	137	134	131	129	127	125	123	122	120	119	117	116	84.9
31	138	135	131	128	125	122	120	119	117	115	114	113	112
32	138	135	131	128	125	122	120	119	117	115	114	113	112
33	138	135	131	128	125	122	120	119	117	115	114	113	112
34	138	135	131	128	125	122	120	119	117	115	114	113	112
35	138	135	131	128	125	122	120	119	117	115	114	113	112

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to [Notes for Tables A-1 and A-1C](#); Specifications Are ASTM Unless Otherwise Indicated

Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]											
	550	575	600	625	650	675	700	725	750	775	800	825
1	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
2	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
3	81.4	40.4	33.2	26.7	21.9	18.2	15.0	12.4	8.87	7.20	6.58	6.21
4	81.4	40.4	33.2	26.7	21.9	18.2	15.0	12.4	8.87	7.20	6.58	6.21
5	81.4	40.4	33.2	26.7	21.9	18.2	15.0	12.4	8.87	7.20	6.58	6.21
6	81.4	40.4	33.2	26.7	21.9	18.2	15.0	12.4	8.87	7.20	6.58	6.21
7	81.4	40.4	33.2	26.7	21.9	18.2	15.0	12.4	8.87	7.20	6.58	6.21
8	80.8	79.3	77.9	58.0	43.6	33.0	25.3	18.8	14.0	10.4	7.99	6.89
9	80.8	79.3	77.9	58.0	43.6	33.0	25.3	18.8	14.0	10.4	7.99	6.89
10	80.8	79.3	77.9	58.0	43.6	33.0	25.3	18.8	14.0	10.4	7.99	6.89
11	80.8	79.3	77.9	58.0	43.6	33.0	25.3	18.8	14.0	10.4	7.99	6.89
12	80.8	73.0	67.9	58.0	43.6	33.0	25.3	18.8	14.0	10.4	7.99	6.89
13	106	101	80.3	65.5	50.4	38.6	29.6	23.0	17.7	13.4	10.4	8.05
14	92.7	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
15	92.7	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
16	92.7	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
17	92.7	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
18	68.1	53.5	42.1	33.2	25.9	20.3	16.5	13.2	10.1	7.35	5.89	5.52
19	73.2	64.4	56.5	49.0	41.0	33.5	25.4	18.3	12.8	9.01	6.59	5.52
20	...	...	...	...	...	...	...	...	...	...	...	...
21	...	...	...	...	...	...	...	...	...	...	...	...
22	...	...	...	...	...	...	...	...	...	...	...	...
23	27.6	...	...	...	...	...	...	...	...	...	...	...
24	37.4	26.3	17.8	11.4	6.89	...	...	...	...	...	...	...
25	38.1	27.6	20.6	15.9	12.4	...	...	...	...	...	...	...
26	...	...	...	...	...	...	...	...	...	...	...	...
27	...	...	...	...	...	...	...	...	...	...	...	...
28	59.0	43.5	31.9	23.6	16.9	10.7	6.10	3.90	2.99	2.36	1.73	1.38
29	59.0	43.5	31.9	23.6	16.9	10.7	6.10	3.90	2.99	2.36	1.73	1.38
30	59.0	43.5	31.9	23.6	16.9	10.7	6.10	3.90	2.99	2.36	1.73	1.38
31	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
32	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
33	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
34	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
35	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
36	23Cr-12Ni	Smls. & wld. pipe	A312	TP309	...	...	...	8
37	23Cr-12Ni	Wld. pipe	A358	309S	S30908	...	...	8
38	18Cr-8Ni	Pipe & tube	A451	CPF8	J92600	...	...	8
39	18Cr-10Ni-Cb	Smls. & wld. pipe	A312	TP347	S34700	...	...	8
40	18Cr-10Ni-Cb	Wld. pipe	A358	347	S34700	...	...	8
41	18Cr-10Ni-Cb	Pipe	A376	TP347	S34700	...	...	8
42	18Cr-10Ni-Cb	Pipe	A409	TP347	S34700	...	...	8
43	18Cr-10Ni-Cb	Smls. & wld. pipe	A312	TP348	S34800	...	...	8
44	18Cr-10Ni-Cb	Wld. pipe	A358	348	S34800	...	...	8
45	18Cr-10Ni-Cb	Pipe	A376	TP348	S34800	...	...	8
46	18Cr-10Ni-Cb	Pipe	A409	TP348	S34800	...	...	8
47	25Cr-12Ni	Pipe & tube	A451	CPH10	J93402	...	...	8
48	25Cr-12Ni	Pipe & tube	A451	CPH20	J93402	...	...	8
49	25Cr-20Ni	Smls. & wld. pipe	A312	TP310H	S31009	...	...	8
50	18Cr-10Ni-Cb	Pipe & tube	A451	CPF8C	J92710	...	...	8
51	18Cr-10Ni-Ti	Smls. pipe	A312	TP321	S32100	...	≤9.50	8
52	18Cr-10Ni-Ti	Wld. pipe	A312	TP321	S32100	...	...	8
53	18Cr-10Ni-Ti	Wld. pipe	A358	321	S32100	...	...	8
54	18Cr-10Ni-Ti	Smls. pipe	A376	TP321	S32100	...	≤9.50	8
55	18Cr-10Ni-Ti	Wld. pipe	A409	TP321	S32100	...	...	8
56	18Cr-10Ni-Ti	Smls. pipe	A312	TP321H	S32109	...	≤9.50	8
57	18Cr-10Ni-Ti	Wld. pipe	A312	TP321H	S32109	...	...	8
58	18Cr-10Ni-Ti	Wld. pipe	A358	321H	S32109	...	...	8
59	18Cr-10Ni-Ti	Smls. pipe	A376	TP321H	S32109	...	≤9.50	8
60	16Cr-12Ni-2Mo	Tube	A213	TP316	S31600	...	...	8
61	16Cr-12Ni-2Mo	Tube	A269	TP316	S31600	...	...	8
62	16Cr-12Ni-2Mo	Tube	A270	TP316	S31600	...	...	8
63	16Cr-12Ni-2Mo	Smls. & wld. pipe	A312	TP316	S31600	...	...	8
64	16Cr-12Ni-2Mo	Wld. pipe	A358	316	S31600	...	...	8
65	16Cr-12Ni-2Mo	Pipe	A376	TP316	S31600	...	...	8
66	16Cr-12Ni-2Mo	Pipe	A409	TP316	S31600	...	...	8
67	18Cr-13Ni-3Mo	Smls. & wld. pipe	A312	TP317	S31700	...	...	8

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]						
						Min. Temp. to 40	65	100	125	150	175	200
36	(28) (35) (39)	-198	517	207	816	138	138	138	138	138	138	138
37	(28) (31) (35) (36)	-198	515	205	816	138	138	138	138	138	138	138
38	(26) (28)	-254	485	205	816	138	138	138	138	138	134	129
39	...	-254	515	205	816	138	138	138	138	138	138	138
40	(30) (36)	-254	515	205	816	138	138	138	138	138	138	138
41	(30) (36)	-254	515	205	816	138	138	138	138	138	138	138
42	(30) (36)	-254	515	205	816	138	138	138	138	138	138	138
43	...	-198	515	205	816	138	138	138	138	138	138	138
44	(30) (36)	-198	515	205	816	138	138	138	138	138	138	138
45	(30) (36)	-198	515	205	816	138	138	138	138	138	138	138
46	(30) (36)	-198	515	205	816	138	138	138	138	138	138	138
47	(12) (14) (28) (35) (39)	-198	485	205	816	138	138	138	138	137	135	134
48	(12) (14) (28) (35) (39)	-198	485	205	816	138	138	138	138	137	135	134
49	(29) (35) (39)	-198	515	205	816	138	138	138	138	138	138	138
50	(28)	-198	485	205	816	138	138	138	138	138	134	129
51	(28) (30)	-254	515	205	816	138	138	138	138	138	138	138
52	(28) (30)	-254	515	205	816	138	138	138	138	138	138	138
53	(28) (30) (36)	-254	515	205	816	138	138	138	138	138	138	138
54	(28) (30) (36)	-254	515	205	816	138	138	138	138	138	138	138
55	(28) (30) (36)	-254	515	205	816	138	138	138	138	138	138	138
56	(30)	-198	515	205	816	138	138	138	138	138	138	138
57	(30)	-198	515	205	816	138	138	138	138	138	138	138
58	(30) (36)	-198	515	205	816	138	138	138	138	138	138	138
59	(30) (36)	-198	515	205	816	138	138	138	138	138	138	138
60	(14) (26) (28) (31) (36)	-254	515	205	816	138	138	138	138	138	138	134
61	(14) (26) (28) (31) (36)	-254	517	207	816	138	138	138	138	138	138	134
62	(14) (26) (28)	-254	517	207	816	138	138	138	138	138	138	134
63	(26) (28)	-254	515	205	816	138	138	138	138	138	138	134
64	(26) (28) (31) (36)	-254	515	205	816	138	138	138	138	138	138	134
65	(26) (28) (31) (36)	-254	515	205	816	138	138	138	138	138	138	134
66	(26) (28) (31) (36)	-254	515	205	816	138	138	138	138	138	138	134
67	(26) (28)	-198	515	205	816	138	138	138	138	138	138	134

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]												
	225	250	275	300	325	350	375	400	425	450	475	500	525
36	138	135	133	131	129	127	125	124	122	121	119	117	108
37	138	135	133	131	129	127	125	124	122	121	119	117	108
38	125	122	119	116	113	111	109	107	105	103	101	99.1	94.4
39	138	138	137	135	132	130	128	127	126	126	125	125	125
40	138	138	137	135	132	130	128	127	126	126	125	125	125
41	138	138	137	135	132	130	128	127	126	126	125	125	125
42	138	138	137	135	132	130	128	127	126	126	125	125	125
43	138	138	137	135	132	130	128	127	126	126	125	125	125
44	138	138	137	135	132	130	128	127	126	126	125	125	125
45	138	138	137	135	132	130	128	127	126	126	125	125	125
46	138	138	137	135	132	130	128	127	126	126	125	125	125
47	133	131	129	128	125	123	120	117	114	111	107	104	100
48	133	131	129	128	125	123	120	117	114	111	107	104	100
49	137	134	131	129	127	125	123	122	120	119	117	116	108
50	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
51	138	135	131	128	125	122	120	119	117	115	114	113	112
52	138	135	131	128	125	122	120	119	117	115	114	113	112
53	138	135	131	128	125	122	120	119	117	115	114	113	112
54	138	135	131	128	125	122	120	119	117	115	114	113	112
55	138	135	131	128	125	122	120	119	117	115	114	113	112
56	138	135	131	128	125	122	120	119	117	115	114	113	112
57	138	135	131	128	125	122	120	119	117	115	114	113	112
58	138	135	131	128	125	122	120	119	117	115	114	113	112
59	138	135	131	128	125	122	120	119	117	115	114	113	112
60	129	125	122	119	116	114	112	111	110	109	108	107	106
61	129	125	122	119	116	114	112	111	110	109	108	107	106
62	129	125	122	119	116	114	112	111	110	109	108	107	106
63	129	125	122	119	116	114	112	111	110	109	108	107	106
64	129	125	122	119	116	114	112	111	110	109	108	107	106
65	129	125	122	119	116	114	112	111	110	109	108	107	106
66	129	125	122	119	116	114	112	111	110	109	108	107	106
67	129	125	122	119	116	114	112	111	110	109	108	107	106

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to [Notes for Tables A-1 and A-1C](#); Specifications Are ASTM Unless Otherwise Indicated

Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1) , (2b) , (3)]											
	550	575	600	625	650	675	700	725	750	775	800	825
36	83.7	64.0	48.5	36.3	27.3	21.0	15.9	12.5	9.87	7.65	5.97	5.17
37	83.7	64.0	48.5	36.3	27.3	21.0	15.9	12.5	9.87	7.65	5.97	5.17
38	75.3	60.4	49.0	40.1	32.8	27.2	23.4	19.6	16.8	14.7	12.8	11.7
39	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
40	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
41	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
42	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
43	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
44	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
45	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
46	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
47	68.1	53.5	42.1	33.2	25.9	20.3	16.5	13.2	10.1	7.35	5.89	5.52
48	68.1	53.5	42.1	33.2	25.9	20.3	16.5	13.2	10.1	7.35	5.89	5.52
49	83.7	64.0	48.5	36.3	27.3	21.0	15.9	12.5	9.87	7.65	5.97	5.17
50	95.5	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
51	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
52	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
53	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
54	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
55	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
56	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
57	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
58	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
59	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
60	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
61	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
62	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
63	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
64	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
65	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
66	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
67	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
68	18Cr-13Ni-3Mo	Wld. pipe	A358	317	S31700	...	...	8
69	18Cr-13Ni-3Mo	Pipe	A409	TP317	S31700	...	...	8
70	16Cr-12Ni-2Mo	Pipe	A376	TP316H	S31609	...	...	8
71	16Cr-12Ni-2Mo	Smls. & wld. pipe	A312	TP316H	S31609	...	...	8
72	18Cr-10Ni-Cb	Pipe	A376	TP347H	S34709	...	...	8
73	18Cr-10Ni-Cb	Smls. & wld. pipe	A312	TP347	S34700	...	...	8
74	18Cr-10Ni-Cb	Wld. pipe	A358	347	S34700	...	...	8
75	18Cr-10Ni-Cb	Pipe	A376	TP347	S34700	...	...	8
76	18Cr-10Ni-Cb	Pipe	A409	TP347	S34700	...	...	8
77	18Cr-10Ni-Cb	Smls. & wld. pipe	A312	TP348	S34800	...	...	8
78	18Cr-10Ni-Cb	Wld. pipe	A358	348	S34800	...	...	8
79	18Cr-10Ni-Cb	Pipe	A376	TP348	S34800	...	...	8
80	18Cr-10Ni-Cb	Pipe	A409	TP348	S34800	...	...	8
81	18Cr-10Ni-Cb	Smls. & wld. pipe	A312	TP347H	S34709	...	...	8
82	18Cr-10Ni-Cb	Smls. & wld. pipe	A312	TP348H	S34809	...	...	8
83	18Cr-8Ni	Tube	A213	TP304	S30400	...	...	8
84	18Cr-8Ni	Tube	A269	TP304	S30400	...	...	8
85	18Cr-8Ni	Tube	A270	TP304	S30400	...	...	8
86	18Cr-8Ni	Smls. & wld. pipe	A312	TP304	S30400	...	...	8
87	18Cr-8Ni	Wld. pipe	A358	304	S30400	...	...	8
88	18Cr-8Ni	Pipe	A376	TP304	S30400	...	...	8
89	18Cr-8Ni	Pipe	A376	TP304H	S30409	...	...	8
90	18Cr-8Ni	Pipe	A409	TP304	S30400	...	...	8
91	18Cr-8Ni	Smls. & wld. pipe	A312	TP304H	S30409	...	...	8
92	18Cr-12Ni-2Mo	Pipe & tube	A451	CPF8M	J92900	...	...	8
93	44Fe-25Ni-21Cr-Mo	Wld. tube	A249	...	N08904	...	...	45
94	44Fe-25Ni-21Cr-Mo	Smls. & wld. pipe	A312	...	N08904	...	...	45
95	20Cr-Cu	Tube	A268	TP443	S44300	...	...	a
96	27Cr	Tube	A268	TP446-1	S44600	...	...	10I
97	12Cr	Wld. pipe	A1053	50	S41003	...	...	7
98	25Cr-8Ni-N	Pipe & tube	A451	CPE20N	J92802	...	...	8
99	23Cr-4Ni-Mo-Cu-N	Smls. & wld. tube	A789	...	S32304	...	...	10H
100	23Cr-4Ni-Mo-Cu-N	Smls. & wld. pipe	A790	2304	S32304	...	...	10H

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]						
						Min. Temp. to 40	65	100	125	150	175	200
68	(26) (28) (31) (36)	−198	515	205	816	138	138	138	138	138	138	134
69	(26) (28) (31) (36)	−198	515	205	816	138	138	138	138	138	138	134
70	(26) (31) (36)	−198	515	205	816	138	138	138	138	138	138	134
71	(26)	−198	515	205	816	138	138	138	138	138	138	134
72	(30) (36)	−198	515	205	816	138	138	138	138	138	138	138
73	(28)	−254	515	205	816	138	138	138	138	138	138	138
74	(28) (30) (36)	−254	515	205	816	138	138	138	138	138	138	138
75	(28) (30) (36)	−254	515	205	816	138	138	138	138	138	138	138
76	(28) (30) (36)	−254	515	205	816	138	138	138	138	138	138	138
77	(28)	−198	515	205	816	138	138	138	138	138	138	138
78	(28) (30) (36)	−198	515	205	816	138	138	138	138	138	138	138
79	(28) (30) (36)	−198	515	205	816	138	138	138	138	138	138	138
80	(28) (30) (36)	−198	515	205	816	138	138	138	138	138	138	138
81	...	−198	515	205	816	138	138	138	138	138	138	138
82	...	−198	515	205	816	138	138	138	138	138	138	138
83	(14) (26) (28) (31) (36)	−254	515	205	816	138	138	138	138	138	134	129
84	(14) (26) (28) (31) (36)	−254	517	207	816	138	138	138	138	138	134	129
85	(14) (26) (28)	−254	517	207	816	138	138	138	138	138	134	129
86	(26) (28)	−254	515	205	816	138	138	138	138	138	134	129
87	(26) (28) (31) (36)	−254	515	205	816	138	138	138	138	138	134	129
88	(20) (26) (28) (31) (36)	−254	515	205	816	138	138	138	138	138	134	129
89	(26) (31) (36)	−198	515	205	816	138	138	138	138	138	134	129
90	(26) (28) (31) (36)	−254	515	205	816	138	138	138	138	138	134	129
91	(26)	−198	515	205	816	138	138	138	138	138	134	129
92	(26) (28)	−254	485	205	816	138	138	138	138	130	124	118
93	(26)	−198	490	215	260	143	143	143	143	141	135	130
94	(26)	−198	490	215	260	143	143	143	143	141	135	130
95	(35)	−29	485	275	538	161	161	161	161	161	161	161
96	(35)	−29	485	275	538	161	161	161	158	155	153	152
97	...	−29	485	350	316	162	162	162	162	162	161	159
98	(35) (39)	−198	550	275	482	184	184	184	184	184	184	184
99	(25)	−51	600	400	316	200	200	191	185	180	175	171
100	(25)	−51	600	400	316	200	200	191	185	180	175	171

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, <i>S</i> , MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]												
	225	250	275	300	325	350	375	400	425	450	475	500	525
68	129	125	122	119	116	114	112	111	110	109	108	107	106
69	129	125	122	119	116	114	112	111	110	109	108	107	106
70	129	125	122	119	116	114	112	111	110	109	108	107	106
71	129	125	122	119	116	114	112	111	110	109	108	107	106
72	138	138	137	135	132	130	129	127	126	126	125	125	125
73	138	138	137	135	132	130	129	127	126	126	125	125	125
74	138	138	137	135	132	130	128	127	126	126	125	125	125
75	138	138	137	135	132	130	128	127	126	126	125	125	125
76	138	138	137	135	132	130	128	127	126	126	125	125	125
77	138	138	137	135	132	130	128	127	126	126	125	125	125
78	138	138	137	135	132	130	128	127	126	126	125	125	125
79	138	138	137	135	132	130	128	127	126	126	125	125	125
80	138	138	137	135	132	130	128	127	126	126	125	125	125
81	138	138	137	135	132	130	129	127	126	126	125	125	125
82	138	138	137	135	132	130	129	127	126	126	125	125	125
83	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
84	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
85	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
86	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
87	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
88	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
89	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
90	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
91	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
92	114	110	107	104	102	101	99.3	98.1	97.0	95.9	94.7	93.3	91.5
93	125	120	116	...	...	...	...	...	...	...	...	...	...
94	125	120	116	...	...	...	...	...	...	...	...	...	...
95	161	161	161	161	161	97.0	84.3	73.3	64.0	55.8	43.9	31.7	21.4
96	150	149	147	145	144	141	139	136	132	128	122	116	109
97	156	154	152	149	146	...	...	...	...	...	...	...	...
98	184	184	184	184	184	184	184	184	184	184	184	184	...
99	166	161	153	143	111	...	...	...	...	...	...	...	...
100	166	161	153	143	111	...	...	...	...	...	...	...	...

Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to [Notes for Tables A-1 and A-1C](#); Specifications Are ASTM Unless Otherwise Indicated

Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]											
	550	575	600	625	650	675	700	725	750	775	800	825
68	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
69	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
70	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
71	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
72	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
73	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
74	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
75	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
76	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
77	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
78	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
79	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
80	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
81	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
82	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
83	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
84	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
85	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
86	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
87	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
88	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
89	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
90	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
91	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
92	89.1	72.7	57.7	46.0	36.9	30.1	24.3	20.3	17.0	14.3	12.1	11.0
93	...	...	...	...	...	...	...	...	...	...	...	...
94	...	...	...	...	...	...	...	...	...	...	...	...
95	17.2	...	...	...	...	...	...	...	...	...	...	...
96	104	...	...	...	...	...	...	...	...	...	...	...
97	...	...	...	...	...	...	...	...	...	...	...	...
98	...	...	...	...	...	...	...	...	...	...	...	...
99	...	...	...	...	...	...	...	...	...	...	...	...
100	...	...	...	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
101	23Cr-4Ni-Mo-Cu-N	Pipe & tube	A928	2304	S32304	...	...	10H
102	20Cr-18Ni-6Mo	Pipe & tube	A813	...	S31254	...	...	8
103	20Cr-18Ni-6Mo	Pipe & tube	A814	...	S31254	...	...	8
104	13Cr	Pipe & tube	A426	CPCA15	J91150	...	...	6
105	20Cr-18Ni-6Mo	Wld. pipe	A358	...	S31254	...	>5.0	8
106	20Cr-18Ni-6Mo	Wld. pipe	A358	...	S31254	...	≤5.0	8
107	22Cr-5Ni-3Mo-N	Smls. & wld. tube	A789	...	S31803	...	...	10H
108	22Cr-5Ni-3Mo-N	Smls. & wld. pipe	A790	...	S31803	...	...	10H
109	22Cr-5Ni-3Mo-N	Pipe & tube	A928	...	S31803	...	...	10H
110	16Cr-4Ni-6Mn	Pipe	A312	TP201LN	S20153	...	...	8
111	16Cr-4Ni-6Mn	Wld. pipe	A358	201LN	S20153	...	...	8
112	20Cr-18Ni-6Mo	Pipe & tube	A249	...	S31254	...	>5.00	8
113	20Cr-18Ni-6Mo	Pipe & tube	A249	...	S31254	...	≤5.00	8
114	20Cr-18Ni-6Mo	Pipe & tube	A312	...	S31254	...	>5.00	8
115	20Cr-18Ni-6Mo	Pipe & tube	A312	...	S31254	...	≤5.00	8
116	26Cr-4Ni-Mo	Smls. & wld. pipe	A790	329	S32900	...	...	10H
117	46Fe-24Ni-21Cr-6Mo-Cu-N	Smls. & wld. pipe	A312	...	N08367	...	>5.0	45
118	46Fe-24Ni-21Cr-6Mo-Cu-N	Wld. pipe	A358	...	N08367	...	>5.0	45
119	46Fe-24Ni-21Cr-6Mo-Cu-N	Wld. pipe	A813	...	N08367	...	>5.0	45
120	46Fe-24Ni-21Cr-6Mo-Cu-N	Wld. pipe	A814	...	N08367	...	>5.0	45
121	46Fe-24Ni-21Cr-6Mo-Cu-N	Smls. & wld. pipe	A312	...	N08367	...	≤5.0	45
122	46Fe-24Ni-21Cr-6Mo-Cu-N	Wld. pipe	A358	...	N08367	...	≤5.0	45
123	46Fe-24Ni-21Cr-6Mo-Cu-N	Wld. pipe	A813	...	N08367	...	≤5.0	45
124	46Fe-24Ni-21Cr-6Mo-Cu-N	Wld. pipe	A814	...	N08367	...	≤5.0	45
125	21Cr-5Mn-1½Ni-Cu-N	Smls. & wld. tube	A789	...	S32101	...	>5.00	10H
126	21Cr-5Mn-1½Ni-Cu-N	Smls. & wld. pipe	A790	...	S32101	...	>5.00	10H
127	22Cr-5Ni-3Mo-N	Smls. & wld. pipe	A790	2205	S32205	...	...	10H
128	22Cr-5Ni-3Mo-N	Smls. & wld. pipe	A789	...	S32205	...	...	10H
129	24Cr-4Ni-3Mn-1.5Mo-N	Smls. & wld. tube	A789	...	S82441	...	≥10	10H
130	24Cr-4Ni-3Mn-1.5Mo-N	Smls. & wld. pipe	A790	...	S82441	...	≥10	10H

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]						
						Min. Temp. to 40	65	100	125	150	175	200
101	(25)	-51	600	400	316	200	200	191	185	180	175	171
102	...	-198	650	300	454	202	202	202	202	199	192	185
103	...	-198	650	300	454	202	202	202	202	199	192	185
104	(35)	-29	620	450	40	207	...	...	...	...	...	...
105	...	-198	655	310	475	207	207	207	207	203	196	189
106	...	-198	690	310	475	207	207	207	207	203	196	189
107	(25)	-51	620	450	316	207	207	207	204	199	196	193
108	(25)	-51	620	450	316	207	207	207	204	199	196	193
109	(25)	-51	620	450	316	207	207	207	204	199	196	193
110	...	-198	655	310	454	207	206	187	177	170	165	162
111	(36)	-198	655	310	454	207	206	187	177	170	165	162
112	...	-198	655	310	454	207	207	207	207	203	196	190
113	...	-198	675	310	454	207	207	207	207	203	196	190
114	...	-198	655	310	454	207	207	207	207	203	196	190
115	...	-198	675	310	454	207	207	207	207	203	196	190
116	(25)	-29	620	485	40	207	...	...	...	...	...	...
117	(26)	-198	655	310	427	207	207	207	207	206	202	198
118	(26)	-198	655	310	427	207	207	207	207	206	202	198
119	(26)	-198	655	310	427	207	207	207	207	206	202	198
120	(26)	-198	655	310	427	207	207	207	207	206	202	198
121	(26)	-198	690	310	427	207	207	207	207	207	207	205
122	(26)	-198	690	310	427	207	207	207	207	207	207	205
123	(26)	-198	690	310	427	207	207	207	207	207	207	205
124	(26)	-198	690	310	427	207	207	207	207	207	207	205
125	(25)	-29	650	450	316	217	217	215	211	206	203	199
126	(25)	-29	650	450	316	217	217	215	211	206	203	199
127	(25)	-51	655	450	316	218	218	218	215	210	206	203
128	(25)	-51	655	485	316	218	218	218	215	210	206	203
129	(25)	-51	680	480	316	227	227	227	227	227	227	227
130	(25)	-51	680	480	316	227	227	227	227	227	227	227

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, <i>S</i> , MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]												
	225	250	275	300	325	350	375	400	425	450	475	500	525
101	166	161	153	143	111	...	...	...	...	...	...	...	...
102	180	175	171	168	165	164	162	161	160	159	158	...	...
103	180	175	171	168	165	164	162	161	160	159	158	...	...
104	...	...	...	...	...	...	...	...	...	...	...	...	...
105	184	179	175	172	169	167	165	165	164	163	161	...	...
106	184	179	175	172	169	167	165	165	164	163	161	...	...
107	190	188	187	186	185	...	...	...	...	...	...	...	...
108	190	188	187	186	185	...	...	...	...	...	...	...	...
109	190	188	187	186	185	...	...	...	...	...	...	...	...
110	160	159	158	158	158	157	156	154	152	149	146	...	...
111	160	159	158	158	158	157	156	154	152	149	146	...	...
112	184	179	175	172	169	167	166	165	164	163	161	...	...
113	184	179	175	172	169	167	166	165	164	163	161	...	...
114	184	179	175	172	169	167	166	165	164	163	161	...	...
115	184	179	175	172	169	167	166	165	164	163	161	...	...
116	...	...	...	...	...	...	...	...	...	...	...	...	...
117	195	192	188	184	179	176	173	170	167	166	...	...	...
118	195	192	188	184	179	176	173	170	167	166	...	...	...
119	195	192	188	184	179	176	173	170	167	166	...	...	...
120	195	192	188	184	179	176	173	170	167	166	...	...	...
121	199	194	188	184	179	176	173	170	167	166	...	...	...
122	199	194	188	184	179	176	173	170	167	166	...	...	...
123	199	194	188	184	179	176	173	170	167	166	...	...	...
124	199	194	188	184	179	176	173	170	167	166	...	...	...
125	199	199	199	199	199	...	...	...	...	...	...	...	...
126	199	199	199	199	199	...	...	...	...	...	...	...	...
127	201	199	197	196	196	...	...	...	...	...	...	...	...
128	201	199	197	196	196	...	...	...	...	...	...	...	...
129	227	227	227	227	227	...	...	...	...	...	...	...	...
130	227	227	227	227	227	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to [Notes for Tables A-1 and A-1C](#); Specifications Are ASTM Unless Otherwise Indicated

Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]											
	550	575	600	625	650	675	700	725	750	775	800	825
101	...	...	...	...	...	...	...	...	...	...	...	...
102	...	...	...	...	...	...	...	...	...	...	...	...
103	...	...	...	...	...	...	...	...	...	...	...	...
104	...	...	...	...	...	...	...	...	...	...	...	...
105	...	...	...	...	...	...	...	...	...	...	...	...
106	...	...	...	...	...	...	...	...	...	...	...	...
107	...	...	...	...	...	...	...	...	...	...	...	...
108	...	...	...	...	...	...	...	...	...	...	...	...
109	...	...	...	...	...	...	...	...	...	...	...	...
110	...	...	...	...	...	...	...	...	...	...	...	...
111	...	...	...	...	...	...	...	...	...	...	...	...
112	...	...	...	...	...	...	...	...	...	...	...	...
113	...	...	...	...	...	...	...	...	...	...	...	...
114	...	...	...	...	...	...	...	...	...	...	...	...
115	...	...	...	...	...	...	...	...	...	...	...	...
116	...	...	...	...	...	...	...	...	...	...	...	...
117	...	...	...	...	...	...	...	...	...	...	...	...
118	...	...	...	...	...	...	...	...	...	...	...	...
119	...	...	...	...	...	...	...	...	...	...	...	...
120	...	...	...	...	...	...	...	...	...	...	...	...
121	...	...	...	...	...	...	...	...	...	...	...	...
122	...	...	...	...	...	...	...	...	...	...	...	...
123	...	...	...	...	...	...	...	...	...	...	...	...
124	...	...	...	...	...	...	...	...	...	...	...	...
125	...	...	...	...	...	...	...	...	...	...	...	...
126	...	...	...	...	...	...	...	...	...	...	...	...
127	...	...	...	...	...	...	...	...	...	...	...	...
128	...	...	...	...	...	...	...	...	...	...	...	...
129	...	...	...	...	...	...	...	...	...	...	...	...
130	...	...	...	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
131	21Cr-5Mn-1 $\frac{1}{2}$ Ni-Cu-N	Smls. & wld. tube	A789	...	S32101	...	≤5.00	10H
132	21Cr-5Mn-1 $\frac{1}{2}$ Ni-Cu-N	Smls. & wld. pipe	A790	...	S32101	...	≤5.00	10H
133	21Cr-3 $\frac{1}{2}$ Ni-1 $\frac{3}{4}$ Mo-N	Smls. & wld. tube	A789	...	S32003	...	>5.00	10H
134	21Cr-3 $\frac{1}{2}$ Ni-1 $\frac{3}{4}$ Mo-N	Smls. & wld. pipe	A790	...	S32003	...	>5.00	10H
135	21Cr-3 $\frac{1}{2}$ Ni-1 $\frac{3}{4}$ Mo-N	Pipe & tube	A928	...	S32003	...	>5.00	10H
136	22Cr-5Ni-3Mo-N	Pipe & tube	A928	2205	S32205	...	...	10H
137	21Cr-3 $\frac{1}{2}$ Ni-1 $\frac{3}{4}$ Mo-N	Smls. & wld. tube	A789	...	S32003	...	≤5.00	10H
138	21Cr-3 $\frac{1}{2}$ Ni-1 $\frac{3}{4}$ Mo-N	Smls. & wld. pipe	A790	...	S32003	...	≤5.00	10H
139	21Cr-3 $\frac{1}{2}$ Ni-1 $\frac{3}{4}$ Mo-N	Pipe & tube	A928	...	S32003	...	≤5.00	10H
140	24Cr-4Ni-3Mn-1.5Mo-N	Smls. & wld. tube	A789	...	S82441	...	<10	10H
141	24Cr-4Ni-3Mn-1.5Mo-N	Smls. & wld. pipe	A790	...	S82441	...	<10	10H
142	25Cr-8Ni-3Mo-W-Cu-N	Smls. & wld. tube	A789	...	S32760	...	...	10H
143	25Cr-8Ni-3Mo-W-Cu-N	Smls. & wld. pipe	A790	...	S32760	...	...	10H
144	29Cr-6.5Ni-2Mo-N	Smls. & wld. tube	A789	...	S32906	...	≥10	10H
145	29Cr-6.5Ni-2Mo-N	Smls. & wld. pipe	A790	...	S32906	...	≥10	10H
146	24Cr-17Ni-6Mn-4 $\frac{1}{2}$ Mo-N	Pipe & tube	A358	...	S34565	...	...	8
147	25Cr-7Ni-4Mo-N	Smls. & wld. tube	A789	...	S32750	...	...	10H
148	25Cr-7Ni-4Mo-N	Smls. & wld. pipe	A790	2507	S32750	...	...	10H
149	25Cr-7Ni-4Mo-N	Pipe & tube	A928	2507	S32750	...	...	10H
150	29Cr-6.5Ni-2Mo-N	Smls. & wld. tube	A789	...	S32906	...	<10	10H
151	29Cr-6.5Ni-2Mo-N	Smls. & wld. pipe	A790	...	S32906	...	<10	10H
152	18Cr-11Ni	Plate & sheet	A240	305	S30500	...	...	8
153	12Cr-Al	Plate & sheet	A240	405	S40500	...	...	7
154	18Cr-8Ni	Plate & sheet	A240	304L	S30403	...	...	8
155	16Cr-12Ni-2Mo	Plate & sheet	A240	316L	S31603	...	...	8
156	18Cr-8Ni	Plate & sheet	A240	302	S30200	...	...	8
157	12Cr-1Ni	Plate, sheet, strip	A1010	40	S41003	...	...	7
158	12Cr-1Ni	Plate, sheet, strip	A1010	50	S41003	...	...	7
159	13Cr	Plate & sheet	A240	410S	S41008	...	...	7
160	13Cr	Plate & sheet	A240	410	S41000	...	...	6
161	15Cr	Plate & sheet	A240	429	S42900	...	...	6
162	17Cr	Plate & sheet	A240	430	S43000	...	...	7

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]						
						Min. Temp. to 40	65	100	125	150	175	200
131	(25)	-29	700	530	316	233	233	231	227	222	219	215
132	(25)	-29	700	530	316	233	233	231	227	222	219	215
133	(25)	-51	655	450	343	218	218	210	203	199	197	197
134	(25)	-51	655	450	343	218	218	210	203	199	197	197
135	(25)	-51	655	450	343	218	218	210	203	199	197	197
136	(25)	-51	655	450	316	218	218	218	215	210	206	203
137	(25)	-51	690	485	343	230	230	221	214	209	207	207
138	(25)	-51	690	485	343	230	230	221	214	209	207	207
139	(25)	-51	690	485	343	230	230	221	214	209	207	207
140	(25)	-51	740	540	316	247	247	247	247	247	247	247
141	(25)	-51	740	540	316	247	247	247	247	247	247	247
142	(25)	-51	750	550	316	250	250	248	243	238	236	234
143	(25)	-51	750	550	316	250	250	248	243	238	236	234
144	(25)	-51	750	550	316	251	251	249	243	238	235	231
145	(25)	-51	750	550	316	251	251	249	243	238	235	231
146	(36)	-198	795	415	427	264	264	260	253	247	242	238
147	(25)	-51	800	550	316	267	265	264	257	251	247	243
148	(25)	-51	800	550	316	267	265	264	257	251	247	243
149	(25)	-51	795	550	316	267	265	264	257	251	247	243
150	(25)	-51	800	650	316	267	267	265	259	253	250	246
151	(25)	-51	800	650	316	267	267	265	259	253	250	246
152	(26) (36) (39)	-198	485	170	40	115	...	...	...	...	...	...
153	(35)	-29	415	170	538	115	109	105	103	102	101	100
154	(36)	-254	485	170	816	115	115	115	115	115	114	110
155	(36)	-254	485	170	816	115	115	115	115	115	113	109
156	(26) (36)	-198	515	205	538	138	138	138	138	138	134	129
157	...	-29	455	275	316	152	152	152	152	152	151	149
158	...	-29	485	345	316	162	162	162	162	162	161	159
159	(35) (50)	-29	415	205	649	138	130	126	124	122	121	120
160	(35)	-29	450	205	649	138	130	126	124	122	121	120
161	(35)	-29	450	205	649	138	130	126	124	122	121	120
162	(35)	-29	450	205	649	138	130	126	124	122	121	120

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]												
	225	250	275	300	325	350	375	400	425	450	475	500	525
131	214	214	214	214	214	...	...	...	...	...	...	...	...
132	214	214	214	214	214	...	...	...	...	...	...	...	...
133	197	197	197	197	197	197	...	...	...	...	...	...	...
134	197	197	197	197	197	197	...	...	...	...	...	...	...
135	197	197	197	197	197	197	...	...	...	...	...	...	...
136	201	199	197	196	196	...	...	...	...	...	...	...	...
137	207	207	207	207	207	207	...	...	...	...	...	...	...
138	207	207	207	207	207	207	...	...	...	...	...	...	...
139	207	207	207	207	207	207	...	...	...	...	...	...	...
140	247	247	247	247	247	...	...	...	...	...	...	...	...
141	247	247	247	247	247	...	...	...	...	...	...	...	...
142	233	233	233	233	233	...	...	...	...	...	...	...	...
143	233	233	233	233	233	...	...	...	...	...	...	...	...
144	230	228	228	228	228	...	...	...	...	...	...	...	...
145	230	228	228	228	228	...	...	...	...	...	...	...	...
146	235	234	232	231	229	228	226	223	221	218	...	...	...
147	241	238	237	237	236	...	...	...	...	...	...	...	...
148	241	238	237	237	236	...	...	...	...	...	...	...	...
149	241	238	237	237	236	...	...	...	...	...	...	...	...
150	245	243	243	242	242	...	...	...	...	...	...	...	...
151	245	243	243	242	242	...	...	...	...	...	...	...	...
152	...	...	...	...	...	...	...	...	...	...	...	...	...
153	99.7	99.1	98.4	97.5	96.2	94.7	92.6	90.1	87.0	83.4	79.2	70.1	38.8
154	106	103	99.9	97.7	95.7	94.1	92.6	91.3	90.0	88.7	87.3	85.6	83.7
155	106	103	100	98.1	96.1	94.3	92.6	90.9	89.3	87.6	85.9	84.2	82.5
156	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
157	147	145	142	140	137	...	...	...	...	...	...	...	...
158	156	154	152	149	146	...	...	...	...	...	...	...	...
159	120	119	118	117	115	114	111	108	104	100	92.5	68.4	51.1
160	120	119	118	117	115	114	111	108	104	100	92.5	68.4	51.1
161	120	119	118	117	115	114	111	108	104	100	88.7	69.8	52.6
162	120	119	118	117	115	114	111	108	104	100	88.7	69.8	52.6

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to [Notes for Tables A-1 and A-1C](#); Specifications Are ASTM Unless Otherwise Indicated

Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]											
	550	575	600	625	650	675	700	725	750	775	800	825
131	...	...	...	...	...	...	...	...	...	...	...	...
132	...	...	...	...	...	...	...	...	...	...	...	...
133	...	...	...	...	...	...	...	...	...	...	...	...
134	...	...	...	...	...	...	...	...	...	...	...	...
135	...	...	...	...	...	...	...	...	...	...	...	...
136	...	...	...	...	...	...	...	...	...	...	...	...
137	...	...	...	...	...	...	...	...	...	...	...	...
138	...	...	...	...	...	...	...	...	...	...	...	...
139	...	...	...	...	...	...	...	...	...	...	...	...
140	...	...	...	...	...	...	...	...	...	...	...	...
141	...	...	...	...	...	...	...	...	...	...	...	...
142	...	...	...	...	...	...	...	...	...	...	...	...
143	...	...	...	...	...	...	...	...	...	...	...	...
144	...	...	...	...	...	...	...	...	...	...	...	...
145	...	...	...	...	...	...	...	...	...	...	...	...
146	...	...	...	...	...	...	...	...	...	...	...	...
147	...	...	...	...	...	...	...	...	...	...	...	...
148	...	...	...	...	...	...	...	...	...	...	...	...
149	...	...	...	...	...	...	...	...	...	...	...	...
150	...	...	...	...	...	...	...	...	...	...	...	...
151	...	...	...	...	...	...	...	...	...	...	...	...
152	...	...	...	...	...	...	...	...	...	...	...	...
153	27.6	...	...	...	...	...	...	...	...	...	...	...
154	81.4	40.4	33.2	26.7	21.9	18.2	15.0	12.4	8.87	7.20	6.58	6.21
155	80.8	73.0	67.9	58.0	43.6	33.0	25.3	18.8	14.0	10.4	7.99	6.89
156	96.4	...	...	...	...	...	...	...	...	...	...	...
157	...	...	...	...	...	...	...	...	...	...	...	...
158	...	...	...	...	...	...	...	...	...	...	...	...
159	37.4	26.3	17.8	11.4	6.89	...	...	...	...	...	...	...
160	37.4	26.3	17.8	11.4	6.89	...	...	...	...	...	...	...
161	38.1	27.6	20.6	15.9	12.4	...	...	...	...	...	...	...
162	38.1	27.6	20.6	15.9	12.4	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
163	18Cr-13Ni-3Mo	Plate & sheet	A240	317L	S31703	...	...	8
164	25Cr-20Ni	Plate & sheet	A240	310S	S31008	...	...	8
165	18Cr-10Ni-Ti	Plate, sheet, strip	A240	321	S32100	...	...	8
166	23Cr-12Ni	Plate & sheet	A240	309S	S30908	...	...	8
167	18Cr-10Ni-Cb	Plate & sheet	A240	347	S34700	...	...	8
168	18Cr-10Ni-Cb	Plate & sheet	A240	348	S34800	...	...	8
169	25Cr-20Ni	Plate & sheet	A240	310H	S31009	...	...	8
170	18Cr-10Ni-Ti	Plate, sheet, strip	A240	321	S32100	...	...	8
171	18Cr-10Ni-Ti	Plate, sheet, strip	A240	321H	S32109	...	...	8
172	16Cr-12Ni-2Mo	Plate & sheet	A240	316	S31600	...	...	8
173	18Cr-13Ni-3Mo	Plate & sheet	A240	317	S31700	...	...	8
174	18Cr-10Ni-Cb	Plate & sheet	A240	347	S34700	...	...	8
175	18Cr-10Ni-Cb	Plate & sheet	A240	348	S34800	...	...	8
176	18Cr-8Ni	Plate & sheet	A240	304	S30400	...	...	8
177	44Fe-25Ni-21Cr-Mo	Plate & sheet	A240	904L	N08904	...	...	45
178	23Cr-4Ni-Mo-Cu-N	Plate & sheet	A240	2304	S32304	...	...	10H
179	22Cr-5Ni-3Mo-N	Plate & sheet	A240	...	S31803	...	...	10H
180	16Cr-4Ni-6Mn	Plate & sheet	A240	201LN	S20153	...	...	8
181	20Cr-18Ni-6Mo	Plate	A240	...	S31254	...	>5.0	8
182	20Cr-18Ni-6Mo	Sheet	A240	...	S31254	...	≤5.0	8
183	46Fe-24Ni-21Cr-6Mo-Cu-N	Plate	A240	...	N08367	...	>5.0	45
184	46Fe-24Ni-21Cr-6Mo-Cu-N	Sheet & strip	A240	...	N08367	...	≤5.0	45
185	21Cr-5Mn-1.5Ni-Cu-N	Plate & sheet	A240	...	S32101	...	>5.0	10H
186	22Cr-5Ni-3Mo-N	Plate & sheet	A240	2205	S32205	...	...	10H
187	24Cr-4Ni-3Mn-1.5Mo-N	Plate & sheet	A240	...	S82441	...	≥10.0	10H
188	21Cr-5Mn-1.5Ni-Cu-N	Plate & sheet	A240	...	S32101	...	≤5.0	10H
189	21Cr-3½Ni-1¾Mo-N	Plate & sheet	A240	...	S32003	...	>5.00	10H
190	21Cr-3½Ni-1¾Mo-N	Plate & sheet	A240	...	S32003	...	≤5.00	10H
191	24Cr-4Ni-3Mn-1.5Mo-N	Plate & sheet	A240	...	S82441	...	<10.0	10H

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]						
						Min. Temp. to 40	65	100	125	150	175	200
163	(36)	−198	515	205	454	138	138	138	138	138	136	131
164	(28) (31) (35) (36)	−198	515	205	816	138	138	138	138	138	138	138
165	(28) (31) (36)	−198	515	205	816	138	138	138	138	138	138	138
166	(28) (35) (36)	−198	515	205	816	138	138	138	138	138	138	138
167	(36)	−254	515	205	816	138	138	138	138	138	138	138
168	(36)	−198	515	205	816	138	138	138	138	138	138	138
169	(29) (35) (39)	−198	515	205	816	138	138	138	138	138	138	138
170	(28) (30) (36)	−198	515	205	816	138	138	138	138	138	138	138
171	(30) (36)	−198	515	205	816	138	138	138	138	138	138	138
172	(26) (28) (36)	−254	515	205	816	138	138	138	138	138	138	134
173	(26) (28) (36)	−198	515	205	816	138	138	138	138	138	138	134
174	(28) (36)	−254	515	205	816	138	138	138	138	138	138	138
175	(28) (36)	−198	515	205	816	138	138	138	138	138	138	138
176	(26) (28) (36)	−254	515	205	816	138	138	138	138	138	134	129
177	(26)	−198	490	220	260	143	143	143	143	141	135	130
178	(25)	−51	600	400	316	200	200	191	185	180	175	171
179	(25)	−51	620	450	316	207	207	207	204	199	196	193
180	...	−198	655	310	454	207	206	187	177	170	165	162
181	...	−254	655	310	454	207	207	207	207	203	196	190
182	...	−254	690	310	454	207	207	207	207	203	196	190
183	(26)	−198	655	310	427	207	207	207	207	206	202	198
184	(26)	−198	690	310	427	207	207	207	207	207	207	205
185	(25)	−29	650	450	316	217	217	217	211	206	203	199
186	(25)	−51	655	450	316	218	218	218	215	210	206	203
187	(25)	−51	680	480	316	227	227	227	227	227	227	227
188	(25)	−29	700	530	316	233	233	231	227	222	219	215
189	(25)	−51	655	450	343	218	218	210	203	199	197	197
190	(25)	−51	690	485	343	230	230	221	214	209	207	207
191	(25)	−51	740	540	316	247	247	247	247	247	247	247

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, <i>S</i> , MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]												
	225	250	275	300	325	350	375	400	425	450	475	500	525
163	127	123	120	118	115	113	111	109	107	105	103	...	...
164	137	134	131	129	127	125	123	122	120	119	117	116	84.9
165	138	135	131	128	125	122	120	119	117	115	114	113	112
166	138	135	133	131	129	127	125	124	122	121	119	117	108
167	138	138	137	135	132	130	128	127	126	126	125	125	125
168	138	138	137	135	132	130	128	127	126	126	125	125	125
169	137	134	131	129	127	125	123	122	120	119	117	116	108
170	138	135	131	128	125	122	120	119	117	115	114	113	112
171	138	135	131	128	125	122	120	119	117	115	114	113	112
172	129	125	122	119	116	114	112	111	110	109	108	107	106
173	129	125	122	119	116	114	112	111	110	109	108	107	106
174	138	138	137	135	132	130	128	127	126	126	125	125	125
175	138	138	137	135	132	130	128	127	126	126	125	125	125
176	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
177	125	120	116	...	...	...	...	...	...	...	...	...	...
178	166	161	153	143	111	...	...	...	...	...	...	...	...
179	190	188	187	186	185	...	...	...	...	...	...	...	...
180	160	159	158	158	158	157	156	154	152	149	146	...	...
181	184	179	175	172	169	167	166	165	164	163	161	...	...
182	184	179	175	172	169	167	166	165	164	163	161	...	...
183	195	192	188	184	179	176	173	170	167	166	...	...	...
184	199	194	188	184	179	176	173	170	167	166	...	...	...
185	199	199	199	199	199	...	...	...	...	...	...	...	...
186	201	199	197	196	196	...	...	...	...	...	...	...	...
187	227	227	227	227	227	...	...	...	...	...	...	...	...
188	214	214	214	214	214	...	...	...	...	...	...	...	...
189	197	197	197	197	197	197	...	...	...	...	...	...	...
190	207	207	207	207	207	207	...	...	...	...	...	...	...
191	247	247	247	247	247	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to [Notes for Tables A-1 and A-1C](#); Specifications Are ASTM Unless Otherwise Indicated

Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]											
	550	575	600	625	650	675	700	725	750	775	800	825
163	...	...	...	...	...	...	...	...	...	...	...	...
164	59.0	43.5	31.9	23.6	16.9	10.7	6.10	3.90	2.99	2.36	1.73	1.38
165	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
166	83.7	64.0	48.5	36.3	27.3	21.0	15.9	12.5	9.87	7.65	5.97	5.17
167	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
168	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
169	83.7	64.0	48.5	36.3	27.3	21.0	15.9	12.5	9.87	7.65	5.97	5.17
170	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
171	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
172	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
173	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
174	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
175	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
176	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
177	...	...	...	...	...	...	...	...	...	...	...	...
178	...	...	...	...	...	...	...	...	...	...	...	...
179	...	...	...	...	...	...	...	...	...	...	...	...
180	...	...	...	...	...	...	...	...	...	...	...	...
181	...	...	...	...	...	...	...	...	...	...	...	...
182	...	...	...	...	...	...	...	...	...	...	...	...
183	...	...	...	...	...	...	...	...	...	...	...	...
184	...	...	...	...	...	...	...	...	...	...	...	...
185	...	...	...	...	...	...	...	...	...	...	...	...
186	...	...	...	...	...	...	...	...	...	...	...	...
187	...	...	...	...	...	...	...	...	...	...	...	...
188	...	...	...	...	...	...	...	...	...	...	...	...
189	...	...	...	...	...	...	...	...	...	...	...	...
190	...	...	...	...	...	...	...	...	...	...	...	...
191	...	...	...	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
192	29Cr-6.5Ni-2Mo-N	Plate & sheet	A240	...	S32906	...	≥10.0	10H
193	29Cr-6.5Ni-2Mo-N	Plate & sheet	A240	...	S32906	...	<10.0	10H
194	25Cr-8Ni-3Mo-W-Cu-N	Plate & sheet	A240	...	S32760	...	...	10H
195	25Cr-7Ni-4Mo-N	Plate & sheet	A240	2507	S32750	...	...	10H
196	18Cr-13Ni-3Mo	Forgings & fittings	A182	F317L	S31703	...	≤125	8
197	18Cr-8Ni	Forgings & fittings	A182	F304L	S30403	...	...	8
198	18Cr-8Ni	Forgings & fittings	A403	WP304L	S30403	...	...	8
199	16Cr-12Ni-2Mo	Forgings & fittings	A182	F316L	S31603	...	...	8
200	16Cr-12Ni-2Mo	Forgings & fittings	A403	WP316L	S31603	...	...	8
201	18Cr-13Ni-3Mo	Forgings & fittings	A403	WP317L	S31703	...	...	8
202	25Cr-20Ni	Forgings & fittings	A182	F310	S31000	...	...	8
203	25Cr-20Ni	Forgings & fittings	A403	WP310S	S31008	...	...	8
204	18Cr-10Ni-Ti	Smls. fittings	A403	WP321	S32100	...	>9.50	8
205	18Cr-10Ni-Ti	Forgings	A182	F321	S32100	...	...	8
206	18Cr-10Ni-Ti	Smls. fittings	A403	WP321	S32100	...	≤9.50	8
207	18Cr-10Ni-Ti	Wld. fittings	A403	WP321	S32100	...	...	8
208	23Cr-12Ni	Forgings & fittings	A403	WP309	S30900	...	...	8
209	25Cr-20Ni	Forgings & fittings	A182	F310H	S31009	...	...	8
210	25Cr-20Ni	Forgings & fittings	A403	WP310H	S31009	...	...	8
211	18Cr-10Ni-Cb	Forgings & fittings	A182	F347	S34700	...	...	8
212	18Cr-10Ni-Cb	Forgings & fittings	A403	WP347	S34700	...	...	8
213	18Cr-10Ni-Cb	Forgings & fittings	A182	F348	S34800	...	...	8
214	18Cr-10Ni-Cb	Forgings & fittings	A403	WP348	S34800	...	...	8
215	18Cr-10Ni-Ti	Smls. fittings	A403	WP321	S32100	...	>9.50	8
216	18Cr-10Ni-Ti	Smls. fittings	A403	WP321H	S32109	...	>9.50	8
217	18Cr-10Ni-Ti	Forgings	A182	F321	S32100	...	...	8
218	18Cr-10Ni-Ti	Forgings	A182	F321H	S32109	...	...	8
219	18Cr-10Ni-Ti	Smls. fittings	A403	WP321	S32100	...	≤9.50	8
220	18Cr-10Ni-Ti	Smls. fittings	A403	WP321H	S32109	...	≤9.50	8
221	18Cr-10Ni-Ti	Wld. fittings	A403	WP321	S32100	...	...	8
222	18Cr-10Ni-Ti	Wld. fittings	A403	WP321H	S32109	...	...	8

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]						
						Min. Temp. to 40	65	100	125	150	175	200
192	(25)	−51	750	550	316	251	251	249	243	238	235	231
193	(25)	−51	800	650	316	267	267	265	259	253	250	246
194	(25)	−51	750	550	316	250	250	248	243	238	236	234
195	(25)	−51	795	550	316	267	265	264	257	251	247	243
196	(9) (23)	−198	485	170	454	115	115	115	115	115	113	109
197	(9) (23)	−254	485	170	816	115	115	115	115	115	114	110
198	(32) (37)	−254	485	170	816	115	115	115	115	115	114	110
199	(9) (23)	−254	485	170	816	115	115	115	115	115	113	109
200	(32) (37)	−254	485	170	816	115	115	115	115	115	113	109
201	(32) (37)	−198	515	205	454	138	138	138	138	138	136	131
202	(9) (35) (39)	−198	515	205	816	138	138	138	138	138	138	138
203	(28) (32) (35) (37)	−198	515	205	816	138	138	138	138	138	138	138
204	(28)	−198	485	170	816	115	115	115	115	115	115	115
205	(9) (21) (28)	−198	515	205	816	138	138	138	138	138	138	138
206	(28)	−198	515	205	816	138	138	138	138	138	138	138
207	(28)	−198	515	205	816	138	138	138	138	138	138	138
208	(28) (32) (35) (37) (39)	−198	515	205	816	138	138	138	138	138	138	138
209	(9) (21) (29) (35) (39)	−198	515	205	816	138	138	138	138	138	138	138
210	(29) (32) (35) (37) (39)	−198	515	205	816	138	138	138	138	138	138	138
211	(9) (21)	−254	515	205	816	138	138	138	138	138	138	138
212	(32) (37)	−254	515	205	816	138	138	138	138	138	138	138
213	(9) (21)	−198	515	205	816	138	138	138	138	138	138	138
214	(32) (37)	−198	515	205	816	138	138	138	138	138	138	138
215	(28) (30)	−198	485	170	816	115	115	115	115	115	115	115
216	(30)	−198	485	170	816	115	115	115	115	115	115	115
217	(9) (21) (28) (30)	−198	515	205	816	138	138	138	138	138	138	138
218	(9) (21)	−198	515	205	816	138	138	138	138	138	138	138
219	(28) (30)	−198	515	205	816	138	138	138	138	138	138	138
220	(30)	−198	515	205	816	138	138	138	138	138	138	138
221	(28) (30)	−198	515	205	816	138	138	138	138	138	138	138
222	(30)	−198	515	205	816	138	138	138	138	138	138	138

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]												
	225	250	275	300	325	350	375	400	425	450	475	500	525
192	230	228	228	228	228	...	...	...	...	...	...	...	...
193	245	243	243	242	242	...	...	...	...	...	...	...	...
194	233	233	233	233	233	...	...	...	...	...	...	...	...
195	241	238	237	237	236	...	...	...	...	...	...	...	...
196	106	103	100	98.1	96.1	94.3	92.6	90.9	89.3	87.6	85.9	...	...
197	106	103	99.9	97.7	95.7	94.1	92.6	91.3	90.0	88.7	87.3	85.6	83.7
198	106	103	99.9	97.7	95.7	94.1	92.6	91.3	90.0	88.7	87.3	85.6	83.7
199	106	103	100	98.1	96.1	94.3	92.6	90.9	89.3	87.6	85.9	84.2	82.5
200	106	103	100	98.1	96.1	94.3	92.6	90.9	89.3	87.6	85.9	84.2	82.5
201	127	123	120	118	115	113	111	109	107	105	103	...	...
202	137	134	131	129	127	125	123	122	120	119	117	116	84.9
203	137	134	131	129	127	125	123	122	120	119	117	116	84.9
204	115	112	109	106	104	102	100	99.0	97.0	96.3	95.3	94.4	93.6
205	138	135	131	128	125	122	120	119	117	115	114	113	112
206	138	135	131	128	125	122	120	119	117	115	114	113	112
207	138	135	131	128	125	122	120	119	117	115	114	113	112
208	138	135	133	131	129	127	125	124	122	121	119	117	108
209	137	134	131	129	127	125	123	122	120	119	117	116	108
210	137	134	131	129	127	125	123	122	120	119	117	116	108
211	138	138	137	135	132	130	128	127	126	126	125	125	125
212	138	138	137	135	132	130	128	127	126	126	125	125	125
213	138	138	137	135	132	130	128	127	126	126	125	125	125
214	138	138	137	135	132	130	128	127	126	126	125	125	125
215	115	112	109	106	104	102	100	99.0	97.0	96.3	95.3	94.4	93.6
216	115	112	109	106	104	102	100	99.0	97.0	96.3	95.3	94.4	93.6
217	138	135	131	128	125	122	120	119	117	115	114	113	112
218	138	135	131	128	125	122	120	119	117	115	114	113	112
219	138	135	131	128	125	122	120	119	117	115	114	113	112
220	138	135	131	128	125	122	120	119	117	115	114	113	112
221	138	135	131	128	125	122	120	119	117	115	114	113	112
222	138	135	131	128	125	122	120	119	117	115	114	113	112

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to [Notes for Tables A-1 and A-1C](#); Specifications Are ASTM Unless Otherwise Indicated

Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]											
	550	575	600	625	650	675	700	725	750	775	800	825
192	...	...	...	...	...	...	...	...	...	...	...	...
193	...	...	...	...	...	...	...	...	...	...	...	...
194	...	...	...	...	...	...	...	...	...	...	...	...
195	...	...	...	...	...	...	...	...	...	...	...	...
196	...	...	...	...	...	...	...	...	...	...	...	...
197	81.4	40.4	33.2	26.7	21.9	18.2	15.0	12.4	8.87	7.20	6.58	6.21
198	81.4	40.4	33.2	26.7	21.9	18.2	15.0	12.4	8.87	7.20	6.58	6.21
199	80.8	73.0	67.9	58.0	43.6	33.0	25.3	18.8	14.0	10.4	7.99	6.89
200	80.8	73.0	67.9	58.0	43.6	33.0	25.3	18.8	14.0	10.4	7.99	6.89
201	...	...	...	...	...	...	...	...	...	...	...	...
202	59.0	43.5	31.9	23.6	16.9	10.7	6.10	3.90	2.99	2.36	1.73	1.38
203	59.0	43.5	31.9	23.6	16.9	10.7	6.10	3.90	2.99	2.36	1.73	1.38
204	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
205	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
206	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
207	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
208	83.7	64.0	48.5	36.3	27.3	21.0	15.9	12.5	9.87	7.65	5.97	5.17
209	83.7	64.0	48.5	36.3	27.3	21.0	15.9	12.5	9.87	7.65	5.97	5.17
210	83.7	64.0	48.5	36.3	27.3	21.0	15.9	12.5	9.87	7.65	5.97	5.17
211	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
212	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
213	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
214	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
215	92.7	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
216	92.7	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
217	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
218	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
219	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
220	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
221	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
222	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
223	16Cr-12Ni-2Mo	Forgings & fittings	A403	WP316H	S31609	...	...	8
224	16Cr-12Ni-2Mo	Forgings & fittings	A182	F316H	S31609	...	...	8
225	18Cr-10Ni-Cb	Forgings & fittings	A403	WP347H	S34709	...	...	8
226	18Cr-10Ni-Cb	Forgings & fittings	A182	F347	S34700	...	...	8
227	18Cr-10Ni-Cb	Forgings & fittings	A403	WP347	S34700	...	...	8
228	18Cr-10Ni-Cb	Forgings & fittings	A182	F348	S34800	...	...	8
229	18Cr-10Ni-Cb	Forgings & fittings	A403	WP348	S34800	...	...	8
230	18Cr-10Ni-Cb	Forgings & fittings	A182	F347H	S34709	...	...	8
231	18Cr-10Ni-Cb	Forgings & fittings	A182	F348H	S34809	...	...	8
232	16Cr-12Ni-2Mo	Forgings & fittings	A182	F316	S31600	...	...	8
233	16Cr-12Ni-2Mo	Forgings & fittings	A403	WP316	S31600	...	...	8
234	18Cr-13Ni-3Mo	Forgings & fittings	A403	WP317	S31700	...	...	8
235	18Cr-8Ni	Forgings & fittings	A182	F304	S30400	...	...	8
236	18Cr-8Ni	Forgings & fittings	A403	WP304	S30400	...	...	8
237	18Cr-8Ni	Forgings & fittings	A403	WP304H	S30409	...	...	8
238	18Cr-8Ni	Forgings & fittings	A182	F304H	S30409	...	...	8
239	44Fe-25Ni-21Cr-Mo	Forgings	A182	F904L	N08904	...	...	45
240	13Cr	Forgings & fittings	A182	F6a	S41000	1	...	6
241	13Cr	Forgings & fittings	A182	F6a	S41000	2	...	6
242	20Cr-18Ni-6Mo	Forgings	A182	F44	S31254	...	...	8
243	20Cr-18Ni-6Mo	Fittings	A403	WPS31254	S31254	...	...	8
244	23Cr-4Ni-Mo-Cu-N	Forgings	A182	F68	S32304	...	...	10H
245	22Cr-5Ni-3Mo-N	Forgings	A182	F51	S31803	...	...	10H
246	22Cr-5Ni-3Mo-N	Fittings	A815	WPS31803	S31803	...	...	10H
247	46Fe-24Ni-21Cr-6Mo-Cu-N	Forgings	A182	F62	N08367	...	...	45
248	46Fe-24Ni-21Cr-6Mo-Cu-N	Fittings	A403	WP6XN	N08367	...	...	45
249	21Cr-5Mn-1½Ni-Cu-N	Fittings	A815	WPS32101	S32101	...	...	10H
250	22Cr-5Ni-3Mo-N	Forgings	A182	F60	S32205	...	...	10H
251	22Cr-5Ni-3Mo-N	Fittings	A815	WPS32205	S32205	...	...	10H
252	25Cr-8Ni-3Mo-W-Cu-N	Forgings & fittings	A182	F55	S32760	...	...	10H

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]						
						Min. Temp. to 40	65	100	125	150	175	200
223	(26) (32) (37)	−198	515	205	816	138	138	138	138	138	138	134
224	(9) (21) (26)	−198	515	205	816	138	138	138	138	138	138	134
225	(32) (37)	−198	515	205	816	138	138	138	138	138	138	138
226	(9) (21) (28)	−254	515	205	816	138	138	138	138	138	138	138
227	(28) (32) (37)	−254	515	205	816	138	138	138	138	138	138	138
228	(9) (21) (28)	−198	515	205	816	138	138	138	138	138	138	138
229	(28) (32) (37)	−198	515	205	816	138	138	138	138	138	138	138
230	(9) (21)	−198	515	205	816	138	138	138	138	138	138	138
231	(9) (21)	−198	515	205	816	138	138	138	138	138	138	138
232	(9) (21) (26) (28)	−254	515	205	816	138	138	138	138	138	138	134
233	(26) (28) (32) (37)	−254	515	205	816	138	138	138	138	138	138	134
234	(26) (28) (32)	−198	515	205	816	138	138	138	138	138	138	134
235	(9) (21) (26) (28)	−254	515	205	816	138	138	138	138	138	134	129
236	(26) (28) (32) (37)	−254	515	205	816	138	138	138	138	138	134	129
237	(26) (32) (37)	−198	515	205	816	138	138	138	138	138	134	129
238	(9) (21) (26)	−198	515	205	816	138	138	138	138	138	134	129
239	(26)	−198	490	215	260	143	143	143	143	141	135	130
240	(35)	−29	485	275	538	161	161	161	160	158	156	155
241	(35)	−29	585	380	649	195	195	195	194	191	190	188
242	...	−198	650	300	454	202	202	202	202	199	192	185
243	...	−198	650	300	454	202	202	202	202	199	192	185
244	(25)	−51	600	400	316	200	200	191	185	180	175	171
245	(25)	−51	620	450	316	207	207	207	204	199	196	193
246	(25)	−51	620	450	316	207	207	207	204	199	196	193
247	(26)	−198	655	310	427	207	207	207	207	206	202	198
248	(26)	−198	655	310	427	207	207	207	207	206	202	198
249	(25)	−29	650	450	316	217	217	215	211	206	203	199
250	(25)	−51	655	450	316	218	218	218	215	210	206	203
251	(25)	−51	655	450	316	218	218	218	215	210	206	203
252	(25)	−51	750	550	316	250	250	248	243	238	236	234

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1) , (2b) , (3)]												
	225	250	275	300	325	350	375	400	425	450	475	500	525
223	129	125	122	119	116	114	112	111	110	109	108	107	106
224	129	125	122	119	116	114	112	111	110	109	108	107	106
225	138	138	137	135	132	130	129	127	126	126	125	125	125
226	138	138	137	135	132	130	128	127	126	126	125	125	125
227	138	138	137	135	132	130	129	127	126	126	125	125	125
228	138	138	137	135	132	130	128	127	126	126	125	125	125
229	138	138	137	135	132	130	128	127	126	126	125	125	125
230	138	138	137	135	132	130	129	127	126	126	125	125	125
231	138	138	137	135	132	130	129	127	126	126	125	125	125
232	129	125	122	119	116	114	112	111	110	109	108	107	106
233	129	125	122	119	116	114	112	111	110	109	108	107	106
234	129	125	122	119	116	114	112	111	110	109	108	107	106
235	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
236	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
237	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
238	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
239	125	120	116	...	...	...	...	...	...	...	...	...	...
240	154	153	152	150	148	145	142	137	133	125	92.5	68.4	51.1
241	187	186	184	182	180	176	172	167	161	125	92.5	68.4	51.1
242	180	175	171	168	165	164	162	161	160	159	158	...	...
243	180	175	171	168	165	164	162	161	160	159	158	...	...
244	166	161	153	143	111	...	...	...	...	...	...	...	...
245	190	188	187	186	185	...	...	...	...	...	...	...	...
246	190	188	187	186	185	...	...	...	...	...	...	...	...
247	195	192	188	184	179	176	173	170	167	166	...	...	...
248	195	192	188	184	179	176	173	170	167	166	...	...	...
249	199	199	199	199	199	...	...	...	...	...	...	...	...
250	201	199	197	196	196	...	...	...	...	...	...	...	...
251	201	199	197	196	196	...	...	...	...	...	...	...	...
252	233	233	233	233	233	...	...	...	...	...	...	...	...

Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to [Notes for Tables A-1 and A-1C](#); Specifications Are ASTM Unless Otherwise Indicated

Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]											
	550	575	600	625	650	675	700	725	750	775	800	825
223	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
224	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
225	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
226	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
227	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
228	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
229	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
230	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
231	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
232	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
233	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
234	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
235	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
236	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
237	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
238	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
239	...	...	...	...	...	...	...	...	...	...	...	...
240	44.1	...	...	...	...	...	...	...	...	...	...	...
241	37.4	26.3	17.8	11.4	6.89	...	...	...	...	...	...	...
242	...	...	...	...	...	...	...	...	...	...	...	...
243	...	...	...	...	...	...	...	...	...	...	...	...
244	...	...	...	...	...	...	...	...	...	...	...	...
245	...	...	...	...	...	...	...	...	...	...	...	...
246	...	...	...	...	...	...	...	...	...	...	...	...
247	...	...	...	...	...	...	...	...	...	...	...	...
248	...	...	...	...	...	...	...	...	...	...	...	...
249	...	...	...	...	...	...	...	...	...	...	...	...
250	...	...	...	...	...	...	...	...	...	...	...	...
251	...	...	...	...	...	...	...	...	...	...	...	...
252	...	...	...	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
253	25Cr-8Ni-3Mo-W-Cu-N	Forgings & fittings	A815	WPS32760	S32760	...	...	10H
254	25Cr-7Ni-4Mo-N	Forgings & fittings	A182	F53	S32750	...	≤50	10H
255	25Cr-7Ni-4Mo-N	Forgings & fittings	A815	WPS32750	S32750	...	...	10H
256	18Cr-8Ni	Bar	A479	304	S30400	...	...	8
257	18Cr-8Ni	Bar	A479	304H	S30409	...	...	8
258	18Cr-8Ni	Bar	A479	304L	S30403	...	...	8
259	16Cr-12Ni-2Mo	Bar	A479	316	S31600	...	...	8
260	16Cr-12Ni-2Mo	Bar	A479	316H	S31609	...	...	8
261	16Cr-12Ni-2Mo	Bar	A479	316L	S31603	...	...	8
262	18Cr-10Ni-Ti	Bar	A479	321	S32100	...	...	8
263	18Cr-10Ni-Ti	Bar	A479	321	S32100	...	...	8
264	18Cr-10Ni-Ti	Bar	A479	321H	S32109	...	...	8
265	18Cr-10Ni-Cb	Bar	A479	347	S34700	...	...	8
266	18Cr-10Ni-Cb	Bar	A479	347	S34700	...	...	8
267	18Cr-10Ni-Cb	Bar	A479	347H	S34709	...	...	8
268	44Fe-25Ni-21Cr-Mo	Bar	A479	904L	N08904	...	...	45
269	22Cr-5Ni-3Mo-N	Bar	A479	...	S31803	...	...	10H
270	20Cr-18Ni-6Mo	Bar	A479	...	S31254	...	...	8
271	46Fe-24Ni-21Cr-6Mo-Cu-N	Bar	A479	...	N08367	...	...	45
272	21Cr-5Mn-1.5Ni-Cu-N	Bar	A479	...	S32101	...	...	10H
273	22Cr-5Ni-3Mo-N	Bar	A479	...	S32205	...	...	10H
274	24Cr-4Ni-3Mn-1.5Mo-N	Bar	A479	...	S82441	...	≥11	10H
275	22Cr-13Ni-5Mn	Bar	A479	XM-19	S20910	Annealed	...	8
276	24Cr-4Ni-3Mn-1.5Mo-N	Bar	A479	...	S82441	...	<11	10H
277	29Cr-6.5Ni-2Mo-N	Bar	A479	...	S32906	...	...	10H
278	25Cr-7Ni-4Mo-N	Bar	A479	...	S32750	...	≤50	10H
279	29Ni-20Cr-3Cu-2Mo	Castings	A351	CN7M	N08007	...	...	45
280	35Ni-15Cr- $\frac{1}{2}$ Mo	Castings	A351	HT30	N08603	...	...	45
281	25Cr-12Ni	Castings	A351	CH8	J93400	...	...	8
282	25Cr-20Ni	Castings	A351	CK20	J94202	...	...	8
283	16Cr-14Ni-2Mo	Castings	A351	CF10MC	...	...	...	8
284	18Cr-8Ni	Castings	A351	CF3	J92500	...	...	8
285	18Cr-12Ni-2Mo	Castings	A351	CF3M	J92800	...	...	8
286	18Cr-8Ni	Castings	A351	CF8	J92600	...	...	8

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]						
						Min. Temp. to 40	65	100	125	150	175	200
253	(25)	−51	750	550	316	250	250	248	243	238	236	234
254	(25)	−51	800	550	316	267	265	264	257	251	247	243
255	(25)	−51	800	550	316	267	265	264	257	251	247	243
256	(26) (28)	−254	515	205	816	138	138	138	138	138	134	129
257	(26)	−198	515	205	816	138	138	138	138	138	134	129
258	...	−254	485	170	816	115	115	115	115	115	114	110
259	(26) (28)	−254	515	205	816	138	138	138	138	138	138	134
260	(26)	−198	515	205	816	138	138	138	138	138	138	134
261	...	−254	485	170	816	115	115	115	115	115	113	109
262	(28)	−198	515	205	816	138	138	138	138	138	138	138
263	(28) (30)	−198	515	205	816	138	138	138	138	138	138	138
264	(30)	−198	515	205	816	138	138	138	138	138	138	138
265	...	−254	515	205	816	138	138	138	138	138	138	138
266	(28) (30)	−254	515	205	816	138	138	138	138	138	138	138
267	...	−198	515	205	816	138	138	138	138	138	138	138
268	(26)	−198	490	220	260	143	143	143	143	141	135	130
269	(25)	−51	620	450	316	207	207	207	204	199	196	193
270	...	−198	655	305	454	207	207	207	207	203	196	190
271	(26)	−198	655	310	427	207	207	207	207	206	202	198
272	(25)	−29	650	450	316	217	217	215	211	206	203	199
273	(25)	−51	655	450	316	218	218	218	215	210	206	203
274	(25)	−51	680	480	316	227	227	227	227	227	227	227
275	...	−198	690	380	649	230	230	227	221	217	213	210
276	(25)	−51	740	540	316	247	247	247	247	247	247	247
277	(25)	−51	750	550	316	251	251	249	243	238	235	231
278	(25)	−51	800	550	316	267	265	264	257	251	247	243
279	(9) (30)	−198	425	170	40	115	...	...	...	...	...	...
280	(36) (39)	−198	450	195	40	129	...	...	...	...	...	...
281	(9) (31)	−198	450	195	816	129	129	129	129	127	125	124
282	(9) (27) (31) (35) (39)	−198	450	195	816	129	129	129	129	127	125	124
283	(30)	−198	485	205	40	138	...	...	...	...	...	...
284	(9)	−254	485	205	427	138	138	138	138	138	134	129
285	(9)	−254	485	205	454	138	138	138	138	138	138	133
286	(9) (26) (27) (31)	−254	485	205	816	138	138	138	138	138	134	129

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, <i>S</i> , MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]												
	225	250	275	300	325	350	375	400	425	450	475	500	525
253	233	233	233	233	233	...	...	...	...	...	...	...	...
254	241	238	237	237	236	...	...	...	...	...	...	...	...
255	241	238	237	237	236	...	...	...	...	...	...	...	...
256	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
257	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
258	106	103	99.9	97.7	95.7	94.1	92.6	91.3	90.0	88.7	87.3	85.6	83.7
259	129	125	122	119	116	114	112	111	110	109	108	107	106
260	129	125	122	119	116	114	112	111	110	109	108	107	106
261	106	103	100	98.1	96.1	94.3	92.6	90.9	89.3	87.6	85.9	84.2	82.5
262	138	135	131	128	125	122	120	119	117	115	114	113	112
263	138	135	131	128	125	122	120	119	117	115	114	113	112
264	138	135	131	128	125	122	120	119	117	115	114	113	112
265	138	138	137	135	132	130	128	127	126	126	125	125	125
266	138	138	137	135	132	130	128	127	126	126	125	125	125
267	138	138	137	135	132	130	128	127	126	126	125	125	125
268	125	120	116	...	...	...	...	...	...	...	...	...	...
269	190	188	187	186	185	...	...	...	...	...	...	...	...
270	184	179	175	172	169	167	166	165	164	163	161	...	...
271	195	192	188	184	179	176	173	170	167	166	...	...	...
272	199	199	199	199	199	...	...	...	...	...	...	...	...
273	201	199	197	196	196	...	...	...	...	...	...	...	...
274	227	227	227	227	227	...	...	...	...	...	...	...	...
275	207	205	204	202	201	200	199	197	195	193	191	188	183
276	247	247	247	247	247	...	...	...	...	...	...	...	...
277	230	228	228	228	228	...	...	...	...	...	...	...	...
278	241	238	237	237	236	...	...	...	...	...	...	...	...
279	...	...	...	...	...	...	...	...	...	...	...	...	...
280	...	...	...	...	...	...	...	...	...	...	...	...	...
281	124	123	121	119	117	115	112	109	106	103	100	96.9	93.7
282	124	123	121	119	117	115	112	109	106	103	100	96.9	93.7
283	...	...	...	...	...	...	...	...	...	...	...	...	...
284	125	122	119	116	113	111	109	107	105	103	...	...	...
285	129	125	122	119	116	114	112	111	109	108	107	...	...
286	125	122	119	116	113	111	109	107	105	103	101	99.1	94.4

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to [Notes for Tables A-1 and A-1C](#); Specifications Are ASTM Unless Otherwise Indicated

Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]											
	550	575	600	625	650	675	700	725	750	775	800	825
253	...	...	...	...	...	...	...	...	...	...	...	...
254	...	...	...	...	...	...	...	...	...	...	...	...
255	...	...	...	...	...	...	...	...	...	...	...	...
256	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
257	95.5	78.9	63.8	51.6	41.6	32.9	26.5	21.3	17.2	14.1	11.2	9.65
258	81.4	40.4	33.2	26.7	21.9	18.2	15.0	12.4	8.87	7.20	6.58	6.21
259	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
260	105	97.8	80.8	65.0	50.4	38.6	29.6	23.0	17.4	13.3	10.4	8.96
261	80.8	73.0	67.9	58.0	43.6	33.0	25.3	18.8	14.0	10.4	7.99	6.89
262	88.7	59.2	44.0	32.9	24.5	18.3	12.5	8.49	6.19	4.28	2.75	1.74
263	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
264	100	76.5	58.7	46.0	36.8	28.7	23.0	18.4	14.5	11.5	9.02	6.83
265	97.6	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
266	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
267	125	112	90.6	69.6	53.8	41.4	31.8	24.0	18.8	14.6	10.9	8.96
268	...	...	...	...	...	...	...	...	...	...	...	...
269	...	...	...	...	...	...	...	...	...	...	...	...
270	...	...	...	...	...	...	...	...	...	...	...	...
271	...	...	...	...	...	...	...	...	...	...	...	...
272	...	...	...	...	...	...	...	...	...	...	...	...
273	...	...	...	...	...	...	...	...	...	...	...	...
274	...	...	...	...	...	...	...	...	...	...	...	...
275	179	174	132	83.6	56.1	...	...	...	...	...	...	...
276	...	...	...	...	...	...	...	...	...	...	...	...
277	...	...	...	...	...	...	...	...	...	...	...	...
278	...	...	...	...	...	...	...	...	...	...	...	...
279	...	...	...	...	...	...	...	...	...	...	...	...
280	...	...	...	...	...	...	...	...	...	...	...	...
281	68.1	53.5	42.1	33.2	25.9	20.3	16.5	13.2	10.1	7.35	5.89	5.52
282	73.2	64.4	56.5	49.0	41.0	33.5	25.4	18.3	12.8	9.01	6.59	5.52
283	...	...	...	...	...	...	...	...	...	...	...	...
284	...	...	...	...	...	...	...	...	...	...	...	...
285	...	...	...	...	...	...	...	...	...	...	...	...
286	75.3	60.4	49.0	40.1	32.8	27.2	23.4	19.6	16.8	14.7	12.8	11.7

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
287	25Cr-12Ni	Castings	A351	CH10	J93401	...	...	8
288	25Cr-12Ni	Castings	A351	CH20	J93402	...	...	8
289	18Cr-10Ni-Cb	Castings	A351	CF8C	J92710	...	...	8
290	18Cr-12Ni-2Mo	Castings	A351	CF8M	J92900	...	...	8
291	25Cr-20Ni- $\frac{1}{2}$ Mo	Castings	A351	HK40	J94204	...	...	8
292	25Cr-20Ni- $\frac{1}{2}$ Mo	Castings	A351	HK30	J94203	...	...	8
293	18Cr-8Ni	Castings	A351	CF3A	J92500	...	...	8
294	18Cr-8Ni	Castings	A351	CF8A	J92600	...	...	8
295	25Cr-8Ni-N	Castings	A351	CE20N	J92802	...	...	8
296	22Cr-5Ni-3Mo-N	Castings	A995	4A	J92205	...	...	10H
297	12Cr	Castings	A217	CA15	J91150	...	...	6
298	24Cr-10Ni-4Mo-N	Castings	A995	2A	J93345	...	...	10H
299	25Cr-8Ni-3Mo-W-Cu-N	Castings	A995	6A	J93380	...	...	10H
300	13Cr-4Ni	Castings	A487	CA6NM	J91540	A	...	6

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]						
						Min. Temp. to 40	65	100	125	150	175	200
287	(27) (31) (35)	−198	485	205	816	138	138	138	138	137	136	134
288	(9) (27) (31) (35) (39)	−198	485	205	816	138	138	138	138	137	136	134
289	(9) (28)	−198	485	205	816	138	138	138	138	138	138	135
290	(9) (26) (27) (30)	−254	485	205	816	138	138	138	138	138	134	129
291	(35) (36) (39)	−198	425	240	40	142	...	...	...	...	...	...
292	(35) (39)	−198	450	240	40	149	...	...	...	...	...	...
293	(9) (56)	−254	530	240	371	161	161	161	160	156	153	151
294	(9) (26) (56)	−254	530	240	371	161	161	161	160	156	153	151
295	(35) (39)	−198	550	275	482	184	184	184	184	184	184	184
296	(9) (25)	−51	620	415	260	207	207	207	202	198	197	196
297	(35)	−29	620	450	649	207	207	207	205	203	201	199
298	(9)	−51	655	448	316	218	218	216	208	201	197	195
299	(9) (25)	−51	690	450	316	230	230	227	221	216	212	209
300	(9) (35)	−29	760	550	371	253	253	253	251	248	245	244

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, <i>S</i> , MPa, at Metal Temperature, °C [Notes (1) , (2b) , (3)]												
	225	250	275	300	325	350	375	400	425	450	475	500	525
287	132	131	130	128	125	123	120	117	114	110	107	104	101
288	132	131	130	128	125	123	120	117	114	110	107	104	101
289	132	130	128	127	127	126	126	126	125	125	124	124	124
290	125	122	119	116	113	111	109	107	105	103	101	99.1	97.3
291	...	...	...	...	...	...	...	...	...	...	...	...	...
292	...	...	...	...	...	...	...	...	...	...	...	...	...
293	146	142	138	135	132	130	127	...	...	...	...	...	...
294	146	142	138	135	132	130	127	...	...	...	...	...	...
295	184	184	184	184	184	184	184	184	184	184	184	184	...
296	196	196	192	...	...	...	...	...	...	...	...	...	...
297	198	197	195	193	190	187	182	120	116	105	82.1	59.9	41.8
298	194	194	194	194	194	...	...	...	...	...	...	...	...
299	207	206	205	204	204	...	...	...	...	...	...	...	...
300	242	240	238	236	232	228	224	...	...	...	...	...	...

Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated												
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b), (3)]											
	550	575	600	625	650	675	700	725	750	775	800	825
287	68.2	53.7	42.1	33.2	25.9	20.3	16.4	13.3	10.2	7.25	5.74	5.33
288	68.2	53.7	42.1	33.2	25.9	20.3	16.4	13.3	10.2	7.25	5.74	5.33
289	98.3	77.2	57.7	39.9	30.0	23.2	16.3	11.2	8.93	7.08	5.77	5.32
290	95.5	75.9	57.2	40.2	30.3	23.2	16.2	11.4	8.97	7.08	5.89	5.52
291	...	...	...	...	...	...	...	...	...	...	...	...
292	...	...	...	...	...	...	...	...	...	...	...	...
293	...	...	...	...	...	...	...	...	...	...	...	...
294	...	...	...	...	...	...	...	...	...	...	...	...
295	...	...	...	...	...	...	...	...	...	...	...	...
296	...	...	...	...	...	...	...	...	...	...	...	...
297	28.7	20.1	14.3	9.77	6.89	...	...	...	...	...	...	...
298	...	...	...	...	...	...	...	...	...	...	...	...
299	...	...	...	...	...	...	...	...	...	...	...	...
300	...	...	...	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated											
Line No.	Nominal Composition	Product Form	Spec. No.	UNS No.	Class/Condition/ Temper	Size, mm	P-No. (5)(7)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa
1	99.95Cu-P	Pipe	B42	C10200	O61	...	31	...	-269	205	60
2	99.9Cu-P	Pipe	B42	C12000	O61	...	31	...	-269	205	60
3	99.9Cu-P	Pipe	B42	C12200	O61	...	31	...	-269	205	60
4	99.95Cu-P	Tube	B75	C10200	O50	...	31	...	-269	205	62
5	99.95Cu-P	Tube	B75	C10200	O60	...	31	...	-269	205	62
6	99.9Cu-P	Tube	B75	C12000	O50	...	31	...	-269	205	62
7	99.9Cu-P	Tube	B75	C12000	O60	...	31	...	-269	205	62
8	99.9Cu-P	Tube	B75	C12200	O50	...	31	...	-269	205	62
9	99.9Cu-P	Tube	B75	C12200	O60	...	31	...	-269	205	62
10	99.9Cu-P	Tube	B68	C12200	O50	...	31	(24)	-269	210	62
11	99.9Cu-P	Tube	B68	C12200	O60	...	31	(24)	-269	210	62
12	99.9Cu-P	Tube	B88	C12200	O50	...	31	(24)	-269	207	62
13	99.9Cu-P	Tube	B88	C12200	O60	...	31	(24)	-269	207	62
14	99.9Cu-P	Tube	B280	C12200	O60	...	31	(24)	-269	207	62
15	85Cu-15Zn	Pipe	B43	C23000	O61	...	32	...	-269	275	85
16	90Cu-10Ni	Pipe & tube	B467	C70600	W050	>114 O.D.	34	(14)	-269	262	90
17	90Cu-10Ni	Pipe & tube	B467	C70600	W061	>114 O.D.	34	(14)	-269	262	90
18	90Cu-10Ni	Pipe & tube	B466	C70600	Annealed	...	34	(14)	-269	262	90
19	90Cu-10Ni	Pipe & tube	B467	C70600	W050	≤114 O.D.	34	(14)	-269	276	103
20	90Cu-10Ni	Pipe & tube	B467	C70600	W061	≤114 O.D.	34	(14)	-269	276	103
21	70Cu-30Ni	Pipe & tube	B467	C71500	W050	>114 O.D.	34	(14)	-269	310	103
22	70Cu-30Ni	Pipe & tube	B467	C71500	W061	>114 O.D.	34	(14)	-269	310	103
23	80Cu-20Ni	Pipe & tube	B466	C71000	Annealed	≤114 O.D.	34	(14)	-269	310	110
24	99.95Cu-P	Pipe	B42	C10200	H55	DN 50 thru 300	31	(14) (34)	-269	250	205
25	99.9Cu-P	Pipe	B42	C12000	H55	DN 50 thru 300	31	(14) (34)	-269	250	205
26	99.9Cu-P	Pipe	B42	C12200	H55	DN 50 thru 300	31	(14) (34)	-269	250	205
27	99.95Cu-P	Tube	B75	C10200	H58	...	31	(14) (34)	-269	250	205
28	99.9Cu-P	Tube	B75	C12000	H58	...	31	(14) (34)	-269	250	205
29	99.9Cu-P	Tube	B75	C12200	H58	...	31	(14) (34)	-269	250	205
30	99.9Cu-P	Tube	B88	C12200	H58	...	31	(14) (24) (34)	-269	248	207
31	70Cu-30Ni	Pipe & tube	B466	C71500	O60	...	34	(14)	-269	359	124
32	70Cu-30Ni	Pipe & tube	B467	C71500	W050	≤114 O.D.	34	(14)	-269	345	138
33	70Cu-30Ni	Pipe & tube	B467	C71500	W061	≤114 O.D.	34	(14)	-269	345	138
34	99.95Cu-P	Pipe	B42	C10200	H80	DN 6 thru 50	31	(14) (34)	-269	310	275
35	99.9Cu-P	Pipe	B42	C12000	H80	DN 6 thru 50	31	(14) (34)	-269	310	275
36	99.9Cu-P	Pipe	B42	C12200	H80	DN 6 thru 50	31	(14) (34)	-269	310	275
37	99.95Cu-P	Tube	B75	C10200	H80	≤102 O.D.	31	(14) (34)	-269	310	275

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated															
Line No.	Max. Temp., °C	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]													
		Min. Temp. to 40	65	100	125	150	175	200	225	250	275	300	325	350	375
1	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
2	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
3	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
4	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
5	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
6	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
7	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
8	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
9	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
10	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
11	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
12	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
13	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
14	260	41.4	34.9	33.5	32.9	32.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
15	232	55.2	54.7	54.7	54.7	54.7	54.7	36.4	17.5	13.8	...	...	...	...	...
16	316	59.8	58.0	56.1	54.9	53.8	52.9	52.0	51.2	50.4	49.7	45.1	39.3	...	...
17	316	59.8	58.0	56.1	54.9	53.8	52.9	52.0	51.2	50.4	49.7	45.1	39.3	...	...
18	316	59.8	58.0	56.1	54.9	53.8	52.9	52.0	51.2	50.4	49.7	45.1	39.3	...	...
19	316	68.9	67.0	65.1	63.7	62.4	61.1	60.0	59.1	58.3	51.3	45.1	39.3	...	...
20	316	68.9	67.0	65.1	63.7	62.4	61.1	60.0	59.1	58.3	51.3	45.1	39.3	...	...
21	371	68.9	66.6	64.6	63.2	61.9	60.7	59.5	58.4	57.4	56.2	55.5	54.9	54.3	53.8
22	371	68.9	66.6	64.6	63.2	61.9	60.7	59.5	58.4	57.4	56.2	55.5	54.9	54.3	53.8
23	371	73.5	72.8	72.1	71.4	70.6	69.6	68.3	66.6	64.7	62.4	60.0	56.2	51.9	48.3
24	204	82.7	80.0	74.1	71.3	69.2	67.4	65.7	65.3	...	...	...	...	...	...
25	204	82.7	80.0	74.1	71.3	69.2	67.4	65.7	65.3	...	...	...	...	...	...
26	204	82.7	80.0	74.1	71.3	69.2	67.4	65.7	65.3	...	...	...	...	...	...
27	204	82.7	80.0	74.1	71.3	69.2	67.4	65.7	65.3	...	...	...	...	...	...
28	204	82.7	80.0	74.1	71.3	69.2	67.4	65.7	65.3	...	...	...	...	...	...
29	204	82.7	80.0	74.1	71.3	69.2	67.4	65.7	65.3	...	...	...	...	...	...
30	204	82.7	80.0	74.1	71.3	69.2	67.4	65.7	65.3	...	...	...	...	...	...
31	371	82.7	79.8	77.5	75.9	74.3	72.9	71.5	70.2	68.9	67.8	66.8	65.9	65.3	64.8
32	371	91.9	88.7	86.1	84.3	82.6	81.0	79.4	78.0	76.6	75.3	74.2	73.2	72.5	71.7
33	371	91.9	88.7	86.1	84.3	82.6	81.0	79.4	78.0	76.6	75.3	74.2	73.2	72.5	71.7
34	204	103	100	92.6	89.1	86.5	84.3	35.8	29.6	...	...	...	...	...	...
35	204	103	100	92.6	89.1	86.5	84.3	35.8	29.6	...	...	...	...	...	...
36	204	103	100	92.6	89.1	86.5	84.3	35.8	29.6	...	...	...	...	...	...
37	204	103	100	92.6	89.1	86.5	84.3	35.8	29.6	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated											
Line No.	Nominal Composition	Product Form	Spec. No.	UNS No.	Class/Condition/ Temper	Size, mm	P-No. (5)(7)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa
38	99.9Cu-P	Tube	B75	C12000	H80	≤102 O.D.	31	(14) (34)	-269	310	275
39	99.9Cu-P	Tube	B75	C12200	H80	≤102 O.D.	31	(14) (34)	-269	310	275
40	99.95Cu-P	Plate & sheet	B152	C10200	O25	...	31	(14) (24)	-269	207	69
41	99.95Cu-Ag	Plate & sheet	B152	C10400	O25	...	31	(14) (24)	-269	207	69
42	99.95Cu-Ag	Plate & sheet	B152	C10500	O25	...	31	(14) (24)	-269	207	69
43	99.95Cu-Ag	Plate & sheet	B152	C10700	O25	...	31	(14) (24)	-269	207	69
44	99.9Cu-P	Plate & sheet	B152	C12200	O25	...	31	(14) (24)	-269	207	69
45	99.9Cu-P	Plate & sheet	B152	C12300	O25	...	31	(14) (24)	-269	207	69
46	90Cu-10Ni	Plate & sheet	B171	C70600	...	≤64 thk.	34	(14)	-269	276	103
47	97Cu-3Si	Plate & sheet	B96	C65500	O61	...	33	...	-269	345	124
48	70Cu-30Ni	Plate & sheet	B171	C71500	...	≤64 thk.	34	(14)	-269	345	138
49	90Cu-7Al-3Fe	Plate & sheet	B169	C61400	O25	≤50 thk.	35	(13)	-269	485	205
50	90Cu-7Al-3Fe	Plate & sheet	B169	C61400	O60	≤50 thk.	35	(13)	-269	485	205
51	99.9Cu	Forgings	B283	C11000	...	...	31	(14)	-269	228	76
52	97Cu-3Si	Forgings	B283	C65500	...	...	33	(14)	-269	359	124
53	60Cu-38Zn-2Pb	Forgings	B283	C37700	M10	>38.1	a	(14)	-198	317	103
54	60Cu-38Zn-2Pb	Forgings	B283	C37700	M10	≤38.1	a	(14)	-198	345	124
55	60Cu-37Zn-2Pb-Sn	Forgings	B283	C48500	...	...	a	(14)	-198	427	165
56	60Cu-39Zn-Sn	Forgings	B283	C46400	...	...	32	(14)	-254	441	179
57	59Cu-39Zn-Fe-Sn	Forgings	B283	C67500	...	...	32	(14)	-198	496	234
58	85Cu-5Sn-5Zn-5Pb	Castings	B62	C83600	...	...	a	(9)	-198	207	97
59	57Cu-20Zn-12Ni-9Pb-2Sn	Castings	B584	C97300	...	...	a	...	-198	207	103
60	64Cu-20Ni-8Zn-4Sn-4Pb	Castings	B584	C97600	...	...	a	...	-198	276	117
61	87Cu-8Sn-4Zn-1Pb	Castings	B584	C92300	...	...	a	...	-198	248	110
62	88Cu-Sn-Zn-Pb	Castings	B584	C92200	...	...	a	...	-198	234	110
63	88Cu-Sn-Zn-Pb	Castings	B61	C92200	...	...	a	(9)	-198	235	110
64	88Cu-8Sn-4Zn	Castings	B584	C90300	...	...	b	...	-198	276	124
65	88Cu-10Sn-2Zn	Castings	B584	C90500	...	...	b	...	-198	276	124
66	58Cu-38Zn-1Sn-1Pb-1Fe	Castings	B584	C86400	...	...	a	(9)	-198	414	138
67	66Cu-25Ni-5Sn-2Pb-2Zn	Castings	B584	C97800	...	...	a	...	-198	345	152
68	58Cu-39Zn-1Fe-1Al-1Mn	Castings	B584	C86500	...	...	b	...	-198	448	172

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated															
Line No.	Max. Temp., °C	Min. Temp. to 40	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]												
			65	100	125	150	175	200	225	250	275	300	325	350	375
38	204	103	100	92.6	89.1	86.5	84.3	35.8	29.6	...	...	...	...	...	...
39	204	103	100	92.6	89.1	86.5	84.3	35.8	29.6	...	...	...	...	...	...
40	260	46.0	39.1	37.1	36.5	34.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
41	260	46.0	39.1	37.1	36.5	34.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
42	260	46.0	39.1	37.1	36.5	34.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
43	260	46.0	39.1	37.1	36.5	34.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
44	260	46.0	39.1	37.1	36.5	34.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
45	260	46.0	39.1	37.1	36.5	34.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
46	316	68.9	67.0	65.1	63.7	62.4	61.1	60.0	59.1	58.3	51.3	45.1	39.3	...	...
47	204	82.7	82.7	82.7	82.7	82.7	75.4	50.8	31.9	...	...	...	...	...	...
48	371	91.9	88.7	86.1	84.3	82.6	81.0	79.4	78.0	76.6	75.3	74.2	73.2	72.5	71.7
49	260	138	137	136	135	135	134	133	131	130	130	...	...	...	...
50	260	138	137	136	135	135	134	133	131	130	130	...	...	...	...
51	260	50.6	42.8	40.8	40.2	34.2	28.0	21.7	16.7	12.9	11.6	...	...	...	...
52	204	82.7	82.7	82.7	82.7	82.7	75.4	50.8	31.9	...	...	...	...	...	...
53	260	68.7	68.7	68.7	55.6	35.7	22.4	13.7	8.0	4.5	2.4	...	...	...	...
54	260	82.7	82.7	82.7	55.6	35.7	22.4	13.7	8.0	4.5	2.4	...	...	...	...
55	204	110	110	110	110	110	110	110	110	...	...	...	...	...	...
56	204	120	120	120	120	118	118	20.0	17.2	...	...	...	...	...	...
57	204	156	156	156	156	156	156	156	156	...	...	...	...	...	...
58	232	64.4	64.4	62.6	59.1	55.7	53.0	51.2	50.3	50.1	...	...	...	...	...
59	40	68.9	...	...	...	...	...	...	...	...	...	...	...	...	...
60	149	78.1	69.6	64.9	62.5	60.3	...	...	...	...	...	...	...	...	...
61	204	73.5	73.5	73.5	73.5	73.5	73.5	73.5	73.5	...	...	...	...	...	...
62	204	73.5	66.4	65.3	64.6	63.3	61.5	59.6	58.1	...	...	...	...	...	...
63	288	73.5	66.4	65.3	64.6	63.3	61.5	59.6	58.1	57.5	57.5	57.3	...	...	...
64	204	82.7	82.7	82.7	82.7	82.7	82.7	82.7	82.7	...	...	...	...	...	...
65	204	82.7	82.7	82.7	82.7	82.7	82.7	82.7	82.7	...	...	...	...	...	...
66	177	91.9	91.9	91.9	91.9	91.9	91.9	91.9	...	...	...	...	...	...	...
67	177	101	101	101	101	101	101	101	...	...	...	...	...	...	...
68	177	115	115	115	115	115	115	115	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated											
Line No.	Nominal Composition	Product Form	Spec. No.	UNS No.	Class/Condition/ Temper	Size, mm	P-No. (5)(7)	Notes	Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa
69	88Cu-9Al-3Fe	Castings	B148	C95200	...	...	35	(9)	-254	450	170
70	89Cu-10Al-1Fe	Castings	B148	C95300	...	...	35	(9)	-254	450	170
71	90Cu-7Al-3Si	Castings	B148	C95600	...	...	35	...	-198	415	195
72	85Cu-11Al-4Fe	Castings	B148	C95400	...	...	35	...	-198	515	205
73	58Cu-34Zn-2Fe-2Al-2Mn	Castings	B584	C86700	...	...	a	...	-198	552	221
74	82Cu-11Al-4Fe-3Mn	Castings	B148	C95500	...	...	35	...	-269	620	275
75	63Cu-27Zn-4Al-3Fe-3Mn	Castings	B584	C86200	...	...	b	...	-198	621	310
76	61Cu-27Zn-6Al-3Fe-3Mn	Castings	B584	C86300	...	...	b	...	-198	758	414
77	75Cu-21.5Zn-3Si	Rod	B371	C69300	H02	≤12	a	...	-198	585	310
78	75Cu-21.5Zn-3Si	Rod	B371	C69300	H02	>12, ≤25	a	...	-198	515	240
79	75Cu-21.5Zn-3Si	Rod	B371	C69300	H02	>25, ≤50	a	...	-198	480	205

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated															
Line No.	Max. Temp., °C	Min. Temp. to 40	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]												
			65	100	125	150	175	200	225	250	275	300	325	350	375
69	316	115	108	104	102	99.8	98.6	97.8	97.5	97.4	97.4	65.9	43.7	...	...
70	316	115	115	115	115	115	115	115	115	115	115	115	115	...	...
71	40	129	...	...	...	...	...	...	...	...	...	...	...	...	...
72	260	138	131	128	128	128	128	128	128	101	95.8	...	...	...	...
73	177	147	147	147	147	147	147	147	...	...	...	...	...	...	...
74	260	184	184	184	184	184	184	184	184	184	184	...	...	...	...
75	177	207	207	207	207	207	207	207	...	...	...	...	...	...	...
76	177	253	253	253	253	253	253	253	...	...	...	...	...	...	...
77	149	195	179	176	176	176	...	...	...	...	...	...	...	...	...
78	149	161	139	137	137	137	...	...	...	...	...	...	...	...	...
79	149	138	119	117	117	117	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
1	99.0Ni-Low C	Pipe & tube	B161	...	N02201	Annealed	>125	41
2	99.0Ni-Low C	Pipe & tube	B725	...	N02201	Annealed	>127	41
3	99.0Ni	Pipe & tube	B161	...	N02200	Annealed	>125	41
4	99.0Ni	Pipe & tube	B725	...	N02200	Annealed	>127	41
5	99.0Ni-Low C	Pipe & tube	B161	...	N02201	Annealed	≤125	41
6	99.0Ni-Low C	Pipe & tube	B725	...	N02201	Annealed	≤127	41
7	99.0Ni	Pipe & tube	B161	...	N02200	Annealed	≤125	41
8	99.0Ni	Pipe & tube	B725	...	N02200	Annealed	≤127	41
9	67Ni-30Cu	Pipe & tube	B165	...	N04400	Annealed	>125	42
10	67Ni-30Cu	Pipe & tube	B725	...	N04400	Annealed	>127	42
11	33Ni-42Fe-21Cr	Pipe & tube	B407	...	N08800	H.F. or H.F. ann.	...	45
12	72Ni-15Cr-8Fe	Pipe & tube	B167	...	N06600	H.F. or H.F. ann.	>125	43
13	33Ni-42Fe-21Cr	Pipe & tube	B407	...	N08810	C.D. sol. ann. or H.F. ann.	...	45
14	33Ni-42Fe-21Cr	Pipe & tube	B514	...	N08810	Annealed	...	45
15	33Ni-42Fe-21Cr-Al-Ti	Pipe & tube	B407	...	N08811	C.D. sol. ann. or H.F. ann.	...	45
16	67Ni-30Cu	Pipe & tube	B165	...	N04400	Annealed	≤125	42
17	67Ni-30Cu	Pipe & tube	B725	...	N04400	Annealed	≤127	42
18	26Ni-22Cr-5Mo-Ti	Pipe & tube	B619	...	N08320	Sol. ann.	...	45
19	26Ni-22Cr-5Mo-Ti	Pipe & tube	B622	...	N08320	Sol. ann.	...	45
20	99.0Ni-Low C	Pipe & tube	B161	...	N02201	Str. rel.	...	41
21	99.0Ni-Low C	Pipe & tube	B725	...	N02201	Str. rel.	...	41
22	33Ni-42Fe-21Cr	Pipe & tube	B514	...	N08800	Annealed	...	45
23	72Ni-15Cr-8Fe	Pipe & tube	B167	...	N06600	H.F. or H.F. ann.	≤125	43
24	72Ni-15Cr-8Fe	Pipe & tube	B167	...	N06600	C.D. ann.	>125	43
25	33Ni-42Fe-21Cr	Pipe & tube	B407	...	N08800	C.D. ann.	...	45
26	31Ni-31Fe-29Cr-Mo	Pipe & tube	B668	...	N08028	Sol. ann.	...	45
27	99.0Ni	Pipe & tube	B161	...	N02200	Str. rel.	...	41
28	99.0Ni	Pipe & tube	B725	...	N02200	Str. rel.	...	41
29	35Ni-35Fe-20Cr-Cb	Pipe & tube	B464	...	N08020	Annealed	...	45
30	35Ni-35Fe-20Cr-Cb	Pipe & tube	B474	...	N08020	Annealed	...	45
31	35Ni-35Fe-20Cr-Cb	Pipe & tube	B729	...	N08020	Annealed	...	45
32	42Ni-21.5Cr-3Mo-2.3Cu	Smls. tube	B163	...	N08825	Annealed	...	45
33	42Ni-21.5Cr-3Mo-2.3Cu	Pipe & tube	B423	...	N08825	C.D. ann.	...	45
34	42Ni-21.5Cr-3Mo-2.3Cu	Pipe & tube	B474	...	N08825	Annealed	...	45

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated														
Line No.	Notes	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]												
		Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Min. Temp. to 40								
							65	100	125	150	175	200	225	250
1	...	-198	345	70	649	46.0	44.7	43.9	43.6	43.3	43.2	43.2	43.2	43.2
2	...	-198	345	70	649	46.0	44.7	43.9	43.6	43.3	43.2	43.2	43.2	43.2
3	...	-198	380	80	316	55.2	55.2	55.2	55.2	55.2	55.2	55.2	55.2	55.2
4	...	-198	380	80	316	55.2	55.2	55.2	55.2	55.2	55.2	55.2	55.2	55.2
5	...	-198	345	80	649	55.2	53.8	52.8	52.3	51.9	51.7	51.6	51.6	51.6
6	...	-198	345	80	649	55.2	53.8	52.8	52.3	51.9	51.7	51.6	51.6	51.6
7	...	-198	380	105	316	68.9	68.9	68.9	68.9	68.9	68.9	68.9	68.9	68.9
8	...	-198	380	105	316	68.9	68.9	68.9	68.9	68.9	68.9	68.9	68.9	68.9
9	...	-198	480	170	482	115	106	99.7	96.2	93.6	91.9	90.9	90.4	90.4
10	...	-198	480	170	482	115	106	99.7	96.2	93.6	91.9	90.9	90.4	90.4
11	...	-198	450	170	816	115	115	115	115	115	115	115	115	115
12	...	-198	515	170	649	115	115	115	115	115	115	115	115	115
13	(62)	-198	450	170	899	115	115	115	115	115	115	115	115	115
14	(62)	-198	450	170	899	115	115	115	115	115	115	115	115	115
15	(62)	-198	450	170	899	115	115	115	115	115	115	115	115	115
16	...	-198	480	195	482	129	119	112	108	105	103	102	101	101
17	...	-198	480	195	482	129	119	112	108	105	103	102	101	101
18	...	-198	517	193	427	129	129	129	129	129	129	129	129	129
19	...	-198	517	193	427	129	129	129	129	129	129	129	129	129
20	...	-198	415	205	316	<i>138</i>	<i>138</i>	<i>138</i>	<i>137</i>	<i>137</i>	<i>137</i>	<i>137</i>	<i>137</i>	<i>136</i>
21	...	-198	415	205	316	<i>138</i>	<i>138</i>	<i>138</i>	<i>137</i>	<i>137</i>	<i>137</i>	<i>137</i>	<i>137</i>	<i>136</i>
22	...	-198	520	207	816	138	138	138	138	138	138	138	138	138
23	...	-198	550	205	649	138	138	138	138	138	138	138	138	138
24	...	-198	550	205	649	138	138	138	138	138	138	138	138	138
25	(61)	-198	520	205	816	138	138	138	138	138	138	138	138	138
26	...	-198	500	214	454	143	<i>143</i>	<i>143</i>	<i>143</i>	<i>143</i>	<i>143</i>	<i>143</i>	<i>143</i>	<i>143</i>
27	...	-198	450	275	316	<i>149</i>	<i>149</i>	<i>149</i>	<i>149</i>	<i>149</i>	<i>149</i>	<i>149</i>	<i>149</i>	<i>148</i>
28	...	-198	450	275	316	<i>149</i>	<i>149</i>	<i>149</i>	<i>149</i>	<i>149</i>	<i>149</i>	<i>149</i>	<i>149</i>	<i>148</i>
29	...	-198	551	241	427	161	161	161	161	161	161	161	161	161
30	...	-198	551	241	427	161	161	161	161	161	161	161	161	161
31	...	-198	551	240	427	161	161	161	161	161	161	161	161	161
32	...	-198	586	241	538	161	161	161	161	161	161	161	161	161
33	...	-198	586	241	538	161	161	161	161	161	161	161	161	161
34	...	-198	586	241	538	161	161	161	161	161	161	161	161	161

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]												
	275	300	325	350	375	400	425	450	475	500	525	550	575
1	43.2	43.2	43.1	42.9	42.6	42.1	41.5	40.7	33.1	27.4	22.8	18.7	15.6
2	43.2	43.2	43.1	42.9	42.6	42.1	41.5	40.7	33.1	27.4	22.8	18.7	15.6
3	55.2	55.2	55.2	...	...	...	...	...	...	...	...	...	...
4	55.2	55.2	55.2	...	...	...	...	...	...	...	...	...	...
5	51.6	51.6	51.6	51.5	51.2	50.7	49.9	41.4	33.1	27.4	22.8	18.7	15.6
6	51.6	51.6	51.6	51.5	51.2	50.7	49.9	41.4	33.1	27.4	22.8	18.7	15.6
7	68.9	68.9	68.9	...	...	...	...	...	...	...	...	...	...
8	68.9	68.9	68.9	...	...	...	...	...	...	...	...	...	...
9	90.4	90.4	90.4	90.4	89.8	89.0	88.0	79.7	59.9	55.2	...	...	...
10	90.4	90.4	90.4	90.4	89.8	89.0	88.0	79.7	59.9	55.2	...	...	...
11	115	115	115	115	115	115	115	115	115	115	115	113	107
12	115	115	115	115	115	115	115	115	115	84.8	58.4	39.7	27.0
13	115	115	113	110	108	106	104	102	100	98.6	97.1	95.7	91.8
14	115	115	113	110	108	106	104	102	100	98.6	97.1	95.7	91.8
15	115	115	113	110	108	106	104	102	100	98.6	97.1	95.7	92.5
16	101	101	101	101	100	99.6	98.6	79.7	59.9	55.2	...	...	...
17	101	101	101	101	100	99.6	98.6	79.7	59.9	55.2	...	...	...
18	129	129	127	125	122	121	119	119	...	...	...	...	...
19	129	129	127	125	122	121	119	119	...	...	...	...	...
20	135	133	130	...	...	...	...	...	...	...	...	...	...
21	135	133	130	...	...	...	...	...	...	...	...	...	...
22	138	138	138	138	138	138	138	138	138	138	138	136	107
23	138	138	138	138	138	138	138	138	138	84.8	58.4	39.7	27.0
24	138	138	138	138	138	138	138	138	138	84.8	58.4	39.7	27.0
25	138	138	138	138	138	138	138	138	138	138	138	136	107
26	140	137	133	130	125	122	119	116	113	...	...	...	...
27	146	144	141	...	...	...	...	...	...	...	...	...	...
28	146	144	141	...	...	...	...	...	...	...	...	...	...
29	161	161	161	161	161	160	157	156	...	...	...	...	...
30	161	161	161	161	161	160	157	156	...	...	...	...	...
31	161	161	161	161	161	160	157	156	...	...	...	...	...
32	161	161	161	161	161	160	159	158	157	156	155	153	...
33	161	161	161	161	161	160	159	158	157	156	155	153	...
34	161	161	161	161	161	160	159	158	157	156	155	153	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1) , (2b)]												
	600	625	650	675	700	725	750	775	800	825	850	875	900
1	12.9	10.0	8.27	...	...	...	...	...	...	...	...	...	...
2	12.9	10.0	8.27	...	...	...	...	...	...	...	...	...	...
3	...	...	...	...	...	...	...	...	...	...	...	...	...
4	...	...	...	...	...	...	...	...	...	...	...	...	...
5	12.9	10.0	8.27	...	...	...	...	...	...	...	...	...	...
6	12.9	10.0	8.27	...	...	...	...	...	...	...	...	...	...
7	...	...	...	...	...	...	...	...	...	...	...	...	...
8	...	...	...	...	...	...	...	...	...	...	...	...	...
9	...	...	...	...	...	...	...	...	...	...	...	...	...
10	...	...	...	...	...	...	...	...	...	...	...	...	...
11	83.8	63.9	44.7	29.8	15.5	11.7	8.68	7.20	6.25	5.11	...	...	...
12	19.2	15.0	13.8	...	...	...	...	...	...	...	...	...	...
13	75.7	62.6	50.6	41.2	33.6	27.7	22.6	18.3	15.0	11.9	9.03	7.35	5.86
14	75.7	62.6	50.6	41.2	33.6	27.7	22.6	18.3	15.0	11.9	9.03	7.35	5.86
15	84.5	69.5	56.7	46.8	38.5	31.5	25.5	20.7	17.0	13.9	11.2	9.33	7.58
16	...	...	...	...	...	...	...	...	...	...	...	...	...
17	...	...	...	...	...	...	...	...	...	...	...	...	...
18	...	...	...	...	...	...	...	...	...	...	...	...	...
19	...	...	...	...	...	...	...	...	...	...	...	...	...
20	...	...	...	...	...	...	...	...	...	...	...	...	...
21	...	...	...	...	...	...	...	...	...	...	...	...	...
22	83.8	63.9	44.7	29.8	15.5	11.7	8.68	7.20	6.25	5.11	...	...	...
23	19.2	15.0	13.8	...	...	...	...	...	...	...	...	...	...
24	19.2	15.0	13.8	...	...	...	...	...	...	...	...	...	...
25	83.8	63.9	44.7	29.8	15.5	11.7	8.68	7.20	6.25	5.11	...	...	...
26	...	...	...	...	...	...	...	...	...	...	...	...	...
27	...	...	...	...	...	...	...	...	...	...	...	...	...
28	...	...	...	...	...	...	...	...	...	...	...	...	...
29	...	...	...	...	...	...	...	...	...	...	...	...	...
30	...	...	...	...	...	...	...	...	...	...	...	...	...
31	...	...	...	...	...	...	...	...	...	...	...	...	...
32	...	...	...	...	...	...	...	...	...	...	...	...	...
33	...	...	...	...	...	...	...	...	...	...	...	...	...
34	...	...	...	...	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
35	42Ni-21.5Cr-3Mo-2.3Cu	Wld. tube	B704	...	N08825	Annealed	...	45
36	42Ni-21.5Cr-3Mo-2.3Cu	Pipe & tube	B705	...	N08825	Annealed	...	45
37	47Ni-22Cr-19Fe-6Mo	Pipe & tube	B619	...	N06007	Sol. ann.	...	45
38	47Ni-22Cr-19Fe-6Mo	Pipe & tube	B622	...	N06007	Sol. ann.	...	45
39	40Ni-29Cr-15Fe-5Mo	Pipe & tube	B619	...	N06030	Sol. ann.	...	45
40	40Ni-29Cr-15Fe-5Mo	Pipe & tube	B622	...	N06030	Sol. ann.	...	45
41	40Ni-29Cr-15Fe-5Mo	Pipe & tube	B626	...	N06030	Sol. ann.	...	45
42	72Ni-15Cr-8Fe	Pipe & tube	B167	...	N06600	C.D. ann.	≤125	43
43	72Ni-15Cr-8Fe	Pipe & tube	B517	...	N06600	C.D. ann.	...	43
44	58Ni-29Cr-9Fe	Tube	B163	...	N06690	Annealed	≤75	43
45	58Ni-29Cr-9Fe	Pipe & tube	B167	...	N06690	C.D. ann.	≤125	43
46	37Ni-33Fe-25Cr	Pipe & tube	B163	...	N08120	Sol. ann.	...	45
47	37Ni-33Fe-25Cr	Pipe & tube	B407	...	N08120	Sol. ann.	...	45
48	37Ni-33Fe-25Cr	Pipe & tube	B514	...	N08120	Sol. ann.	...	45
49	37Ni-33Fe-25Cr	Pipe & tube	B515	...	N08120	Sol. ann.	...	45
50	61Ni-16Mo-16Cr	Pipe & tube	B619	...	N06455	Sol. ann.	...	43
51	47Ni-22Cr-9Mo-18Fe	Pipe & tube	B619	...	N06002	Sol. ann.	...	43
52	47Ni-22Cr-9Mo-18Fe	Pipe & tube	B622	...	N06002	Sol. ann.	...	43
53	31Ni-33Fe-27Cr-6.5Mo-Cu-N	Pipe & tube	B619	...	N08031	Annealed	...	45
54	31Ni-33Fe-27Cr-6.5Mo-Cu-N	Pipe & tube	B622	...	N08031	Annealed	...	45
55	61Ni-16Mo-16Cr	Pipe & tube	B622	...	N06455	Sol. ann.	...	43
56	54Ni-16Mo-15Cr	Pipe & tube	B619	...	N10276	Sol. ann.	...	43
57	54Ni-16Mo-15Cr	Pipe & tube	B622	...	N10276	Sol. ann.	...	43
58	54Ni-16Mo-15Cr	Pipe & tube	B626	...	N10276	Sol. ann.	...	43
59	67Ni-30Cu	Pipe & tube	B165	...	N04400	Str. rel.	...	42
60	67Ni-30Cu	Pipe & tube	B725	...	N04400	Str. rel.	...	42
61	46Fe-24Ni-21Cr-6Mo-Cu-N	Pipe & tube	B675	...	N08367	Annealed	>5	45
62	46Fe-24Ni-21Cr-6Mo-Cu-N	Pipe & tube	B690	...	N08367	Annealed	>5	45
63	46Fe-24Ni-21Cr-6Mo-Cu-N	Pipe & tube	B804	...	N08367	Annealed	>5	45
64	46Fe-24Ni-21Cr-6Mo-Cu-N	Pipe & tube	B675	...	N08367	Annealed	≤5	45
65	46Fe-24Ni-21Cr-6Mo-Cu-N	Pipe & tube	B690	...	N08367	Annealed	≤5	45
66	46Fe-24Ni-21Cr-6Mo-Cu-N	Pipe & tube	B804	...	N08367	Annealed	≤5	45

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated														
Line No.	Notes	Basic Allowable Stress, S, MPa, at Metal Temperature, °C												
		[Notes (1), (2b)]												
		Min. Temp., °C (6)	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Max. Temp., °C	Min. Temp. to 40	65	100	125	150	175	200	225	250
35	...	−198	586	241	538	161	161	161	161	161	161	161	161	161
36	...	−198	586	241	538	161	161	161	161	161	161	161	161	161
37	...	−198	621	241	538	161	161	161	161	161	161	161	161	161
38	...	−198	621	241	427	161	161	161	161	161	161	161	161	161
39	...	−198	586	241	427	161	161	161	161	161	161	161	157	154
40	...	−198	586	241	427	161	161	161	161	161	161	161	157	154
41	...	−198	586	241	427	161	161	161	161	161	161	161	157	154
42	...	−198	550	240	649	161	161	161	161	161	161	161	161	161
43	...	−198	550	240	649	161	161	161	161	161	161	161	161	161
44	...	−198	586	241	482	160	160	160	160	160	160	160	160	160
45	...	−198	586	240	482	160	160	160	160	160	160	160	160	160
46	...	−198	620	276	899	184	184	184	184	184	184	184	180	175
47	...	−198	621	276	899	184	184	184	184	184	184	184	180	175
48	...	−198	621	276	899	184	184	184	184	184	184	184	180	175
49	...	−198	621	276	899	184	184	184	184	184	184	184	180	175
50	...	−198	690	276	427	184	184	184	184	184	184	184	184	184
51	...	−198	690	276	816	184	184	184	184	184	184	184	184	178
52	...	−198	689	276	816	184	184	184	184	184	184	184	184	178
53	...	−198	650	276	427	184	184	184	184	184	177	171	166	162
54	...	−198	650	276	427	184	184	184	184	184	177	171	166	162
55	...	−198	690	276	427	184	184	184	184	184	184	184	184	184
56	...	−198	689	283	677	188	188	188	188	188	188	188	188	187
57	...	−198	689	283	677	188	188	188	188	188	188	188	188	187
58	...	−198	690	283	677	188	188	188	188	188	188	188	188	187
59	(54)	−198	585	380	260	195	195	195	195	195	195	195	195	195
60	(54)	−198	585	380	260	195	195	195	195	195	195	195	195	195
61	...	−198	655	310	427	207	207	207	207	206	202	198	195	192
62	...	−198	655	310	427	207	207	207	207	206	202	198	195	192
63	...	−198	655	310	427	207	207	207	207	206	202	198	195	192
64	...	−198	690	310	427	207	207	207	207	206	202	198	195	192
65	...	−198	690	310	427	207	207	207	207	206	202	198	195	192
66	...	−198	690	310	427	207	207	207	207	206	202	198	195	192

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]												
	275	300	325	350	375	400	425	450	475	500	525	550	575
35	161	161	161	161	161	160	159	158	157	156	155	153	...
36	161	161	161	161	161	160	159	158	157	156	155	153	...
37	160	158	156	154	153	152	151	150	149	136	132	130	...
38	160	158	156	154	153	152	151	150	...	...	...	...	...
39	151	148	146	143	141	138	136	136	...	...	...	...	...
40	151	148	146	143	141	138	136	136	...	...	...	...	...
41	151	148	146	143	141	138	136	136	...	...	...	...	...
42	161	161	161	161	161	161	161	161	161	84.8	58.4	39.7	27.0
43	161	161	161	161	161	161	161	161	161	84.8	58.4	39.7	27.0
44	160	160	160	160	160	160	160	160	160	160	...	...	...
45	160	160	160	160	160	160	160	160	160	160	...	...	...
46	170	166	163	160	158	156	154	153	153	152	151	151	151
47	170	166	163	160	158	156	154	153	153	152	151	151	151
48	170	166	163	160	158	156	154	153	153	152	151	151	151
49	170	166	163	160	158	156	154	153	153	152	151	151	151
50	184	184	184	184	183	180	178	178	...	...	...	...	...
51	173	169	165	162	160	158	157	155	154	135	134	133	129
52	173	169	165	162	160	158	157	155	154	135	134	133	129
53	158	155	152	149	74.2	61.1	50.3	49.6	...	...	...	...	...
54	158	155	152	149	74.2	61.1	50.3	49.6	...	...	...	...	...
55	184	184	184	184	182	180	178	178	...	...	...	...	...
56	183	177	172	169	165	162	159	157	156	155	154	143	118
57	183	177	172	169	165	162	159	157	156	155	154	143	118
58	183	177	172	169	165	162	159	157	156	155	154	143	118
59	195	...	...	...	...	...	...	...	...	...	...	...	...
60	195	...	...	...	...	...	...	...	...	...	...	...	...
61	188	183	179	176	173	170	168	167	...	...	...	...	...
62	188	183	179	176	173	170	168	167	...	...	...	...	...
63	188	183	179	176	173	170	168	167	...	...	...	...	...
64	188	183	179	176	173	170	168	167	...	...	...	...	...
65	188	183	179	176	173	170	168	167	...	...	...	...	...
66	188	183	179	176	173	170	168	167	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated													
Line No.	Basic Allowable Stress, S, MPa, at Metal Temperature, °C [Notes (1), (2b)]												
	600	625	650	675	700	725	750	775	800	825	850	875	900
35	...	...	...	...	...	...	...	...	...	...	...	...	...
36	...	...	...	...	...	...	...	...	...	...	...	...	...
37	...	...	...	...	...	...	...	...	...	...	...	...	...
38	...	...	...	...	...	...	...	...	...	...	...	...	...
39	...	...	...	...	...	...	...	...	...	...	...	...	...
40	...	...	...	...	...	...	...	...	...	...	...	...	...
41	...	...	...	...	...	...	...	...	...	...	...	...	...
42	19.2	15.0	13.8	...	...	...	...	...	...	...	...	...	...
43	19.2	15.0	13.8	...	...	...	...	...	...	...	...	...	...
44	...	...	...	...	...	...	...	...	...	...	...	...	...
45	...	...	...	...	...	...	...	...	...	...	...	...	...
46	117	96.2	79.3	65.6	54.4	45.0	37.2	30.6	25.1	20.5	16.5	13.0	9.8
47	117	96.2	79.3	65.6	54.4	45.0	37.2	30.6	25.1	20.5	16.5	13.0	9.8
48	117	96.2	79.3	65.6	54.4	45.0	37.2	30.6	25.1	20.5	16.5	13.0	9.8
49	117	96.2	79.3	65.6	54.4	45.0	37.2	30.6	25.1	20.5	16.5	13.0	9.8
50	...	...	...	...	...	...	...	...	...	...	...	...	...
51	115	94.2	77.3	64.9	54.7	44.7	36.1	29.2	23.6	20.7	...	...	...
52	115	94.2	77.3	64.9	54.7	44.7	36.1	29.2	23.6	20.7	...	...	...
53	...	...	...	...	...	...	...	...	...	...	...	...	...
54	...	...	...	...	...	...	...	...	...	...	...	...	...
55	...	...	...	...	...	...	...	...	...	...	...	...	...
56	99.1	81.6	67.0	54.6	42.2	...	...	...	...	...	...	...	...
57	99.1	81.6	67.0	54.6	42.2	...	...	...	...	...	...	...	...
58	99.1	81.6	67.0	54.6	42.2	...	...	...	...	...	...	...	...
59	...	...	...	...	...	...	...	...	...	...	...	...	...
60	...	...	...	...	...	...	...	...	...	...	...	...	...
61	...	...	...	...	...	...	...	...	...	...	...	...	...
62	...	...	...	...	...	...	...	...	...	...	...	...	...
63	...	...	...	...	...	...	...	...	...	...	...	...	...
64	...	...	...	...	...	...	...	...	...	...	...	...	...
65	...	...	...	...	...	...	...	...	...	...	...	...	...
66	...	...	...	...	...	...	...	...	...	...	...	...	...

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Table A-1
Basic Allowable Stresses in Tension for Metals — SI Units (Cont'd)

Numbers in Parentheses Refer to Notes for Tables A-1 and A-1C ; Specifications Are ASTM Unless Otherwise Indicated								
Line No.	Nominal Composition	Product Form	Spec. No.	Type/ Grade	UNS No.	Class/ Condition/ Temper	Size, mm	P-No. (5)(7)
67	55Ni-21Cr-13.5Mo	Pipe & tube	B619	...	N06022	Sol. ann.	...	43
68	55Ni-21Cr-13.5Mo	Pipe & tube	B622	...	N06022	Sol. ann.	...	43
69	58Ni-33Cr-8Mo	Pipe & tube	B619	...	N06035	Sol. ann.	...	43
70	58Ni-33Cr-8Mo	Pipe & tube	B622	...	N06035	Sol. ann.	...	43
71	58Ni-33Cr-8Mo	Pipe & tube	B626	...	N06035	Sol. ann.	...	43
72	59Ni-23Cr-16Mo	Pipe & tube	B619	...	N06059	Sol. ann.	...	43
73	59Ni-23Cr-16Mo	Pipe & tube	B622	...	N06059	Sol. ann.	...	43
74	59Ni-23Cr-16Mo	Wld. tube	B626	...	N06059	Sol. ann.	All	43
75	59Ni-23Cr-16Mo-1.6Cu	Pipe & tube	B619	...	N06200	Sol. ann.	All	43
76	59Ni-23Cr-16Mo-1.6Cu	Pipe & tube	B622	...	N06200	Sol. ann.	All	43
77	59Ni-23Cr-16Mo-1.6Cu	Pipe & tube	B626	...	N06200	Sol. ann.	All	43
78	62Ni-22Mo-15Cr	Pipe & tube	B619	...	N10362	Sol. ann.	All	43
79	62Ni-22Mo-15Cr	Pipe & tube	B622	...	N10362	Sol. ann.	All	43
80	62Ni-22Mo-15Cr	Pipe & tube	B626	...	N10362	Sol. ann.	All	43
81	62Ni-28Mo-5Fe	Pipe & tube	B619	...	N10001	Sol. ann.	...	44
82	62Ni-28Mo-5Fe	Pipe & tube	B622	...	N10001	Sol. ann.	...	44
83	65Ni-28Mo-2Fe	Pipe & tube	B619	...	N10665	Sol. ann.	...	44
84	65Ni-28Mo-2Fe	Pipe & tube	B622	...	N10665	Sol. ann.	...	44
85	65Ni-29.5Mo-2Fe-2Cr	Pipe & tube	B619	...	N10675	Sol. ann.	...	44
86	65Ni-29.5Mo-2Fe-2Cr	Pipe & tube	B622	...	N10675	Sol. ann.	...	44
87	65Ni-29.5Mo-2Fe-2Cr	Pipe & tube	B626	...	N10675	Sol. ann.	...	44
88	60Ni-22Cr-9Mo-3.5Cb	Pipe & tube	B444	1	N06625	Annealed	...	43
89	60Ni-22Cr-9Mo-3.5Cb	Pipe & tube	B705	1	N06625	Annealed	...	43
90	57Ni-22Cr-14W-2Mo-La	Pipe & tube	B619	...	N06230	Sol. ann.	...	43
91	57Ni-22Cr-14W-2Mo-La	Pipe & tube	B622	...	N06230	Sol. ann.	...	43
92	57Ni-22Cr-14W-2Mo-La	Pipe & tube	B626	...	N06230	Sol. ann.	...	43
93	33Cr-31Ni-32Fe-1.5Mo-0.6Cu-N	Pipe	B619	...	R20033	Sol. ann.	...	45
94	33Cr-31Ni-32Fe-1.5Mo-0.6Cu-N	Pipe & tube	B622	...	R20033	Sol. ann.	...	45
95	33Cr-31Ni-32Fe-1.5Mo-0.6Cu-N	Tube	B626	...	R20033	Sol. ann.	...	45
96	99.0Ni-Low C	Plate	B162	...	N02201	H.R. ann.	...	41
97	99.0Ni-Low C	Plate	B162	...	N02201	H.R. as R.	...	41
98	99.0Ni	Plate	B162	...	N02200	H.R. ann.	...	41
99	99.0Ni	Plate	B162	...	N02200	H.R. as R.	...	41
100	33Ni-42Fe-21Cr	Pl. & sht.	B409	...	N08810	Annealed	All	45
101	33Ni-42Fe-21Cr-Al-Ti	Pl. & sht.	B409	...	N08811	Annealed	All	45